

# INTEGRATION GUIDE



## XiIntercept eCommerce with XiSecure



## XiIntercept eCommerce with XiSecure Integration Guide

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| Version          | Description of Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20200422-1       | Updated resource domain URLs through-out.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 20200330-1       | Removed empty section heading. This entire guide supports XiSecure token integration only. See XiIntercept eComm with TMS Integration guide for the option to integrate one or many token types.                                                                                                                                                                                                                                                                                                                                                                                       |
| 20200326-1       | Corrected JS Plugin samples for Form Post and Ajax. Changed Intro and title to reflect this document supports integration to XiSecure only.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 20190701-1       | Removed xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" from all XML samples.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 20190621-1       | <ul style="list-style-type: none"> <li>Added 3DS 2 content and samples under the Get Access Token sections for <a href="#">Form Post</a><sup>[17]</sup>, <a href="#">Ajax</a><sup>[25]</sup>, <a href="#">Iframe</a><sup>[35]</sup>, <a href="#">Hosted</a><sup>[51]</sup> and <a href="#">3DS with Token</a><sup>[62]</sup>.</li> <li>Updated the XIPlugin URL path to the new version that supports 3DS2 for <a href="#">Form Post</a><sup>[21]</sup> and <a href="#">Ajax</a><sup>[29]</sup>.</li> <li>All samples were reviewed against the XSD and modified as needed.</li> </ul> |
| 20190403-1       | In <a href="#">Child Element: &lt;cardTypeIndicator&gt;</a> <sup>[13]</sup> section, added reference to the new unselected class that gives further control of styling the unselected card type logos.                                                                                                                                                                                                                                                                                                                                                                                 |
| 20190215-1       | Added mask-number attribute details in Custom HTML Reference section, <a href="#">Textbox Elements: &lt;tbxx{XXX}&gt;</a> <sup>[135]</sup> topic.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 20181128-1       | In <a href="#">Common Attributes:</a> <sup>[116]</sup> section, changed is-default value from "true" to "1" to make that value the default.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 20181113-1       | Corrected typo in intro sentence for <a href="#">iFrame sequence diagram</a> <sup>[33]</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 20181023-1       | Correct <a href="#">AliPay sequence diagram</a> <sup>[69]</sup> . Correct Iframe JavaScript URL in <a href="#">Iframe: On-Submit Javascript Plugin</a> <sup>[49]</sup> topic.                                                                                                                                                                                                                                                                                                                                                                                                          |
| 20181004-1       | Corrected <a href="#">Ajax Access Token with 3DS XML</a> <sup>[28]</sup> example. Added variable placeholders to each Access Token XML example.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 20181001-1       | Added <a href="#">3DS with Token</a> <sup>[61]</sup> section. Updated Ajax example in <a href="#">Ajax: On-Submit Javascript Plugin</a> <sup>[29]</sup> section.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 20180906-1       | Formatting and wording changes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 20180824-1       | Added <a href="#">AliPay</a> <sup>[69]</sup> and <a href="#">Visa Checkout</a> <sup>[76]</sup> sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 20180823-1       | New structure and format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Jun-2018-3110-R3 | In <a href="#">\$XIframe.onload</a> <sup>[106]</sup> section, corrected description for onCardTypeChange and wording in the first paragraph.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| May-2018-3110-R2 | Added 150 to <a href="#">Error Codes</a> <sup>[97]</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Jan-2018-3100-R2 | Corrected verbiage in Integration Type Overviews to clarify Post call is made from Client's browser. Edited Form Post workflow to include Post call in Client browser swimlane.                                                                                                                                                                                                                                                                                                                                                                                                        |

PAYMETRIC® XiIntercept™ for eCommerce (XiIntercept eComm) utilizes XML packets to submit sensitive data directly to Paymetric's SOC 1 compliant data center where it is encrypted and securely stored by XiSecure®. A Token is returned for use by the Merchant's application; thereby, removing sensitive data from the Merchant's environment.

XiIntercept eComm is used to tokenize data; authorization and settlement functionality is implemented directly through Paymetric's XiPay® web service.

### Purpose

The purpose of the document is to provide an understanding of how to integrate to XiIntercept eComm for XiSecure tokens only. If you want to integrate to the Token Management Service which provides the flexibility of using all Worldpay token types, please refer to the *XiIntercept eCommerce XiSecure Token Integration* guide.

### Audience

This document is intended for use by Web Developers and assumes a familiarity with XML and XSD.

### Browser Compatibility

XiIntercept eComm is compatible with the following browsers:

- Internet Explorer 10.0 and above
- Firefox (latest version)
- Chrome (latest version)
- Safari (latest version)

### Mobile Device Support

XiIntercept eComm is compatible with both IOS and Android devices. Use the same integration steps as documented in this guide for these devices.

Onboarding for the following standard raw card integration types only requires submitting the environment request via the Onboarding page:

- Form Post
- Ajax
- Iframe
- Hosted

If you are implementing 3D Secure or AliPay with XiIntercept eComm, you must obtain the following from Cardinal Commerce before you can submit your XiIntercept eComm environment requests.

- Cardinal Commerce Merchant ID
- Processor ID
- Password

If you are implementing Visa Checkout, you must obtain the following from Visa before you can submit your XiIntercept eComm environment requests.

- Visa API Key
- Visa Shared Secret

The identify of the system making calls to the XiIntercept eComm platform is authenticated using the following security elements and URL endpoints:

Table 1: Security Elements for XiIntercept eComm

| Security Element | Description                                                                                                                                                                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Merchant GUID    | A unique ID assigned by Paymetric that is included in the Request packet. Each Merchant has a single GUID.                                                                                                                                                                                                                                                          |
| Shared Key       | A string generated and provided by Paymetric used to "sign" the Request Packet. Paymetric uses the same shared key to confirm the signature.                                                                                                                                                                                                                        |
| URL Endpoint     | <p>Following are the URL endpoints used to initiate the first application server call and obtain an AccessToken.</p> <p>Dev/QA:</p> <pre>https://cert-xiecomm.paymetric.com/DIeComm/AccessToken</pre> <p>Production:</p> <pre>https://xiecomm.paymetric.com/DIeComm/AccessToken</pre>                                                                               |
| Access Token     | <p>A string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key.</p> <p>The returned AccessToken is then used in all subsequent calls within the given session. The AccessToken is valid for 30 minutes.</p> |

Review this section to determine which type(s) are applicable for your implementation.

### Raw Value Tokenization XiSecure Tokens Only

Form Post, Ajax, Iframe and Hosted are the XiIntercept eComm "Standard Integration Types" used to tokenize directly-entered information (i.e. not from some sort of wallet functionality), such as credit cards, account numbers for echeck payments, and other non-payment related sensitive data like Social Security Numbers. These standard integration types all use the same XiIntercept eComm URL endpoints to trigger each of the calls.

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form Post | <ul style="list-style-type: none"><li>• A POST request is made to XiIntercept eComm from Client's browser.</li><li>• XiIntercept eComm redirects to the URL defined by the Merchant; additional fields aside from what is being tokenized can be sent in the XML packet.</li><li>• The page has a high-level of page responsiveness since it is completely hosted by Merchant.</li><li>• Merchant is responsible for all card validations.</li><li>• It is PCI v3.0 - SAQ A-EP compliant (see PCI Notes below).</li></ul> |
| Ajax      | <ul style="list-style-type: none"><li>• An Ajax call to XiIntercept eComm from the Client's browser.</li><li>• User never leaves page.</li><li>• The page has a high-level of page responsiveness since it is completely hosted by Merchant.</li><li>• Merchant is responsible for all card validations.</li><li>• It is PCI v3.0 - SAQ A-EP compliant (see PCI Notes below).</li></ul>                                                                                                                                   |
| Iframe    | <ul style="list-style-type: none"><li>• Paymetric hosts the content of the Iframe.</li><li>• Merchant can style and control what fields display; there are multiple content templates or Merchant can completely customize.</li><li>• The page has a potential delay while loading the Iframe; Merchant could hide all contents of page while Iframe loads.</li><li>• It is PCI v3.0 - SAQ A compliant (see PCI Notes below).</li></ul>                                                                                   |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hosted | <ul style="list-style-type: none"><li>• Paymetric hosts the entire page</li><li>• When initiating payment, Merchant's customer is redirected to the Paymetric URL</li><li>• XiIntercept eComm redirects to the URL defined by the Merchant after processing the request</li><li>• The page has a high-level of page responsiveness since it is completely hosted by Paymetric</li><li>• It is PCI v3.0 - SAQ A compliant (see PCI Notes below)</li></ul> |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**PCI Notes:**

- PCI v3.0 - SAQ A Compliant - Iframe and Hosted
  - For Iframe and Hosted methods, the Merchant has no way to access the raw card data in support of PCI v3.0 SAQ A compliance. Either the Iframe or the entire payment page is hosted within the Paymetric data center depending upon which method you choose.
- PCI v3.0 - SAQ A-EP Compliant - Form Post and Ajax
  - For Form Post and Ajax integrations, while no card numbers are entered within the Merchant's environment in support of PCI v3.0 SAQ A-EP compliance, the Merchant does have access to the raw card number through javascript and therefore must ensure it is not sent to the server.



When implementing the Form Post or Ajax integration type, the Developer must ensure they do NOT post nor send raw card numbers in their environment.

**3DS with Token Integration**

XiIntercept eComm supports a separate 3DS workflow using an XiSecure Token. There is a separate URL endpoint to trigger this call.

- An Ajax call is made to XiIntercept eComm with the Token from the customer's browser.
- The user never leaves page.
- It is PCI v3.0 - SAQ A compliant.

Note that Ajax call in this workflow uses a different Paymetric JS plugin than the Ajax call mentioned under the Standard Raw Value Tokenization section.

### Alternative Payment Method Tokenization

XiIntercept eComm also provides tokenization functionality for [AliPay](#)<sup>[69]</sup> and [Visa Checkout](#)<sup>[76]</sup> alternative payment methods, each with their own workflow and URL endpoint to trigger the calls.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AliPay        | <ul style="list-style-type: none"> <li>Implemented as an alternative payment type with either the Form Post, Ajax or Iframe integration type (not supported with Hosted).</li> <li>A POST request is made to XiIntercept eComm from Client's browser.</li> <li>XiIntercept eComm redirects to Cardinal Commerce to verify enrollment then to AliPay if enrolled.</li> <li>Xi-eComm redirects to URL defined by the Merchant after processing the request.</li> </ul>                                                                               |
| Visa Checkout | <ul style="list-style-type: none"> <li>Implemented as an alternative payment type with either the Form Post, Ajax or Iframe integration type (not supported with Hosted).</li> <li>Merchant makes a call directly to Visa Checkout platform to obtain call ID.</li> <li>Merchant performs a POST request to XiIntercept eComm from the application server.</li> <li>XiIntercept eComm obtains full payment details from Visa Checkout and sends decrypted response packet with token in either XML or JSON as specified by the Merchant</li> </ul> |

### Securelink Integration

- Used to trigger an email with a secure link to XiIntercept eComm hosted page to obtain credit card information. Can alternatively choose to copy/paste the link into a chat window.
- Paymetric hosts the entire page.
- Hosted page is launched via a link sent in an email to the specified address or from a chat window.
- XiIntercept eComm redirects to the URL defined by the Merchant after processing the request.

### 6.1 Raw Value Tokenization XiSecure Tokens Only

Enter topic text here.

---

! Ensure that you do not send raw card numbers to your environment.

---

Form Post integration consists of the following high-level steps:

- A. Initiate XiIntercept eComm session with a POST call to get the access token.
  - I. If implementing 3DS, there are extra fields that may be provided in the PacketXML. If 3DS2 there is also an attribute to indicate the 3DS version.
- B. Render page as desired and upon user submission of credit card data, perform POST call using the Paymetric JS plugin.
  - I. If implementing 3DS 2.0, XileComm automatically performs the additional device data collection (DDC) used by Cardinal and the Banks.
- C. Perform GET call to obtain the response packet containing the token.

The following sequence diagram illustrates the Form Post integration flow. Review the subsections below for more details.

NOTE: If implementing 3DS2, the user is only redirected to the bank to enter 3DS credentials if the DDC or Merchant-provided risk information is determined to be insufficient.

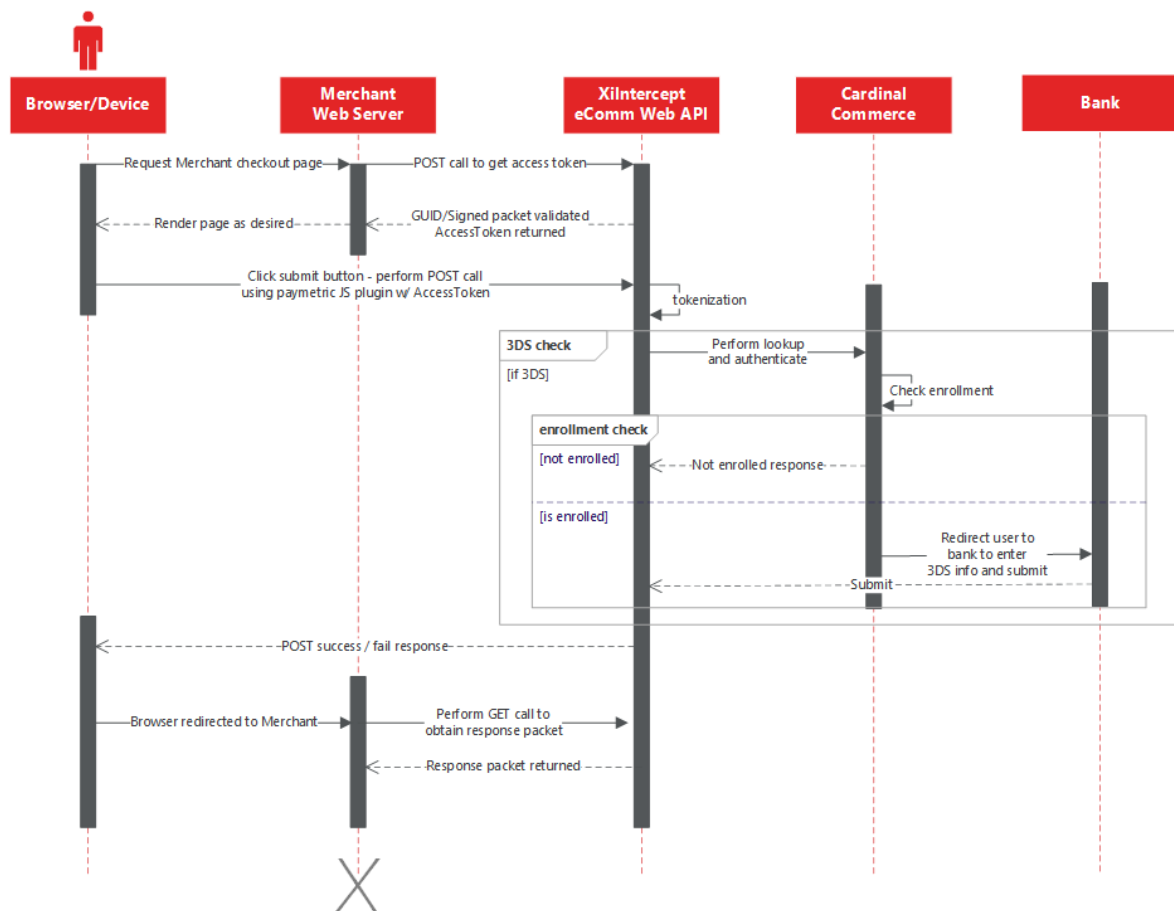


Figure 1: Form Post Integration Sequence Diagram

## 7.1 Form Post: Get Access Token

An AccessToken is required to initiate an XiIntercept eComm session.

Reminder that the AccessToken is a string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key.

The AccessToken is valid for 30 minutes. If the AccessToken expires before the process is complete, you must restart the process and obtain a new AccessToken.

## Request Method

A direct server-side POST is required to retrieve the Access Token. How this POST is performed depends on the language you are using.

## Content Type

The content type must be "application/x-www-form-urlencoded".

## Endpoints

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/AccessToken`

Production:

`https://xiecomm.paymetric.com/DIEComm/AccessToken`

## POST Call Parameters

`MerchantGuid={Your-GUID}&SessionRequestType={Int-Type-Code}&Signature={Signature}&MerchantDevelopmentEnvironment={code language e.g. asp.net}&Packet={PACKET_XML}`

Table 2: Form Post Get Access Token Parameters

| Parameter                      | Description                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MerchantGuid                   | The Global Unique Identifier assigned by Paymetric                                                                                                                                            |
| SessionRequestType             | Indicates the integration type. Use 3 for Form Post.                                                                                                                                          |
| Signature                      | Sign the Packet XML; encode it with HMAC-SHA-256 using the shared key then convert it to a base 64 string.<br>The Signature must be URL encoded.                                              |
| MerchantDevelopmentEnvironment | Indicate the language in which you are coding your integration to XiIntercept eComm, for example, asp.net or php.                                                                             |
| Packet XML                     | The Packet XML must be URL encoded. Following are the elements: <ul style="list-style-type: none"><li>RedirectUri = the redirect URL for the Response Packet from XiIntercept eComm</li></ul> |

Table 2: Form Post Get Access Token Parameters

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>If implementing 3DS, include the following fields as well within &lt;Cardinal3DSRequest&gt; container:</p> <ul style="list-style-type: none"><li>• Amount = amount of transaction</li><li>• CurrencyCode = currency code for transaction</li><li>• CardNumberFieldName = the Merchant's field name that will contain the card number value entered by the User</li><li>• ExpirationMonthFieldName = the Merchant's field name that will contain the expiration month value entered by the User</li><li>• ExpirationYearFieldName = the Merchant's field name that will contain the expiration year value entered by the User</li><li>• OrderNumber = the Merchant's order number</li><li>• AdditionalFields = optional container with additional fields for 3DS2</li></ul> <p>See samples below.</p> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.1.1 Form Post XML Sample

#### Form Post XML Sample

```
<?xml version="1.0"?>
<PostPacketModel>
  <RedirectUri>{http://Yourdomain.here}</RedirectUri>
</PostPacketModel>
```

### 7.1.2 Form Post with 3DS 1 XML Sample

#### Form Post with 3DS 1 XML Sample

```
<?xml version="1.0"?>
<PostPacketModel>
  <RedirectUri>{http://Yourdomain.here}</RedirectUri>
  <Cardinal3DSRequest>
    <Amount>{AmountValue}</Amount>
    <CurrencyCode>{CurrencyCodeValue}</CurrencyCode>
    <CardNumberFieldName>{CCNumValue}</CardNumberFieldName>
    <ExpirationMonthFieldName>{ExpMonthValue}</ExpirationMonthFieldName>
    <ExpirationYearFieldName>{ExpYearValue}</ExpirationYearFieldName>
    <OrderNumber>{OrderNumValue}</OrderNumber>
  </Cardinal3DSRequest>
</PostPacketModel>
```

### 7.1.3 Form Post with 3DS 2 XML Sample

The 3DS fields that differ from the prior example and are specific to version 2 are highlighted blue in the example below. These include `threeDSVersion` attribute within the `<Cardinal3DSecure>` container and the `<AdditionalFields>` container and child elements.

The `threeDSVersion` attribute must be set to "2" or can be a specific version up to 3 digits. If there is a dot ( . ), then there must be a following number. E.g. "2" or "2.0" or "2.2", and so on.

Any Cardinal 3DS field can be included in `<AdditionalFields>` container. See the following URL for a complete list of the additional Cardinal fields:

<https://cardinaldocs.atlassian.net/wiki/spaces/CCen/pages/905478187/Lookup+Request+Response>

NOTE: Cardinal `<AdditionalFields>` container cannot have duplicate field elements.

### Form Post with 3DS 2 XML Sample

```
<?xml version="1.0"?>
<PostPacketModel>
  <RedirectUri>{http://Yourdomain.here}</RedirectUri>
  <Cardinal3DSRequest threeDSVersion="2.2.0">
    <Amount>{AmountValue}</Amount>
    <CurrencyCode>{CurrencyCodeValue}</CurrencyCode>
    <CardNumberFieldName>{CCNumValue}</CardNumberFieldName>
    <ExpirationMonthFieldName>{ExpMonthValue}</ExpirationMonthFieldName>
    <ExpirationYearFieldName>{ExpYearValue}</ExpirationYearFieldName>
    <OrderNumber>{OrderNumValue}</OrderNumber>
    <AdditionalFields>
      <Field name="UserAccountAge" value="04" />
      <Field name="UserPasswordChange" value="20190325" />
      <Field name="DeliveryTimeframe" value="02" />
      <Field name="GiftCardAmount" value="100" />
    </AdditionalFields>
  </Cardinal3DSRequest>
</PostPacketModel>
```

## 7.2 Form Post: On-Submit Javascript Plugin

When the Customer performs the submit action by clicking the merchant-created submit button, the data will be posted to Paymetric using JavaScript functions.

Following is an example of pulling the values from the text boxes and populating the data variable.

### Populating Data Variable Example

```
var cc = document.getElementById('PaymentCreditCard').value;
var name = document.getElementById('Name').value;
var myData = $XIPlugin.createJSRequestPacket(merchantGuid,
accessToken);
myData.addField($XIPlugin.createField('PaymentCreditCard', true,
cc));
myData.addField($XIPlugin.createField('name', false, name));
myData.addAdditional3DSField("UserAccountAge", "04");
```

Either reference or download the following JavaScript plugin:

<https://xiecomm.paymetric.com/DIEComm/Scripts/XIPlugin/XIPlugin-1.2.0.js>



Existing integrations can still use `XiPlugin01.1.0.js` UNLESS you are adding 3DS2 support, then you must use `XiPlugin-1.2.0.js` version.

Include the Paymetric JavaScript. Reference the Submit Example below to handle the posting of the APM details directly to XiIntercept eComm.

#### Form Post: QA JS Submit Example

```
$XIPlugin.submit({  
  
  url: "https://cert-xiecomm.paymetric.com/DIEComm/Form",  
  data: {Formatted JS Data - See above example myData},  
  validation: function(a)  
    {  
      //Run custom validation  
      return true; //should return true or false.  
  
      // return true will go through with post.  
      // return false will not go through with post.  
    }  
  
  });
```

#### Form Post: QA JS Submit 3DS2 Example

```
$XIPlugin.submit({  
  
  url: "https://cert-xiecomm.paymetric.comom/DIEComm/Form",  
  data: {Formatted JS Data - See above example myData},  
  validation: function(a)  
    {  
      //Run custom validation  
      return true; //should return true or false.  
  
      // return true will go through with post.  
      // return false will not go through with post.  
    },  
  threeDSVersion: "2.2.0"  
  
  });
```

## 7.3 Form Post: Get Response Packet

### Request Method

A direct server-side GET is required to retrieve the Response Packet. How this GET is performed depends on the language you are using.

### Content Type

The content type must be "application/x-www-form-urlencoded".

### Endpoints

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/ResponsePacket`

Production:

`https://xiecomm.paymetric.com/DIEComm/ResponsePacket`

### GET Call Parameters

`MerchantGUID={MerchantGUID}&Signature={Signature}&AccessToken={AccessToken}`

| Table 3: Get Response Packet Parameters |                                                                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                               | Description                                                                                                                                                                                                          |
| MerchantGuid                            | The Global Unique Identifier assigned by Paymetric                                                                                                                                                                   |
| Signature                               | Sign the Access Token; encode it with HMAC-SHA-256 using the shared key then covert it to a base 64 string.<br><br>The Signature must be URL encoded.                                                                |
| AccessToken                             | A string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key. |

---

 You must ensure that you do not send raw card numbers to your environment.

---

Ajax integration consists of the following high-level steps:

- A. Initiate XiIntercept eComm session with a POST call to get the access token.
  - I. If implementing 3DS, there are extra fields that may be provided in the PacketXML. If 3DS2 there is also an attribute to indicate the 3DS version.
- B. Render page as desired and upon user submission of Credit Card data, perform POST call using the Paymetric JS plugin.
  - I. If implementing 3DS 2.0, XileComm automatically performs the additional device data collection (DDC) used by Cardinal and the Banks.
- C. Perform GET call to get the response packet containing the token. If implementing 3DS, all response fields from Cardinal Commerce will be included as well.

The following sequence diagram illustrates the Ajax integration flow. Review the subsections below for more details.

NOTE: If implementing 3DS2, the user is only redirected to the bank to enter 3DS credentials if the DDC or Merchant-provided risk information is determined to be insufficient.

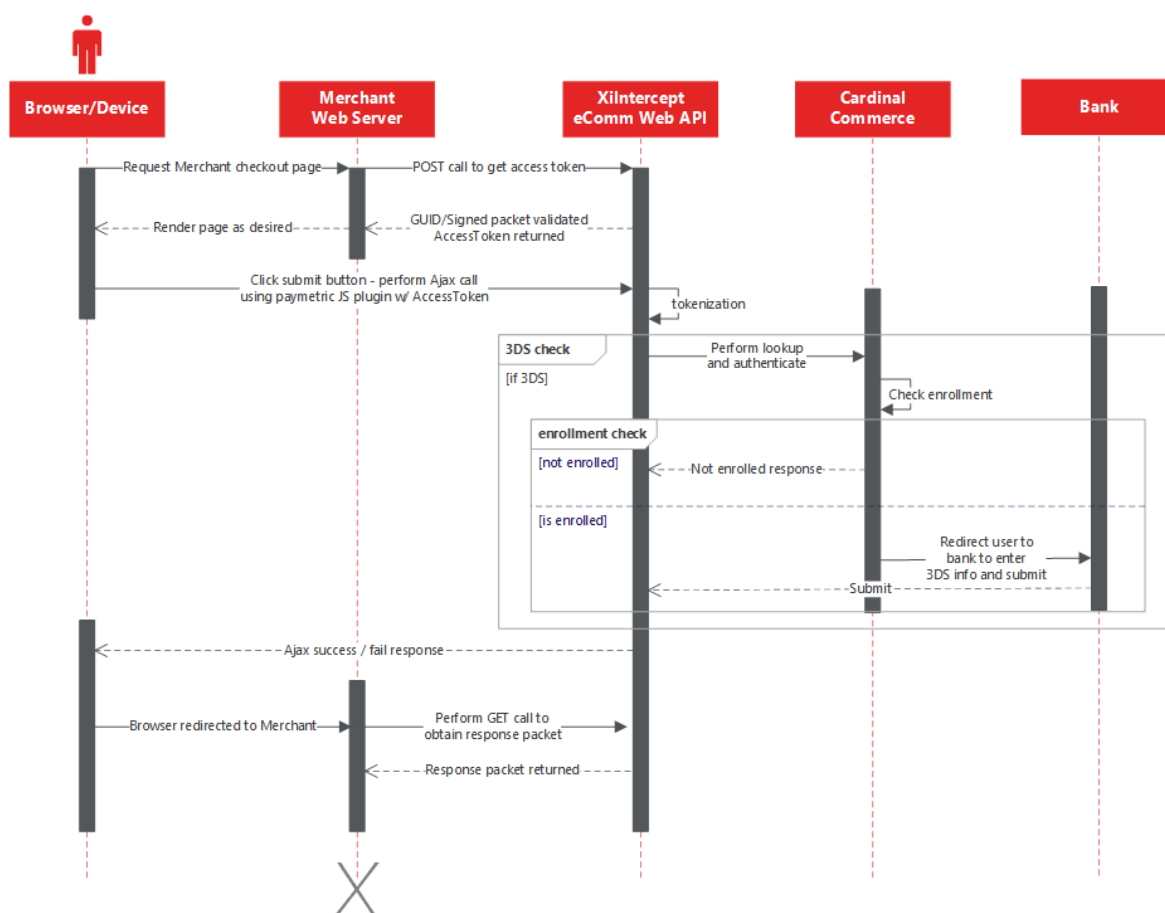


Figure 2: Ajax Integration Sequence Diagram

## 8.1 Ajax: Get Access Token

An AccessToken is required to initiate an XiIntercept eComm session.

Reminder that the AccessToken is a string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key.

The AccessToken is valid for 30 minutes. If the AccessToken expires before the process is complete, you must restart the process and obtain a new AccessToken.

### Request Method

A direct server-side POST is required to retrieve the Access Token. How this POST is performed depends on the language you are using.

## Content Type

The content type must be "application/x-www-form-urlencoded".

## Endpoints

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/AccessToken`

Production:

`https://xiecomm.paymetric.com/DIEComm/AccessToken`

## POST Call Parameters

`MerchantGuid={Your-GUID}&SessionRequestType={Int-Type-Code}&Signature={Signature}&MerchantDevelopmentEnvironment={code language e.g. asp.net}&Packet={PACKET_XML}`

| Table 4: Ajax Get Access Token Parameters |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| MerchantGuid                              | The Global Unique Identifier assigned by Paymetric                                                                                                                                                                                                                                                                                                                                                                                                           |
| SessionRequestType                        | Indicates the integration type. Use 2 for Ajax.                                                                                                                                                                                                                                                                                                                                                                                                              |
| Signature                                 | Sign the Packet XML; encode it with HMAC-SHA-256 using the shared key then convert it to a base 64 string.<br><br>The Signature must be URL encoded.                                                                                                                                                                                                                                                                                                         |
| MerchantDevelopmentEnvironment            | Indicate the language in which you are coding your integration to XiIntercept eComm, for example, asp.net or php.                                                                                                                                                                                                                                                                                                                                            |
| Packet XML                                | For non-3DS implementations, the Packet XML parameter for the Ajax call contains no elements.<br><br>NOTE: If you have a previous integration that sends HostUri, this field is not used. It will have no effect.<br><br>If implementing 3DS, include the following fields as well within <Cardinal3DSRequest> container: <ul style="list-style-type: none"><li>• RedirectUri = the URI used to redirect back to the merchant after 3DS processing</li></ul> |

Table 4: Ajax Get Access Token Parameters

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"><li>• Amount = amount of transaction</li><li>• CurrencyCode = currency code for transaction</li><li>• CardNumberFieldName = the Merchant's field name that will contain the card number value entered by the User</li><li>• ExpirationMonthFieldName = the Merchant's field name that will contain the expiration month value entered by the User</li><li>• ExpirationYearFieldName = the Merchant's field name that will contain the expiration year value entered by the User</li><li>• OrderNumber = the Merchant's order number</li><li>• AdditionalFields = optional container with additional fields for 3DS2</li></ul> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 8.1.1 Ajax XML Sample

#### Ajax XML Sample

```
<?xml version="1.0"?>  
<AjaxPacketModel/>
```

### 8.1.2 Ajax with 3DS 1 XML Sample

#### Ajax with 3DS 1 with Existing XML Sample

```
<?xml version="1.0"?>
<AjaxPacketModel
  xmlns:xsd=http://www.w3.org/2001/XMLSchema
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <Cardinal3DSRequest>
    <RedirectUri>{http://YourRedirectURLHere}</RedirectUri>
    <Amount>{AmountValue}</Amount>
    <CardNumberFieldName>{CCNumValue}</CardNumberFieldName>
    <CurrencyCode>{CurrencyCodeValue}</CurrencyCode>
    <ExpirationMonth>{ExpMonthValue}</ExpirationMonth>
    <ExpirationYear>{ExpYearValue}</ExpirationYear>
    <OrderNumber>{OrderNumValue}</OrderNumber>
  </Cardinal3DSRequest>
</AjaxPacketModel>
```

### 8.1.3 Ajax with 3DS 2 XML Sample

The 3DS fields that differ from the prior example and are specific to version 2 are highlighted **blue** in the example below. These include `threeDSVersion` attribute within the `<cardinal3DSecure>` container and the `<AdditionalFields>` container and child elements.

The `threeDSVersion` attribute must be set to "2" or can be a specific version up to 3 digits. If there is a dot ( . ), then there must be a following number. E.g. "2" or "2.0" or "2.2", and so on.

Any Cardinal 3DS field can be included in `<AdditionalFields>` container. See the following URL for a complete list of the additional Cardinal fields:

<https://cardinaldocs.atlassian.net/wiki/spaces/CCen/pages/905478187/Lookup+Request+Response>

NOTE: Cardinal `<AdditionalFields>` container cannot have duplicate field elements.

### Ajax with 3DS 2 with Existing XML Sample

```
<?xml version="1.0"?>
<AjaxPacketModel
  xmlns:xsd=http://www.w3.org/2001/XMLSchema
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <Cardinal3DSRequest threeDSVersion="2.2.0">
    <RedirectUri>{http://YourRedirectURLHere}</RedirectUri>
    <Amount>{AmountValue}</Amount>
    <CardNumberFieldName>{CCNumValue}</CardNumberFieldName>
    <CurrencyCode>{CurrencyCodeValue}</CurrencyCode>
    <ExpirationMonth>{ExpMonthValue}</ExpirationMonth>
    <ExpirationYear>{ExpYearValue}</ExpirationYear>
    <OrderNumber>{OrderNumValue}</OrderNumber>
    <AdditionalFields>
      <Field name="UserAccountAge" value="04" />
      <Field name="UserPasswordChange" value="20190325" />
      <Field name="DeliveryTimeframe" value="02" />
      <Field name="GiftCardAmount" value="100" />
    </AdditionalFields>
  </Cardinal3DSRequest>
</AjaxPacketModel>
```

## 8.2 Ajax: On-Submit Javascript Plugin

When the Customer performs the submit action by clicking the merchant-created submit button, the data will be posted to Paymetric using JavaScript functions.

Following is an example of pulling the values from the text boxes and populating the data variable.



**For 3DS2**, you can add the Cardinal additional fields via the JavaScript plugin as well. The following "Populating Data Variables Example" includes one of the 3DS2 AdditionalFields.

If you send the same additional field that you sent when getting the access token, it will be overwritten.

If you accidentally add more than one field with the same name and different values via the plugin, it will take the value of the first entry. So, for example:

- Access token field "UserAccountAge" = 01
- JavaScript field "UserAccountAge" = 02
- JavaScript field "UserAccountAge" = 03

The value used will be "02".

### Populating Data Variables Example

```
var cc = document.getElementById('PaymentCreditCard').value;
var name = document.getElementById('Name').value;
var myData = $XIPlugin.createJSRequestPacket(merchantGuid,
accessToken);
myData.addField($XIPlugin.createField('PaymentCreditCard', true,
cc));
myData.addField($XIPlugin.createField('name', false, name));
myData.addAdditional3DSField("UserAccountAge", "04");
```

You can reference or download the following JavaScript plugin:

<https://xiecomm.paymetric.com/DIEComm/Scripts/XIPlugin/XIPlugin-1.2.0.js>



Existing integrations can still use `XiPlugin01.1.0.js` **UNLESS** you are adding 3DS2 support, then you must use `XiPlugin-1.2.0.js` version.

Include the Paymetric JavaScript and reference the Submit Example below to handle the posting of the credit card details directly to XiIntercept eComm.

### Ajax: QA JS Submit Example

```
$XIPlugin.ajax({

  url: '<% XiIntercept Url%>' + "/Ajax",
  type: "POST", //<GET/POST>
  data: {Formatted JS Data - See above example myData},
  success: function(a)
  {
    //function triggered on success.
  },
  error: function(a)
  {
    //function triggered on failure.
  },
  beforeSend: function(a)
  {
    //function triggered before ajax call.
  },
  complete: function(a)
  {
    //function triggered after ajax call.
    // After success or failure
  }
  validation: function(a)
  {
    //Run custom validation
```

```
        return true; //should return true or false.

        // return true will go through with ajax call.
        // return false will not go through with ajax call.
    }

});
```

#### Ajax: QA JS Submit 3DS2 Example

```
$XIPlugin.ajax({

    url: '<% XiIntercept Url%>' + "/Ajax",
    type: "POST", //<GET/POST>
    data: {Formatted JS Data - See above example myData},
    success: function(a)
    {
        //function triggered on success.
    },
    error: function(a)
    {
        //function triggered on failure.
    },
    beforeSend: function(a)
    {
        //function triggered before ajax call.
    },
    complete: function(a)
    {
        //function triggered after ajax call.
        // After success or failure
    }
    validation: function(a)
    {
        //Run custom validation
        return true; //should return true or false.

        // return true will go through with ajax call.
        // return false will not go through with ajax call.
    },
    threeDSVersion: "2.2.0"

    intcp3DSecure: function(a)
    {
        //Optional function for custom 3DS challenge prompt;
        set to null or omit for default prompt; default is full
        redirect to a Paymetric hosted page
    }

});
```

## 8.3 Ajax: Get Response Packet

### Request Method

A direct server-side GET is required to retrieve the Response Packet. How this GET is performed depends on the language you are using.

### Content Type

The content type must be "application/x-www-form-urlencoded".

### Endpoints

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/ResponsePacket`

Production:

`https://xiecomm.paymetric.com/DIEComm/ResponsePacket`

### GET Call Parameters

MerchantGUID={**MerchGUID**}&Signature={**Signature**}&AccessToken={**AccessToken**}

| Table 5: Get Response Packet Parameters |                                                                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                               | Description                                                                                                                                                                                                          |
| MerchantGuid                            | The Global Unique Identifier assigned by Paymetric                                                                                                                                                                   |
| Signature                               | Sign the Access Token; encode it with HMAC-SHA-256 using the shared key then covert it to a base 64 string.<br>The Signature must be URL encoded.                                                                    |
| AccessToken                             | A string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key. |

Iframe integration consists of the following high-level steps:

- A. Initiate XiIntercept eComm session with a POST call to get the access token.
  - I. If implementing 3DS, there are extra fields that may be provided in the PacketXML including an attribute to indicate the 3DS version.
- B. Embed Iframe in web page.
- C. Render page as desired and upon user submission of Credit Card data, perform POST call using the Paymetric JS plugin.
  - I. If implementing 3DS 2.0, XileComm automatically performs the additional device data collection (DDC) used by Cardinal and the Banks.
- D. Perform GET call to get the response packet containing the token. If implementing 3DS, all response fields from Cardinal Commerce will be included as well.

The following sequence diagram illustrates the Iframe integration flow. Review the subsections below for more details.

NOTE: If implementing 3DS2, the user is only redirected to the bank to enter 3DS credentials if the DDC or Merchant-provided risk information is determined to be insufficient.

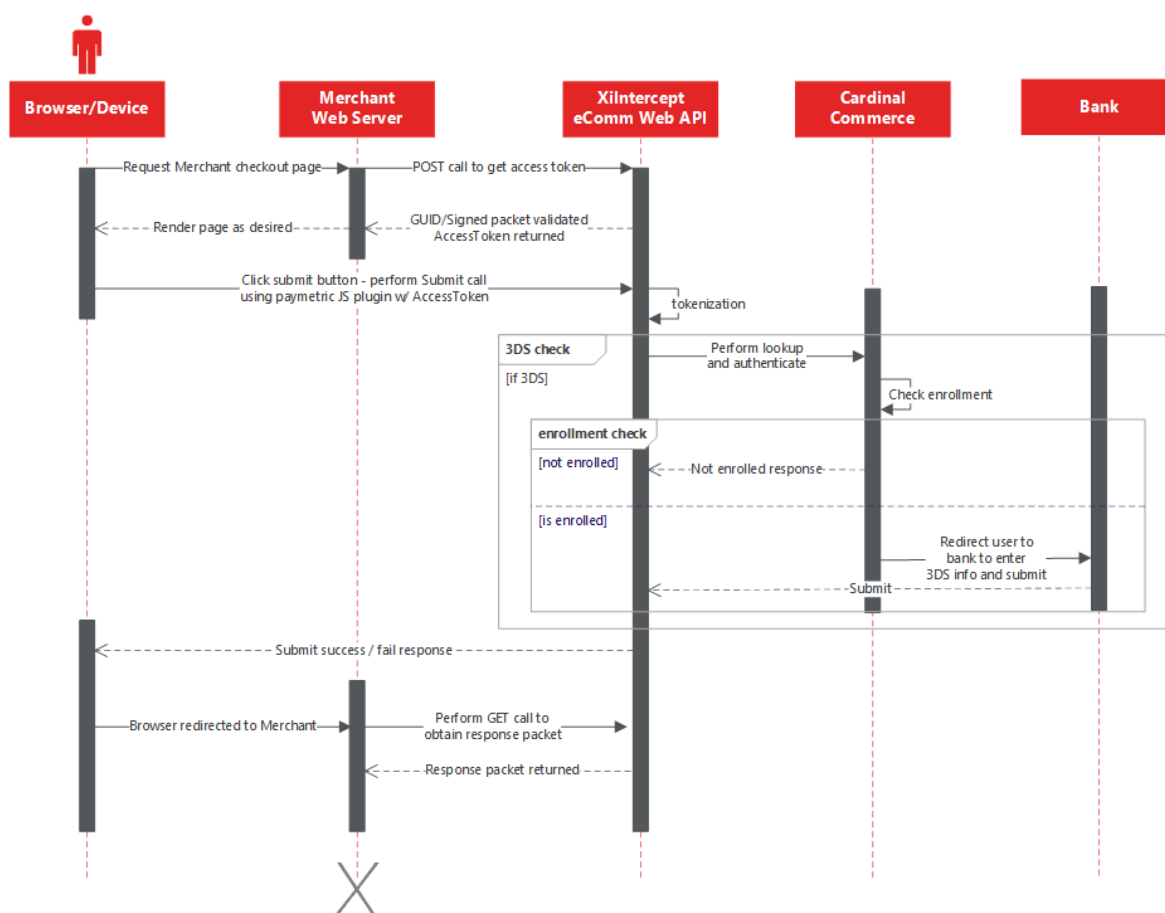


Figure 3: Iframe Integration Sequence Diagram

## Templates and Customizations

When using the Iframe integration type, there are templates available for the following:

- Card
- Card, CVV
- Card, CVV and Address

Alternatively, you can customize the fields that display. Refer to [Custom HTML Reference](#)<sup>115</sup> for complete details on the Custom Iframe option.

Whether using an Iframe template or the Custom Iframe option, you can use your own Cascading Style Sheet (CSS) to control the look and feel of your web page.

## Localization

You have the ability to localize standard labels and messaging for Iframe integrations; see the [Localization](#)<sup>92</sup> section for more information.

## 9.1 Iframe: Get Access Token

An AccessToken is required to initiate an XiIntercept eComm session.

Reminder that the AccessToken is a string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key.

The AccessToken is valid for 30 minutes. If the AccessToken expires before the process is complete, you must restart the process and obtain a new AccessToken.

### Request Method

A direct server-side POST is required to retrieve the Access Token. How this POST is performed depends on the language you are using.

### Content Type

The content type must be "application/x-www-form-urlencoded".

### Endpoints

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/AccessToken`

Production:

`https://xiecomm.paymetric.com/DIEComm/AccessToken`

### POST Call Parameters

`MerchantGuid={Your-GUID}&SessionRequestType={Int-Type-Code}&Signature={Signature}  
&MerchantDevelopmentEnvironment={code language e.g. asp.net}&Packet={PACKET_XML}`

| Table 6: Form Post Get Access Token Parameters |                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Parameter                                      | Description                                                                                                |
| MerchantGuid                                   | The Global Unique Identifier assigned by Paymetric                                                         |
| SessionRequestType                             | Indicates the integration type. Use 1 for Iframe.                                                          |
| Signature                                      | Sign the Packet XML; encode it with HMAC-SHA-256 using the shared key then convert it to a base 64 string. |

Table 6: Form Post Get Access Token Parameters

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | The Signature must be URL encoded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| MerchantDevelopmentEnvironment | Indicate the language in which you are coding your integration to XiIntercept eComm, for example, asp.net or php.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Packet XML                     | <p>The Packet XML must be URL encoded. Following are the elements:</p> <ul style="list-style-type: none"> <li>• Iframe Templates <ul style="list-style-type: none"> <li>○ hostUri = the domain of the Iframe parent page</li> <li>○ cssUri = the URL of the Merchant's CSS file</li> <li>○ templateHtml = of three templates (creditcard, creditcardCVV, creditcardCVVAddress)</li> <li>○ paymentTypes = create separate payment type for each one supported; these display in the card type/payment type dropdown</li> </ul> </li> <li>• Iframe Custom: <ul style="list-style-type: none"> <li>○ hostUri = the domain of the Iframe parent page</li> <li>○ cssUri = the URL of the Merchant's CSS file</li> <li>○ merchantHTML = see <a href="#">Custom HTML Reference</a><sup>115</sup> for details</li> </ul> </li> <li>• If implementing 3DS, include the following attributes for the &lt;cardinal3DSecure&gt; element for either the Iframe Template or Iframe Custom: <ul style="list-style-type: none"> <li>○ redirectUri = the redirect URL back to the merchant after 3DS processing</li> <li>○ amount = amount of transaction</li> <li>○ currencyCode = currency code for transaction</li> </ul> </li> </ul> |

Table 6: Form Post Get Access Token Parameters

|  |                                                                                                                                                                                                                                    |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>○ <code>orderNumber</code> = the Merchant's order number</li> <li>○ <code>additionalFields</code> = optional container with additional fields for 3DS2</li> </ul> <p>See samples below.</p> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 9.1.1 Iframe Template XML Sample

#### Iframe Template XML Sample

```
<?xml version="1.0" encoding="utf-8"?>
<merchantHtmlPacketModel>
  <iFramePacket>
    <hostUri>{http://Yourdomain.here}</hostUri>
    <cssUri>{http://Yourcssurlhere.here}</cssUri>
  </iFramePacket>
  <templateHtml name="CreditCard">
    <paymentTypes>
      <paymentType type="american express" />
      <paymentType type="mastercard" />
      <paymentType type="maestro" />
      <paymentType type="visa" />
      <paymentType type="custom" value="HD" text="Home Depot" />
    </paymentTypes>
  </templateHtml>
</merchantHtmlPacketModel>
```

### 9.1.2 Iframe Custom XML Sample

#### Iframe Custom XML Sample

```
<merchantHtmlPacketModel xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <iFramePacket>
    <hostUri>{http://Yourdomain.here}</hostUri>
    <cssUri>{http://Yourcssurlhere.here}</cssUri>
  </iFramePacket>
  <htmlFieldValues>
    <field for="cardholderName" value="John Smith" />
    <additionalField for="foo" value="My custom value" />
  </htmlFieldValues>
  <merchantHtml>
    <htmlSection class="merchant_paycontent">
      <xiProperties error-class="xiInputError" >
        <errorTooltip class="errorTooltip" show-effect="fadeIn" show-
duration="5000" hide-effect="fadeOut" hide-duration="5000" />
      </xiProperties>
    </htmlSection>
  </merchantHtml>
</merchantHtmlPacketModel>
```

## Iframe Custom XML Sample

```

<paymentOptionRadioSection>
  <tag name="div" class="PaymentDetailHeader">Select a payment option</tag>
  <label for="echeck" />
  <radPaymentOption for="echeck" />
  <label for="card" />
  <radPaymentOption for="card" />
</paymentOptionRadioSection>
<cardDropdownSection>
  <tag name="div" class="PaymentDetailHeader">
    Enter your credit card information
  </tag>
  <tag name="div">
    <label for="cardType" />
    <ddlCardType id="cd">
      <items>
        <item for="american express" />
        <item for="mastercard" />
        <item for="visa" />
        <item for="maestro" />
        <customItem for="CA" text="Company A"></customItem>
      </items>
    </ddlCardType>
  </tag>
  <tag name="div">
    <label for="cardNumber" />
    <tboxCardNumber tokenize="true" />
    <validationMsg for="cardNumber" class="valmsg" />
  </tag>
  <tag name="div">
    <label for="cardholderName" />
    <tboxCardHolderName />
    <validationMsg for="cardholderName" class="valmsg" />
  </tag>
  <tag name="div">
    <label for="startMonth" />
    <ddlStartMonth default-text="month" display-format="MMM"
class="merchant_combos" required="false" />
    <ddlStartYear default-text="year" class="merchant_combos" years-to-
display="10" required="false" start-date="true" />
    <validationMsg for="startYear" class="valmsg" />
  </tag>
  <tag name="div">
    <label for="expMonth" />
    <ddlExpMonth default-text="month" class="merchant_combos"
required="false" />
    <ddlExpYear default-text="year" class="merchant_combos" years-to-
display="10" required="false" exp-date="true" />
    <validationMsg for="expYear" class="valmsg" />
  </tag>
  <tag name="div">
    <label for="cvv" />
    <tboxCvv />
    <validationMsg for="cvv" class="valmsg" />
  </tag>

```

**Iframe Custom XML Sample**

```

</cardDropDownSection>
<echeckRadioSection>
  <tag name="div" class="PaymentDetailHeader">Enter your bank
information</tag>
  <tag name="div">
    <validationMsg for="accountType" class="valmsg-ac-type" />
    <label for="checking" />
    <radAccountType for="checking" />
    <label for="savings" />
    <radAccountType for="savings" />
  </tag>
  <tag name="div">
    <label for="accountNumber" />
    <tboxAccountNumber tokenize="true" error-msg-style="imageTooltip" />
    <validationMsg for="accountNumber" error-class="valmsg-img" />
  </tag>
  <tag name="div">
    <label for="routingNumber" />
    <tboxRoutingNumber error-msg-style="imageTooltip" />
    <validationMsg for="routingNumber" error-class="valmsg-img" />
  </tag>
  <tag name="div">
    <label for="nameOnAccount" />
    <tboxNameOnAccount error-msg-style="imageTooltip" />
    <validationMsg for="nameOnAccount" error-class="valmsg-img" />
  </tag>
  <tag name="div">
    <label for="bankName" />
    <tboxBankName error-msg-style="imageTooltip" />
    <validationMsg for="bankName" error-class="valmsg-img" />
  </tag>
</echeckRadioSection>
</htmlSection>
</merchantHtml>
</merchantHtmlPacketModel>

```

**9.1.3 Iframe Custom with 3DS 1 XML Sample****Iframe Custom with 3DS 1.0 XML Sample**

```

<merchantHtmlPacketModel xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <iFramePacket>
    <hostUri>{http://Yourdomain.here}</hostUri>
    <cssUri>{http://Yourcssurihere.here}</cssUri>
    <cardinal3DSecure amount="13.42" currencyCode="840" orderNumber="12345"
redirectUri="{http://Yourredirecturi.here}" />
  </iFramePacket>
  <merchantHtml>
    <htmlSection class="merchant_paycontent"
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
      <xiProperties error-class="xiInputError">

```

## Iframe Custom with 3DS 1.0 XML Sample

```

        <errorTooltip class="errorTooltip" show-effect="fadeIn" show-
duration="5000" hide-effect="fadeOut" hide-duration="5000" />
    </xiProperties>
    <paymentOptionRadioSection>
        <tag name="div" class="PaymentDetailHeader">Select a payment option</tag>
        <label text="echeck" for="echeck" />
        <radPaymentOption for="echeck" />
        <label text="credit card" for="card" />
        <radPaymentOption for="card" />
    </paymentOptionRadioSection>
    <cardDropdownSection>
        <tag name="div" class="PaymentDetailHeader">Enter your credit card
information</tag>
        <tag name="div">
            <label for="cardType" text="card type" />
            <ddlCardType id="cd">
                <items>
                    <item for="american express" />
                    <item for="mastercard" />
                    <item for="visa" />
                    <item for="maestro" />
                    <customItem for="myCard" text="My Card" />
                    <customItem for="myCard2" text="My Card 2" />
                </items>
            </ddlCardType>
        </tag>
        <tag name="div">
            <label for="cardNumber" text="card number" />
            <tboxCardNumber tokenize="true" />
            <validationMsg for="cardNumber" class="valmsg" />
        </tag>
        <tag name="div">
            <label for="cardholderName" text="name on card" />
            <tboxCardHolderName />
            <validationMsg for="cardholderName" class="valmsg" />
        </tag>
        <tag name="div">
            <label for="startMonth" text="start date" />
            <ddlStartMonth default-text="month" display-format="MMM"
class="merchant_combos" required="false" />
            <ddlStartYear default-text="year" class="merchant_combos" years-to-
display="10" required="false" start-date="true" />
            <validationMsg for="startYear" class="valmsg" />
        </tag>
        <tag name="div">
            <label for="expMonth" text="exp date" />
            <ddlExpMonth default-text="month" class="merchant_combos"
required="false" />
            <ddlExpYear default-text="year" class="merchant_combos" years-to-
display="10" required="false" exp-date="true" />
            <validationMsg for="expYear" class="valmsg" />
        </tag>
        <tag name="div">
            <label for="cvv" text="card cvv" />

```

## Iframe Custom with 3DS 1.0 XML Sample

```

        <tbxCvv />
        <validationMsg for="cvv" class="valmsg" />
    </tag>
    <tag name="div">
        <label for="cardAmount" />
        <tbxCardAmount />
        <validationMsg for="cardAmount" class="valmsg" />
    </tag>
</cardDropdownSection>
<echeckRadioSection>
    <tag name="div" class="PaymentDetailHeader">Enter your bank
information</tag>
    <tag name="div">
        <validationMsg for="accountType" class="valmsg-ac-type" />
        <label for="checking" />
        <radAccountType for="checking" />
        <label for="savings" />
        <radAccountType for="savings" />
    </tag>
    <tag name="div">
        <label for="accountNumber" text="account number" />
        <tbxAccountNumber tokenize="true" error-msg-style="imageTooltip" />
        <validationMsg for="accountNumber" error-class="valmsg-img" />
    </tag>
    <tag name="div">
        <label for="routingNumber" text="routing number" />
        <tbxRoutingNumber error-msg-style="imageTooltip" />
        <validationMsg for="routingNumber" error-class="valmsg-img" />
    </tag>
    <tag name="div">
        <label for="nameOnAccount" text="name on account" />
        <tbxNameOnAccount error-msg-style="imageTooltip" />
        <validationMsg for="nameOnAccount" error-class="valmsg-img" />
    </tag>
    <tag name="div">
        <label for="bankName" text="bank name" />
        <tbxBankName error-msg-style="imageTooltip" />
        <validationMsg for="bankName" error-class="valmsg-img" />
    </tag>
</echeckRadioSection>
</htmlSection>
</merchantHtml>
</merchantHtmlPacketModel>

```

## 9.1.4 Iframe Custom with 3DS 2 XML Sample

The 3DS fields that differ from the prior example and are specific to version 2 are highlighted **blue** in the example below. These include `threeDSVersion` attribute within the `<cardinal3DSecure>` container and the `<additionalFields>` container and child elements.

The `threeDSVersion` attribute must be set to "2" or can be a specific version up to 3 digits. If there is a dot ( . ), then there must be a following number. E.g. "2" or "2.0" or "2.2", and so on.

Any Cardinal 3DS field can be included in `<additionalFields>` container. See the following URL for a complete list of the additional Cardinal fields:

<https://cardinaldocs.atlassian.net/wiki/spaces/CCen/pages/905478187/Lookup+Request+Response>

NOTE: Cardinal `<additionalFields>` container cannot have duplicate field elements.

#### Iframe Custom with 3DS 2.0 XML Sample

```
<merchantHtmlPacketModel xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <iFramePacket>
    <hostUri>{http://Yourdomain.here}</hostUri>
    <cssUri>{http://Yourcssurl.here}</cssUri>
    <cardinal3DSecure amount="13.42" currencyCode="840" orderNumber="12345"
    redirectUri="{http://Yourredirecturi.here}" threeDSVersion="2.2.0">
      <additionalFields>
        <field name="UserAccountAge" value="04" />
        <field name="UserPasswordChange" value="20190325" />
        <field name="DeliveryTimeframe" value="02" />
        <field name="GiftCardAmount" value="100" />
      </additionalFields>
    </cardinal3DSecure>
  </iFramePacket>
</merchantHtml>

<htmlSection class="merchant_paycontent"
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <xiProperties error-class="xiInputError">
    <errorTooltip class="errorTooltip" show-effect="fadeIn" show-
    duration="5000" hide-effect="fadeOut" hide-duration="5000" />
  </xiProperties>
  <paymentOptionRadioSection>
    <tag name="div" class="PaymentDetailHeader">Select a payment option</tag>
    <label text="echeck" for="echeck" />
    <radPaymentOption for="echeck" />
    <label text="credit card" for="card" />
    <radPaymentOption for="card" />
  </paymentOptionRadioSection>
  <cardDropdownSection>
    <tag name="div" class="PaymentDetailHeader">Enter your credit card
    information</tag>
    <tag name="div">
      <label for="cardType" text="card type" />
      <ddlCardType id="cd">
        <items>
          <item for="american express" />
          <item for="mastercard" />
          <item for="visa" />
          <item for="maestro" />
          <customItem for="myCard" text="My Card" />
          <customItem for="myCard2" text="My Card 2" />
        </items>
      </ddlCardType>
    </tag>
  </cardDropdownSection>
</htmlSection>
```

## Iframe Custom with 3DS 2.0 XML Sample

```

        </ddlCardType>
    </tag>
    <tag name="div">
        <label for="cardNumber" text="card number" />
        <tboxCardNumber tokenize="true" />
        <validationMsg for="cardNumber" class="valmsg" />
    </tag>
    <tag name="div">
        <label for="cardholderName" text="name on card" />
        <tboxCardHolderName />
        <validationMsg for="cardholderName" class="valmsg" />
    </tag>
    <tag name="div">
        <label for="startMonth" text="start date" />
        <ddlStartMonth default-text="month" display-format="MMM"
class="merchant_combos" required="false" />
        <ddlStartYear default-text="year" class="merchant_combos" years-to-
display="10" required="false" start-date="true" />
        <validationMsg for="startYear" class="valmsg" />
    </tag>
    <tag name="div">
        <label for="expMonth" text="exp date" />
        <ddlExpMonth default-text="month" class="merchant_combos"
required="false" />
        <ddlExpYear default-text="year" class="merchant_combos" years-to-
display="10" required="false" exp-date="true" />
        <validationMsg for="expYear" class="valmsg" />
    </tag>
    <tag name="div">
        <label for="cvv" text="card cvv" />
        <tboxCvv />
        <validationMsg for="cvv" class="valmsg" />
    </tag>
    <tag name="div">
        <label for="cardAmount" />
        <tboxCardAmount />
        <validationMsg for="cardAmount" class="valmsg" />
    </tag>
</cardDropdownSection>
<echeckRadioSection>
    <tag name="div" class="PaymentDetailHeader">Enter your bank
information</tag>
    <tag name="div">
        <validationMsg for="accountType" class="valmsg-ac-type" />
        <label for="checking" />
        <radAccountType for="checking" />
        <label for="savings" />
        <radAccountType for="savings" />
    </tag>
    <tag name="div">
        <label for="accountNumber" text="account number" />
        <tboxAccountNumber tokenize="true" error-msg-style="imageTooltip" />
        <validationMsg for="accountNumber" error-class="valmsg-img" />
    </tag>

```

## Iframe Custom with 3DS 2.0 XML Sample

```

<tag name="div">
  <label for="routingNumber" text="routing number" />
  <tbxRoutingNumber error-msg-style="imageTooltip" />
  <validationMsg for="routingNumber" error-class="valmsg-img" />
</tag>
<tag name="div">
  <label for="nameOnAccount" text="name on account" />
  <tbxNameOnAccount error-msg-style="imageTooltip" />
  <validationMsg for="nameOnAccount" error-class="valmsg-img" />
</tag>
<tag name="div">
  <label for="bankName" text="bank name" />
  <tbxBankName error-msg-style="imageTooltip" />
  <validationMsg for="bankName" error-class="valmsg-img" />
</tag>
</echeckRadioSection>
</htmlSection>
</merchantHtml>
</merchantHtmlPacketModel>

```

## 9.2 Iframe: Embed Iframe in Web Page

To set the source URL of the Iframe:

```
Iframe source = {DieCommUrl}/view/iframe/{MerchantGuid}/
{AccessTokenPacket}/{BubbleUpErrorToParentWindow(boolean Type)}
```

For example:

```
https://cert-xiecomm.paymetric.com/diecomm/View/Iframe/8c9aea21-6063-
4f30-b09d-e44706fedb05/942fb4bf-x546-4b32-942d-76aadg3087e2/True
```

Iframe renders using fields defined in the [Iframe: Get Access Token](#)<sup>35</sup> request.

Iframe sizing can be handled by HTML or CSS.

Resizing of the Iframe can be handled by CSS, but if there is a need to handle resizing depending upon content that is inside Iframe (e.g. different fields for different payment methods or validation error messages), then include the following function.

#### Resize Iframe Function Sample

```
$XIFrame.onload({
    iFrameId: {Iframe ID},
    targetUrl: {Iframe source}
    autosizewidth: true,
    onSuccess: function (e) {
        //Successfully Iframe Loaded
    },
    onError: function (e) {
        //Error on Iframe Loading
    }
});
```

### 9.3 Iframe: On-Submit Javascript Plugin

When the Customer performs the submit action by clicking the merchant-created submit button, the data will be posted to Paymetric using JavaScript functions.

You can reference or download the following JavaScript plugin:

<https://xiecomm.paymetric.com/DIEComm/Scripts/XIFrame/XIFrame-1.2.0.js>



Existing integrations can still use XIFrame-1.1.0.js UNLESS you are adding 3DS2 support, then you must use XIFrame-1.2.0.js version.

Include the Paymetric JavaScript and reference the Submit Example below to handle the posting of the credit card details directly to XiIntercept eComm.

#### QA JS Submit Example

```
$XIFrame.submit({
    iFrameId: {Iframe ID},
    targetUrl: {Iframe source}
    onSuccess: function (msg)
    {
        //function triggered on success.
    },
    onError: function (msg)
    {
        //function triggered on failure
    }
});
```

If validation passes and tokenization is successful, then the onSuccess function is called; otherwise, onError function is called.

## QA JS Submit 3DS 2 Example

```
$XIFrame.submit({
  iFrameId: {Iframe ID},
  targetUrl: {Iframe source}
  onSuccess: function (msg)
  {
    //function triggered on success.
  },
  onError: function (msg)
  {
    //function triggered on failure
  },
  threeDSVersion: "2.2.0"

});
```

## 9.4 Iframe: Get Response Packet

### Request Method

A direct server-side GET is required to retrieve the Response Packet. How this GET is performed depends on the language you are using.

### Content Type

The content type must be "application/x-www-form-urlencoded".

### Endpoint

Dev/QA:

<https://cert-xiecomm.paymetric.com/DIEComm/ResponsePacket>

Production:

<https://xiecomm.paymetric.com/DIEComm/ResponsePacket>

### GET Call Parameters

Following is the GET data:

MerchantGUID={**MerchGUID**}&Signature={**Signature**}&AccessToken={**AccessToken**}

Table 7: Form Post Get Response Packet Parameters

| Parameter | Description |
|-----------|-------------|
|-----------|-------------|

|              |                                                                                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MerchantGuid | The Global Unique Identifier assigned by Paymetric                                                                                                                                                                   |
| Signature    | Sign the Access Token; encode it with HMAC-SHA-256 using the shared key then covert it to a base 64 string.<br><br>The Signature must be URL encoded.                                                                |
| AccessToken  | A string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key. |

## 9.5 Iframe: Additional Information

The section contains additional information related to Iframe integration as follows:

- [Multiple IFrames](#)<sup>47</sup>
- [Display Content Based on Selected Payment Type](#)<sup>48</sup>
- [Embed 3DS in IFrame](#)<sup>49</sup>

### 9.5.1 Multiple IFrames

You can implement multiple IFrames on your web page, for example, if you want to allow multiple payment methods.

The workflow for multiple IFrames is the same as documented above in IFrame Workflow Step 3.b -- for the Java IFrame Submit function, you define the IFrame instance and then reference that, e.g. instead of `$XIFrame.submit` it could be `iframe1Plugin.submit` and then `iframe2Plugin.submit`.

You can find JavaScript examples under JavaScript Plugin API Reference | XIFrame - IFrame API Reference | [Multiple IFrames](#)<sup>[11]</sup> in this document.

### 9.5.2 Display Content Based on Selected Payment Type

Part of what makes the XiIntercept eComm IFrame solution SAQ A compliant is the inability to make changes based on content selected within the IFrame fields. XiIntercept eComm has a feature that allows a change to the Payment Type field within the IFrame to fire an event at the parent level.

The `onCardTypeChange` element is used to trigger the on-change event.

```

//Create 2 IFrames. hold on to the XIFrame instance
iframe1Plugin = $XIFrame({
  iframeId: iframeName1,
  targetUrl: $(iframe1).attr("src"),
  onCardTypeChange: IFrame1CardChange, //register event for card type change
  autosizeheight: true,
  autosizewidth: true,
  onError: function(msg) { alert(msg); },
  onValidate: Validation,
  storeCardAmount: true,
  intcp3DSecure: Custom1Handling
});

```

### 9.5.3 Embed 3DS in IFrame

In the standard workflow defined above, the 3DS call redirects to the Bank's website. You can choose to embed the 3DS fields within the IFrame. The following screenshot displays an example of 3DS fields embedded within the IFrame.

The screenshot shows a credit card payment form with a header featuring a gold card icon and a blue globe. The card number 0000 1234 5678 0001 is displayed. The form is divided into two main sections: 'Billing Information' and 'Split credit card payment'. The 'Billing Information' section includes a text input for 'Amount outside of IFrame' with the value '25', and a 'Enter your credit card information' section with dropdowns for 'card type' (VISA), text inputs for 'card number' (4000000000000002), 'name on card' (test), 'exp date' (Mar 2018), and 'card cvv' (123). The 'Split credit card payment' section includes a text input for 'Amount inside of IFrame', a 'Split credit card payment' section with the same card details as above, and an 'amount' input with the value '75'. A red arrow points from the 'Split credit card payment' section to a red-bordered box at the bottom. This box contains the 'VERIFIED by VISA' logo, 'YourBank' logo, and an 'Added Protection' section. The 'Added Protection' section includes the text 'Please submit your Verified by Visa password.', 'Merchant: paymenttest', 'Amount: \$75.00USD', 'Date: 04/01/2015', 'Card Number: \*\*\*\*\*0002', 'Personal Message: Password is "1234"', 'User Name: testt', and a 'Password' input field. At the bottom of the box are 'Submit', 'Help', and 'Exit' buttons.

Hosted integration consists of the following high-level steps:

- A. Initiate XiIntercept eComm session with a POST call to get the access token upon page load then render page as desired.
  - I. If implementing 3DS, there are extra fields that should be provided in the PacketXML including an attribute to indicate the 3DS version.
- B. Redirect to Hosted page.
  - I. If implementing 3DS 2.0, XileComm automatically performs the additional device data collection (DDC) used by Cardinal and the Banks.
- C. Perform GET call to get the response packet containing the token. If implementing 3DS, all response fields from Cardinal Commerce will be included as well.

NOTE: If implementing 3DS2, the user is only redirected to the bank to enter 3DS credentials if the DDC or Merchant-provided risk information is determined to be insufficient.

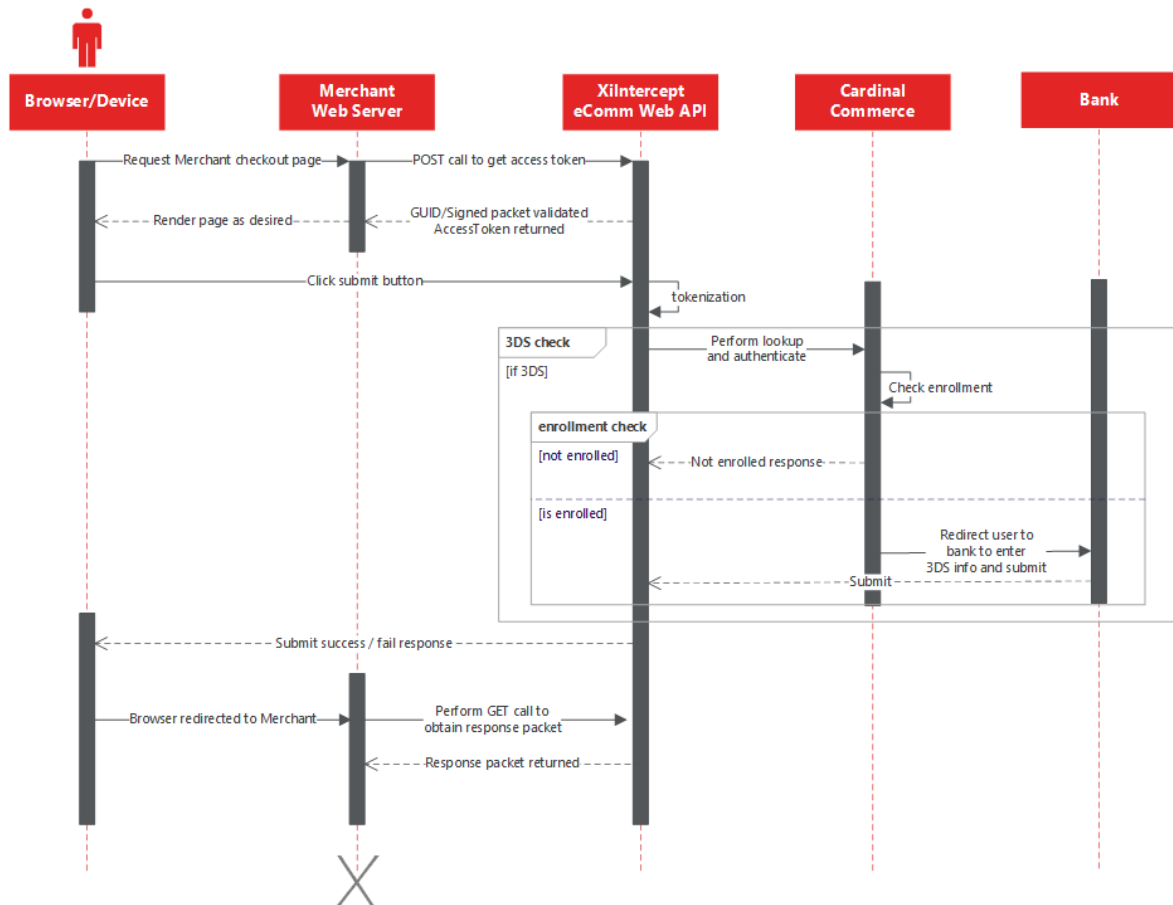


Figure 4: Hosted Integration Sequence Diagram

## Localization

You have the ability to localize standard labels and messaging for hosted integrations; see the [Localization](#)<sup>92</sup> section for more information.

## 10.1 Hosted: Get Access Token

An AccessToken is required to initiate an XiIntercept eComm session.

Reminder that the AccessToken is a string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key.

The AccessToken is valid for 30 minutes. If the AccessToken expires before the process is complete, you must restart the process and obtain a new AccessToken.

## Request Method

A direct server-side POST is required to retrieve the Access Token. How this POST is performed depends on the language you are using.

## Content Type

The content type must be "application/x-www-form-urlencoded".

## Endpoint

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/AccessToken`

Production:

`https://xiecomm.paymetric.com/DIEComm/AccessToken`

## POST Call Parameters

Following is the POST data:

```
MerchantGuid={Your-GUID}&SessionRequestType={Int-Type-Code}&Signature={Signature}
&MerchantDevelopmentEnvironment={code language e.g. asp.net}&Packet={PACKET_XML}
```

Table 8: Form Post Get Access Token Parameters

| Parameter                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MerchantGuid                   | The Global Unique Identifier assigned by Paymetric                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| SessionRequestType             | Indicates the integration type. Use 4 for Hosted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Signature                      | Sign the Packet XML; encode it with HMAC-SHA-256 using the shared key then convert it to a base 64 string.<br>The Signature must be URL encoded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| MerchantDevelopmentEnvironment | Indicate the language in which you are coding your integration to XiIntercept eComm, for example, asp.net or php.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Packet XML                     | <p>The Packet XML must be URL encoded. Following are the elements:</p> <ul style="list-style-type: none"> <li>• redirectUri = the redirect URL for the Response Packet from XiIntercept eComm</li> <li>• cssUri = the URL of the Merchant's CSS file</li> <li>• merchantHtml = see <a href="#">Custom HTML Reference</a> <sup>115</sup> for details (and XML sample below)</li> </ul> <p>If implementing 3DS, include the following attributes for the &lt;cardinal3DSecure&gt; element for either the Iframe Template or Iframe Custom:</p> <ul style="list-style-type: none"> <li>• redirectUri = the redirect URL back to the merchant after 3DS processing</li> <li>• amount = amount of transaction</li> <li>• currencyCode = currency code for transaction</li> <li>• orderNumber = the Merchant's order number</li> </ul> |

Table 8: Form Post Get Access Token Parameters

|                     |                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <ul style="list-style-type: none"> <li>additionalFields = optional container with additional fields for 3DS2</li> </ul> <p>See samples below.</p> |
| customerRedirectUri | Applicable only for the Hosted page when implementing Securelink. Defines the Merchant redirect URL after the User clicks submit.                 |

### 10.1.1 Hosted Custom XML Sample

#### Hosted Custom XML Sample

```
<?xml version="1.0" encoding="utf-8"?><merchantHtmlPacketModel
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <paymentPagePacket>
    <redirectUri>{http://Yourdomain.here}</redirectUri>
    <cssUri>http://dasvml21/DIEComm Sample/Content/PaymentStyleSheet.css</cssUri>
  </paymentPagePacket>
  <merchantHtml>
    <htmlSection class="payment-content"
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
      <tag name="div" class="merchant-logo"></tag>
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header">Billing Information</tag>
        <tag name="div" class="billing-info">
          <paymentOptionRadioSection>
            <tag name="div">
              <tag name="div">choose payment type</tag>
            </tag>
            <label text="echeck" for="echeck" />
            <radPaymentOption for="echeck" />
            <label text="credit card" for="card" />
            <radPaymentOption for="card" />
          </paymentOptionRadioSection>
          <cardDropdownSection>
            <tag name="div">Enter your credit card information</tag>
            <tag name="div">
              <label for="cardType" text="select a card" />
              <ddlCardType id="cd">
                <items>
                  <item for="american express" />
                  <item for="mastercard" />
                  <item for="visa" />
                </items>
              </ddlCardType>
            </tag>
            <tag name="div">
```

## Hosted Custom XML Sample

```

        <label for="cardNumber" text="card number" />
        <tboxCardNumber tokenize="true" />
    </tag>
    <tag name="div">
        <label for="cardholderName" text="name on card" />
        <tboxCardHolderName />
    </tag>
    <tag name="div">
        <label for="expMonth" text="exp date" />
        <ddlExpMonth default-text="month" class="merchant_combos" />
        <ddlExpYear default-text="year" class="merchant_combos" years-to-
display="10" />
    </tag>
    <tag name="div">
        <label for="cvv" text="card cvv:" />
        <tboxCvv />
    </tag>
</cardDropDownSection>
<echeckRadioSection>
    <tag name="div">
        <label for="checking" />
        <radAccountType for="checking" />
        <label for="savings" />
        <radAccountType for="savings" />
    </tag>
    <tag name="div">
        <label for="accountNumber" text="account number" />
        <tboxAccountNumber tokenize="true" />
    </tag>
    <tag name="div">
        <label for="routingNumber" text="routing number" />
        <tboxRoutingNumber />
    </tag>
    <tag name="div">
        <label for="nameOnAccount" text="name on account" />
        <tboxNameOnAccount />
    </tag>
    <tag name="div">
        <label for="bankName" text="bank name" />
        <tboxBankName />
    </tag>
</echeckRadioSection>
</tag>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button" />
    <submitButton text="submit" class="submit-button" />
</tag>
</tag>
</htmlSection>
</merchantHtml>
</merchantHtmlPacketModel>

```

## 10.1.2 Hosted Custom with 3DS 1 XML Sample

## Hosted Custom with 3DS XML Sample

```

<?xml version="1.0" encoding="utf-8"?><merchantHtmlPacketModel
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <paymentPagePacket>
    <redirectUri>{http://Yourdomain.here}</redirectUri>
    <cssUri>http://dasvm121/DIEComm Sample/Content/PaymentStyleSheet.css</cssUri>
    <cardinal3DSecure amount="13.42" currencyCode="840" orderNumber="12345" />
  </paymentPagePacket>
  <merchantHtml>
    <htmlSection class="payment-content"
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
      <tag name="div" class="merchant-logo"></tag>
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header">Billing Information</tag>
        <tag name="div" class="billing-info">
          <paymentOptionRadioSection>
            <tag name="div">
              <tag name="div">choose payment type</tag>
            </tag>
            <label text="echeck" for="echeck" />
            <radPaymentOption for="echeck" />
            <label text="credit card" for="card" />
            <radPaymentOption="" for="card" />
          </paymentOptionRadioSection>
          <cardDropdownSection>
            <tag name="div">Enter your credit card information</tag>
            <tag name="div">
              <label for="cardType" text="select a card" />
              <ddlCardType id="cd">
                <items>
                  <item for="american express" />
                  <item for="mastercard" />
                  <item for="visa" />
                </items>
              </ddlCardType>
            </tag>
            <tag name="div">
              <label for="cardNumber" text="card number" />
              <tboxCardNumber tokenize="true" />
            </tag>
            <tag name="div">
              <label for="cardholderName" text="name on card" />
              <tboxCardHolderName />
            </tag>
            <tag name="div">
              <label for="expMonth" text="exp date" />
              <ddlExpMonth default-text="month" class="merchant_combos" />
              <ddlExpYear default-text="year" class="merchant_combos" years-to-
display="10" />
            </tag>
            <tag name="div">
              <label for="cvv" text="card cvv:" />
              <tboxCvv />
            </tag>
          </div>
        </tag>
      </htmlSection>
    </merchantHtml>
  </paymentPagePacket>
</merchantHtmlPacketModel>

```

## Hosted Custom with 3DS XML Sample

```

    </tag>
  </cardDropdownSection>
  <echeckRadioSection>
    <tag name="div">
      <label for="checking" />
      <radAccountType for="checking" />
      <label for="savings" />
      <radAccountType for="savings" />
    </tag>
    <tag name="div">
      <label for="accountNumber" text="account number" />
      <tboxAccountNumber tokenize="true" />
    </tag>
    <tag name="div">
      <label for="routingNumber" text="routing number" />
      <tboxRoutingNumber />
    </tag>
    <tag name="div">
      <label for="nameOnAccount" text="name on account" />
      <tboxNameOnAccount />
    </tag>
    <tag name="div">
      <label for="bankName" text="bank name" />
      <tboxBankName />
    </tag>
  </echeckRadioSection>
</tag>
<tag name="div" class="payment-button">
  <cancelButton text="cancel" class="cancel-button" />
  <submitButton text="submit" class="submit-button" />
</tag>
</htmlSection>
</merchantHtml>
</merchantHtmlPacketModel>

```

## 10.1.3 Hosted Custom with 3DS 2 XML Sample

The 3DS fields that differ from the prior example and are specific to version 2 are highlighted **blue** in the example below. These include `threeDSVersion` attribute within the `<cardinal3DSecure>` container and the `<additionalFields>` container and child elements.

The `threeDSVersion` attribute must be set to "2" or can be a specific version up to 3 digits. If there is a dot ( . ), then there must be a following number. E.g. "2" or "2.0" or "2.2", and so on.

Any Cardinal 3DS field can be included in `<additionalFields>` container. See the following URL for a complete list of the additional Cardinal fields:

<https://cardinaldocs.atlassian.net/wiki/spaces/CCen/pages/905478187/Lookup+Request+Response>

NOTE: Cardinal `<additionalFields>` container cannot have duplicate field elements.

## Hosted Custom with 3DS XML Sample

```

<?xml version="1.0" encoding="utf-8"?><merchantHtmlPacketModel
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <paymentPagePacket>
    <redirectUri>{http://Yourdomain.here}</redirectUri>
    <cssUri>http://dasvm121/DIeComm Sample/Content/PaymentStyleSheet.css</cssUri>
    <cardinal3DSecure amount="13.42" currencyCode="840" orderNumber="12345"
threeDSVersion="2.2.0">
      <additionalFields>
        <field name="UserAccountAge" value="04" />
        <field name="UserPasswordChange" value="20190325" />
        <field name="DeliveryTimeframe" value="02" />
        <field name="GiftCardAmount" value="100" />
      </additionalFields>
    </paymentPagePacket>
  <merchantHtml>
    <htmlSection class="payment-content"
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
      <tag name="div" class="merchant-logo"></tag>
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header">Billing Information</tag>
        <tag name="div" class="billing-info">
          <paymentOptionRadioSection>
            <tag name="div">
              <tag name="div">choose payment type</tag>
            </tag>
            <label text="echeck" for="echeck" />
            <radPaymentOption for="echeck" />
            <label text="credit card" for="card" />
            <radPaymentOption="" for="card" />
          </paymentOptionRadioSection>
          <cardDropdownSection>
            <tag name="div">Enter your credit card information</tag>
            <tag name="div">
              <label for="cardType" text="select a card" />
              <ddlCardType id="cd">
                <items>
                  <item for="american express" />
                  <item for="mastercard" />
                  <item for="visa" />
                </items>
              </ddlCardType>
            </tag>
            <tag name="div">
              <label for="cardNumber" text="card number" />
              <tboxCardNumber tokenize="true" />
            </tag>
            <tag name="div">
              <label for="cardholderName" text="name on card" />
              <tboxCardHolderName />
            </tag>
            <tag name="div">
              <label for="expMonth" text="exp date" />
              <ddlExpMonth default-text="month" class="merchant_combos" />
            </tag>
          </cardDropdownSection>
        </div>
      </htmlSection>
    </merchantHtml>
  </paymentPagePacket>
</merchantHtmlPacketModel>

```

## Hosted Custom with 3DS XML Sample

```

        <ddlExpYear default-text="year" class="merchant_combos" years-to-
display="10" />
    </tag>
    <tag name="div">
        <label for="cvv" text="card cvv:" />
        <tboxCvv />
    </tag>
</cardDropdownSection>
<echeckRadioSection>
    <tag name="div">
        <label for="checking" />
        <radAccountType for="checking" />
        <label for="savings" />
        <radAccountType for="savings" />
    </tag>
    <tag name="div">
        <label for="accountNumber" text="account number" />
        <tboxAccountNumber tokenize="true" />
    </tag>
    <tag name="div">
        <label for="routingNumber" text="routing number" />
        <tboxRoutingNumber />
    </tag>
    <tag name="div">
        <label for="nameOnAccount" text="name on account" />
        <tboxNameOnAccount />
    </tag>
    <tag name="div">
        <label for="bankName" text="bank name" />
        <tboxBankName />
    </tag>
</echeckRadioSection>
</tag>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button" />
    <submitButton text="submit" class="submit-button" />
</tag>
</htmlSection>
</merchantHtml>
</merchantHtmlPacketModel>

```

## 10.2 Hosted: Redirect to Hosted Page

After receiving the AccessToken response packet, redirect Customer to the XiIntercept eComm hosted page at the following URL:

QA: `https://cert-xiecomm.paymetric.com/DIEComm/paymentpage/index/{Merchant GUID}/{Access Token}/`

Production: `https://xiecomm.paymetric.com/DIEComm/paymentpage/index/{Merchant GUID}/{Access Token}/`

User enters information and submits.

If user cancels on the rendered page,



Figure 5: Hosted Page Cancel URL Example

### 10.3 Hosted: Get Response Packet

#### Request Method

A direct server-side GET is required to retrieve the Response Packet. How this GET is performed depends on the language you are using.

#### Content Type

The content type must be "application/x-www-form-urlencoded".

#### Endpoint

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/ResponsePacket`

Production:

`https://xiecomm.paymetric.com/DIEComm/ResponsePacket`

## GET Call Parameters

Following is the GET data:

MerchantGUID=<MerchGUID>&Signature=<SIGNATURE>&AccessToken=<AccessToken>

| Table 9: Form Post Get Response Packet Parameters |                                                                                                                                                                                                                      |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                                         | Description                                                                                                                                                                                                          |
| MerchantGuid                                      | The Global Unique Identifier assigned by Paymetric                                                                                                                                                                   |
| Signature                                         | Sign the Access Token; encode it with HMAC-SHA-256 using the shared key then covert it to a base 64 string.<br>The Signature must be URL encoded.                                                                    |
| AccessToken                                       | A string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key. |

XiIntercept eComm supports a workflow that allows you to submit a token for the 3DS call if, for example, you have an existing customer with a token on file.

Note that the Ajax call in this workflow does not refer to the Ajax XiIntercept eComm integration type for raw card numbers. This is a completely separate workflow that is implemented independently of the raw card section.

3DS with token integration consists of the following high-level steps:

- Initiate XiIntercept eComm session with a POST call to get the access token.
- Merchant renders page as desired and upon user initiation of 3DS lookup call, perform POST call using the Paymetric JS plugin.
- Perform GET call to get the response packet with token. All response fields from Cardinal Commerce are included in the response.

The following sequence diagram illustrates the 3DS with Token integration flow. Review the subsections below for more details.

NOTE: If implementing 3DS2, the user is only redirected to the bank to enter 3DS credentials if the DDC or Merchant-provided risk information is determined to be insufficient.

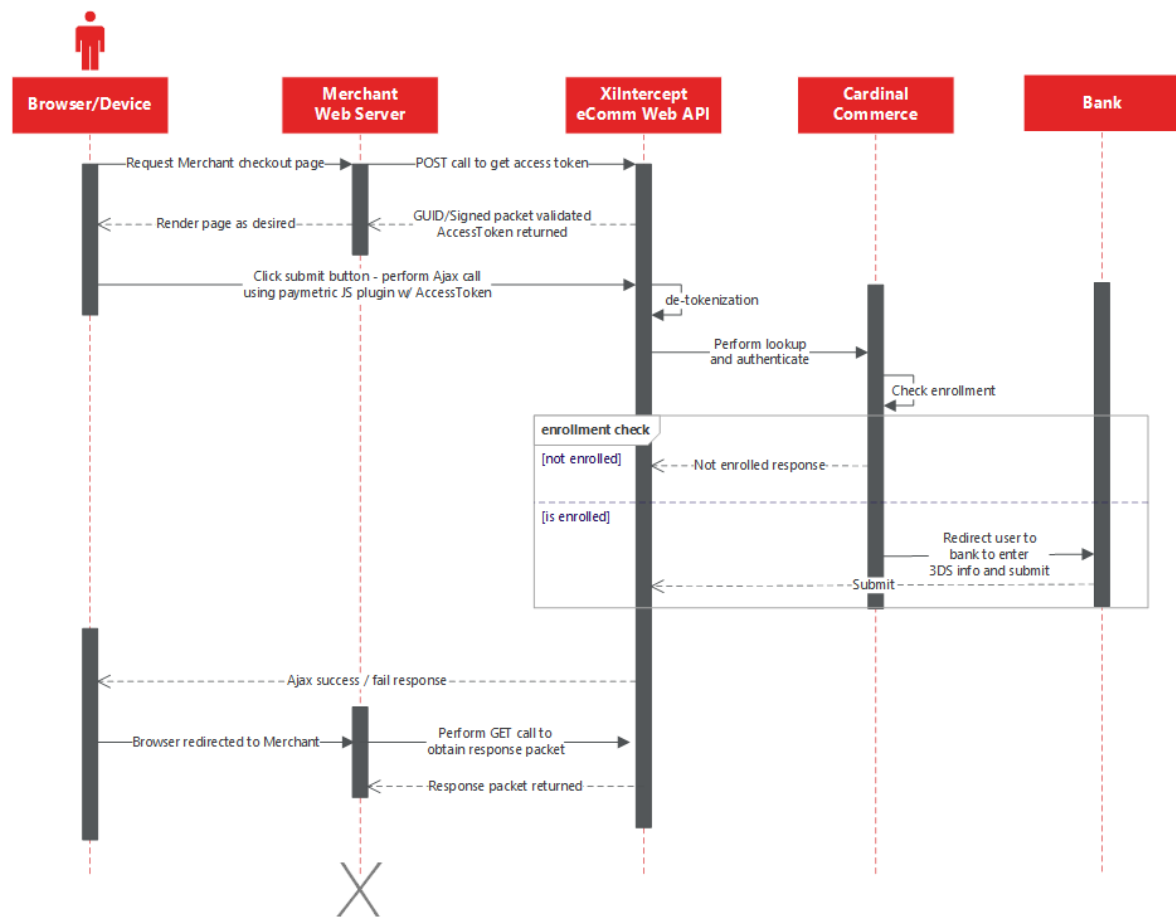


Figure 6: 3DS with Token Integration Sequence Diagram

## 11.1 3DS Token: Get Access Token

An AccessToken is required to initiate an XiIntercept eComm session.

Reminder that the AccessToken is a string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key.

The AccessToken is valid for 30 minutes. If the AccessToken expires before the process is complete, you must restart the process and obtain a new AccessToken.

### Request Method

A direct server-side POST is required to retrieve the Access Token. How this POST is performed depends on the language you are using.

### Content Type

The content type must be "application/x-www-form-urlencoded".

### Endpoints

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/AccessToken`

Production:

`https://xiecomm.paymetric.com/DIEComm/AccessToken`

### POST Call Parameters

`MerchantGuid={Your-GUID}&SessionRequestType={Int-Type-Code}&Signature={Signature}  
&MerchantDevelopmentEnvironment={code language e.g. asp.net}&Packet={PACKET_XML}`

| Table 10: 3DS with Token Get Access Token Parameters |                                                                                                            |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Parameter                                            | Description                                                                                                |
| MerchantGuid                                         | The Global Unique Identifier assigned by Paymetric                                                         |
| SessionRequestType                                   | Indicates the integration type. Use 2 for Ajax.                                                            |
| Signature                                            | Sign the Packet XML; encode it with HMAC-SHA-256 using the shared key then convert it to a base 64 string. |

Table 10: 3DS with Token Get Access Token Parameters

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | The Signature must be URL encoded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| MerchantDevelopmentEnvironment | Indicate the language in which you are coding your integration to XiIntercept eComm, for example, asp.net or php.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Packet XML                     | <p>The Packet XML must be URL encoded. Following are the elements:</p> <ul style="list-style-type: none"> <li>• RedirectUri = the redirect URL for the Response Packet from XiIntercept eComm</li> <li>• Within the &lt;Cardinal3DSWithToken&gt; container: <ul style="list-style-type: none"> <li>○ XiSecureToken = the token representing the appropriate card number</li> <li>○ Amount = amount of transaction</li> <li>○ CurrencyCode = currency code for transaction</li> <li>○ ExpirationMonth = the expiration month value entered by the User</li> <li>○ ExpirationYear = the expiration year value entered by the User</li> <li>○ OrderNumber = the Merchant's order number</li> <li>○ AdditionalFields = optional container with additional fields for 3DS2</li> </ul> </li> </ul> <p>See samples below.</p> |

### 11.1.1.1 3DS 1 with Token XML Sample

#### 3DS with Token XML Sample

```
<?xml version="1.0"?>
<Cardinal3DSWithTokenPacketModel>
  <RedirectUri>http://dasvm121/DIEComm Sample/Status.aspx</RedirectUri>
  <Cardinal3DSRequestWithToken>
    <XiSecureToken>Token</XiSecureToken>
    <Amount>1.42</Amount>
    <CurrencyCode>840</CurrencyCode>
    <ExpirationMonth>06</ExpirationMonth>
    <ExpirationYear>2018</ExpirationYear>
    <OrderNumber>12345</OrderNumber>
  </Cardinal3DSRequestWithToken>
</Cardinal3DSWithTokenPacketModel>
```

### 11.1.1.2 3DS 2 with Token XML Sample

The 3DS fields that differ from the prior example and are specific to version 2 are highlighted **blue** in the sample below. These include `threeDSVersion` attribute within the `<Cardinal3DSRequestWithToken>` container and the `<AdditionalFields>` container and child elements.

The `threeDSVersion` attribute must be set to "2" or can be a specific version up to 3 digits. If there is a dot ( . ), then there must be a following number. E.g. "2" or "2.0" or "2.2", and so on.

Any Cardinal 3DS field can be included in `<AdditionalFields>` container. See the following URL for a complete list of the additional Cardinal fields:

<https://cardinaldocs.atlassian.net/wiki/spaces/CCen/pages/905478187/Lookup+Request+Response>

NOTE: Cardinal `<AdditionalFields>` container cannot have duplicate field elements.

#### 3DS 2 with Token XML Sample

```
<?xml version="1.0"?>
<Cardinal3DSWithTokenPacketModel>
  <RedirectUri>http://dasvm121/DIEComm Sample/Status.aspx</RedirectUri>
  <Cardinal3DSRequestWithToken threeDSVersion="2.2.0">
    <XiSecureToken>Token</XiSecureToken>
    <Amount>1.42</Amount>
    <CurrencyCode>840</CurrencyCode>
    <ExpirationMonth>06</ExpirationMonth>
    <ExpirationYear>2018</ExpirationYear>
    <OrderNumber>12345</OrderNumber>
    <AdditionalFields>
      <Field name="UserAccountAge" value="04" />
      <Field name="UserPasswordChange" value="20190325" />
      <Field name="DeliveryTimeframe" value="02" />
      <Field name="GiftCardAmount" value="100" />
    </AdditionalFields>
  </Cardinal3DSRequestWithToken>
```

```
</Cardinal3DSWithTokenPacketModel>
```

## 11.2 3DS Token: On-Submit Javascript Plugin

When the Customer performs the submit action by clicking the merchant-created submit button, the data will be posted to Paymetric using JavaScript functions.

Following is an example of pulling the values from the text boxes and populating the data variable.



**For 3DS2**, you can add the Cardinal additional fields via the JavaScript plugin as well. The following "Populating Data Variables Example" includes one of the 3DS2 AdditionalFields.

If you send the same additional field that you sent when getting the access token, it will be overwritten.

If you accidentally add more than one field with the same name and different values via the plugin, it will take the value of the first entry.

So, for example:

- Access token field "UserAccountAge" = 01
- JavaScript field "UserAccountAge" = 02
- JavaScript field "UserAccountAge" = 03

The value used will be "02".

### Populating Data Variables Example

```
var myData = $XIPlugin.createJSRequestPacket(merchantGuid, accessToken);

myData.addField($XIPlugin.createField('Amount', false,
$('#textbox').val()));
myData.addAdditional3DSField("UserAccountAge", "04");
```

NOTE: Amount is optional from the browser.

You can reference or download the following JavaScript plugin:

<https://xiecomm.paymetric.com/DIEComm/Scripts/XIPlugin/XIPlugin-1.2.0.js>

Include the Paymetric JavaScript and reference the Submit Example below to handle the posting of the credit card details directly to XiIntercept eComm.

### QA JS Submit Example

```
$XIPlugin.ajax({
```

```

url: '<% XiIntercept Url%>' + "/Cardinal3DSWithToken",
type: "POST", //<GET/POST>
data: {Formatted JS Data - See above example myData},
success: function(a)
{
    //function triggered on success.
},
error: function(a)
{
    //function triggered on failure.
},
beforeSend: function(a)
{
    //function triggered before ajax call.
},
complete: function(a)
{
    //function triggered after ajax call.
    // After success or failure
}

validation: function(a)
{
    //Run custom validation
    return true; //should return true or false.

    // return true will go through with ajax call.
    // return false will not go through with ajax call.
}

});

```

#### QA JS Submit 3DS 2 Example

```

$XIPlugin.ajax({

    url: '<% XiIntercept Url%>' + "/Cardinal3DSWithToken",
    type: "POST", //<GET/POST>
    data: {Formatted JS Data - See above example myData},
    success: function(a)
    {
        //function triggered on success.
    },
    error: function(a)
    {
        //function triggered on failure.
    },
    beforeSend: function(a)
    {

```

```
        //function triggered before ajax call.
    },
    complete: function(a)
    {
        //function triggered after ajax call.
        // After success or failure
    }

    validation: function(a)
    {
        //Run custom validation
        return true; //should return true or false.

        // return true will go through with ajax call.
        // return false will not go through with ajax call.
    },
    threeDSVersion: "2.2.0"

    });
```

### 11.3 3DS Token: Get Response Packet

#### Request Method

A direct server-side GET is required to retrieve the Response Packet. How this GET is performed depends on the language you are using.

#### Content Type

The content type must be "application/x-www-form-urlencoded".

#### Endpoints

Dev/QA:

<https://cert-xiecomm.paymetric.com/DIEComm/ResponsePacket>

Production:

<https://xiecomm.paymetric.com/DIEComm/ResponsePacket>

### GET Call Parameters

MerchantGUID={**MerchGUID**}&Signature={**Signature**}&AccessToken={**AccessToken**}

Table 11: Get Response Packet Parameters

| Parameter    | Description                                                                                                                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MerchantGuid | The Global Unique Identifier assigned by Paymetric                                                                                                                                                                   |
| Signature    | Sign the Access Token; encode it with HMAC-SHA-256 using the shared key then covert it to a base 64 string.<br>The Signature must be URL encoded.                                                                    |
| AccessToken  | A string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key. |

XiIntercept eComm is used to integrate the AliPay alternative payment method through the Cardinal Commerce platform.

## Requirements

- AliPay workflow requires Cardinal Commerce Credentials.
  - You will see AliPay listed on the XiIntercept eComm page in Merchant Portal if your Merchant Profile is properly configured. You choose the edit option to enter your Cardinal Credentials.
  - The same credentials are used for both AliPay and 3DS workflows. If you already have credentials entered in Merchant Portal for 3DS, you do not need to re-enter them; you will just now see AliPay listed on the XiIntercept eComm page as well .
- AliPay configuration also requires a Notification URL to which XiIntercept eComm will post the new AccessToken should an AliPay transaction result in a Pending status. See workflow below for more information.
  - For additional Onboarding information related to AliPay, search on "alipay" in the Merchant Portal help or XiAssist sites.

## Additional Information

- The reservation and settlement of funds occurs during the XiIntercept eComm call through Cardinal to AliPay.
- The authorization and settlement calls to XiPay Cardinal Centinel Cartridge are for reporting purposes; a call is not made out to AliPay at this time since the funds have already been moved, unless a credit is being performed.
- Credits are not performed through XiIntercept eComm; they are submitted directly through XiPay via the Authorization and Settlement operations with a negative amount. Cardinal requires a reference to the original transaction being credited. The original OrderId should be submitted in the CardNumber field.

## AliPay High-Level Integration Steps

- A. Initiate XiIntercept eComm session with a POST call to get the access token upon page load then render page as desired.
- B. Upon user submit, perform POST call using the Paymetric JS plugin.
- C. Perform GET call to get the response packet with token. If implementing 3DS, all response fields from Cardinal Commerce will be included as well.

The following sequence diagram illustrates the Form Post integration flow. Review the subsections below for more details.

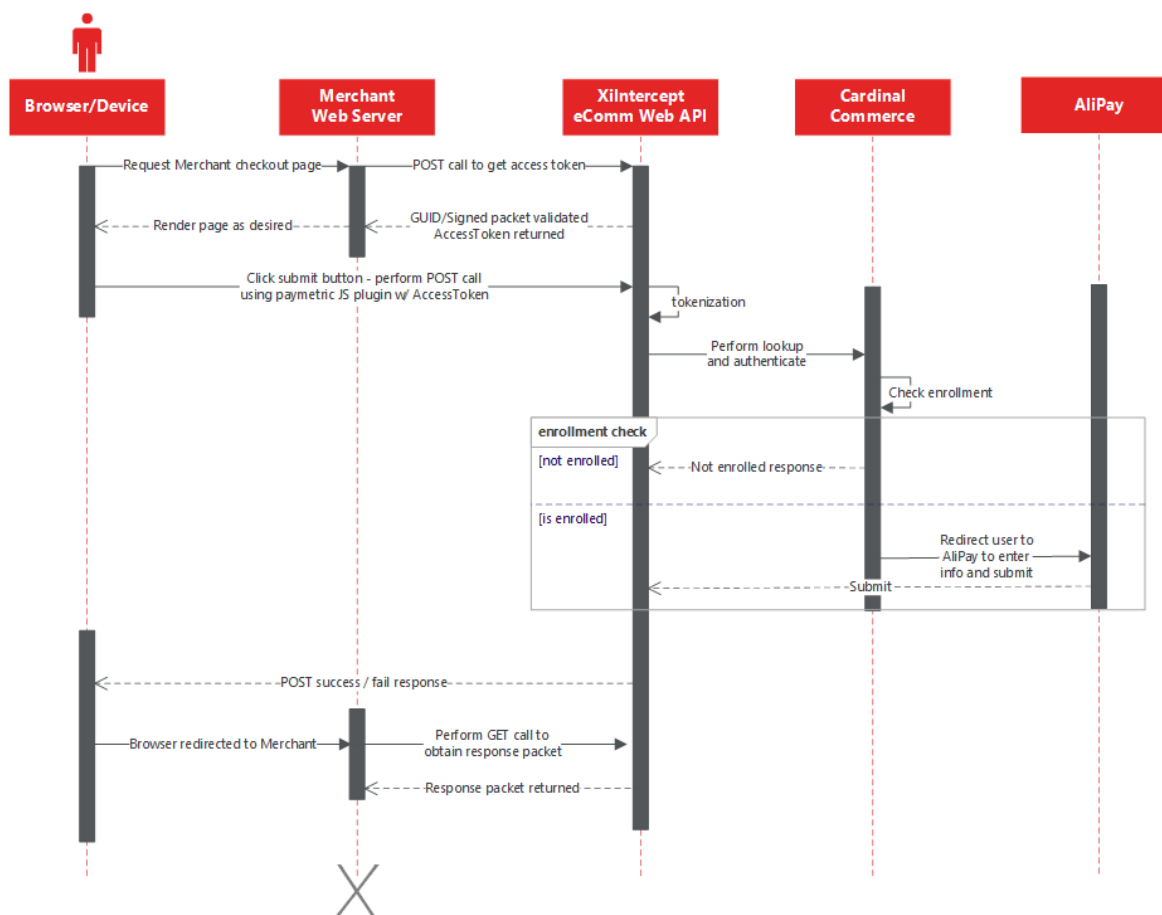


Figure 7: AliPay Integration Sequence Diagram

## Pending Status

Note that if the status from AliPay is Pending when the Response Packet is returned, the following occurs:

1. AliPay initiates a notification through CardinalCommerce when approved.
2. XiComm and Cardinal perform the status request and verification.
3. XiIntercept eComm creates new AccessToken and posts to Notification URL defined in configuration in Merchant Portal.
4. Merchant performs a server-side call to get the Response Packet using new AccessToken.

### 12.1 AliPay: Get Access Token

An AccessToken is required to initiate an XiIntercept eComm session.

Reminder that the AccessToken is a string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key.

### Request Method

A direct server-side POST is required to retrieve the Access Token. How this POST is performed depends on the language you are using.

### Content Type

The content type must be "application/x-www-form-urlencoded".

### Endpoint

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/AccessToken`

Production:

`https://xiecomm.paymetric.com/DIEComm/AccessToken`

### POST Call Parameters

Following is the POST data:

MerchantGuid={Your-GUID}&SessionRequestType={Int-Type-Code}&Signature={Signature}  
&MerchantDevelopmentEnvironment={code language e.g. asp.net}&Packet={PACKET\_XML}

| Table 12: Form Post Get Access Token Parameters |                                                                                                                                                    |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter                                       | Description                                                                                                                                        |
| MerchantGuid                                    | The Global Unique Identifier assigned by Paymetric                                                                                                 |
| SessionRequestType                              | Indicates the integration type. Use 7 for AliPay.                                                                                                  |
| Signature                                       | Sign the Packet XML; encode it with HMAC-SHA-256 using the shared key then convert it to a base 64 string.<br>The Signature should be URL encoded. |
| MerchantDevelopmentEnvironment                  | Indicate the language in which you are coding your integration to XiIntercept eComm, for example, asp.net or php.                                  |

Table 12: Form Post Get Access Token Parameters

## Packet XML

The Packet XML should be URL encoded. Following are the elements:

## Required:

- RedirectUri = the redirect URL from AliPay to Merchant

Optional: (see the Cardinal Centinel data mapping for more details on optional fields)

- Amount = amount is optional to support partial payments (allow for client-side ability to change the amount)
- Tax Amount
- ShippingAmount
- GiftCardAmount
- CurrencyCode
- OrderNumber
- OrderDescription
- MerchantData
- OrderChannel
- ProductCode
- TransactionMode
- CategoryCode
- RecurringAmount
- Email
- BillingFirstName
- Items (can have multiple instances of Items -- see AliPay XML example)
- Item\_Name
- Item\_Desc
- Item\_Price

Table 12: Form Post Get Access Token Parameters

|                     |                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|                     | <ul style="list-style-type: none"> <li>Item_Quantity</li> <li>Item_SKU</li> </ul> <p>See samples below.</p>                       |
| customerRedirectUri | Applicable only for the Hosted page when implementing Securelink. Defines the Merchant redirect URL after the User clicks submit. |

### AliPay XML Sample

```
<?xml version="1.0"?>
<AliPayPostPacketModel>
  <RedirectUri>{http://Yourdomain.here}</RedirectUri>
  <Amount>20.00</Amount>
  <TaxAmount>1.65</TaxAmount>
  <Items>
    <Item_Desc>Blue fuzzy socks</Item_Desc>
    <Item_Name>Womens Socks</Item_Name>
    <Item_Price>2.00</Item_Price>
    <Item_Quantity>5</Item_Quantity>
    <Item_SKU>123</Item_SKU>
  </Items>
  <Items>
    <Item_Desc>Red fuzzy socks</Item_Desc>
    <Item_Name>Womens Socks</Item_Name>
    <Item_Price>2.00</Item_Price>
    <Item_Quantity>5</Item_Quantity>
    <Item_SKU>122</Item_SKU>
  </Items>
</AliPayPostPacketModel>
```

## 12.2 AliPay: On-Submit Javascript Plugin

When Customer performs the submit action (e.g. Pay Now button), a POST call should be made to XiI-eComm using a JavaScript (JS) plugin provided by Paymetric.

You can reference or download the following JavaScript plugin:

<https://xiecomm.paymetric.com/DIEComm/Scripts/AliPayPlugin/AliPayPlugin-1.0.0.js>

Include the Paymetric JavaScript and reference the Submit Example below to handle the posting of the credit card details directly to XiIntercept eComm.

### QA JS Submit Example

```
<script src="https://cert-xiecomm.paymetric.com/DIEComm/Form/Scripts/AliPayPlugin/AliPay-1.0.0.js"></script>

...

var diecommUrl = 'https://cert-xiecomm.paymetric.com/DIEComm/Form/AliPay';
$AliPlugin.submit({
  url: diecommUrl,
  data: myData,
  validation: function () { return true; }
});
```

The appropriate lookup and verification calls are made to check enrollment.

If enrolled, redirect to AliPay, User enters AliPay information and submits payment. XiI-eComm and CardinalCommerce make the authenticate and verification calls.

## 12.3 AliPay: Get Response Packet

### Request Method

A direct server-side GET is required to retrieve the Response Packet. How this GET is performed depends on the language you are using.

### Content Type

The content type must be "application/x-www-form-urlencoded".

### Endpoints

Dev/QA:

`https://cert-xiecomm.paymetric.com/DIEComm/ResponsePacket`

Production:

`https://xiecomm.paymetric.com/DIEComm/ResponsePacket`

### GET Call Parameters

MerchantGUID=<MerchGUID>&Signature=<SIGNATURE>&AccessToken=<AccessToken>

Table 13: Form Post Get Response Packet Parameters

| Parameter    | Description                                        |
|--------------|----------------------------------------------------|
| MerchantGuid | The Global Unique Identifier assigned by Paymetric |

|             |                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signature   | Sign the Access Token; encode it with HMAC-SHA-256 using the shared key then covert it to a base 64 string.<br><br>The Signature should be URL encoded.                                                              |
| AccessToken | A string that is dynamically generated by XiIntercept eComm during the first application server call. The call includes the Merchant GUID and the XiIntercept eComm Request Packet that is signed by the Shared Key. |

### Pending Status

Note that if the status from AliPay is Pending when the Response Packet is returned, the following occurs:

1. AliPay initiates a notification through CardinalCommerce when approved.
2. XieComm and Cardinal perform the status request and verification.
3. XiIntercept eComm creates new AccessToken and posts to Notification URL defined in configuration in Merchant Portal.
4. Merchant performs a server-side call to get the Response Packet using new AccessToken.

XiIntercept eComm can be used to keep the raw credit number from entering the Merchant's website and environment when using the Visa Checkout alternative payment method.

## Visa Checkout with XiIntercept Highlights

- Visa Checkout can be added as an alternative payment with either the Form Post, Ajax or Iframe integration type.
- Merchants provide their customers with a Visa checkout button. When their customer clicks the button to make a payment, a unique 'callid' for the transaction is generated by Visa and given back to the Merchant. Merchant provides the unique 'callid' to allow Paymetric to interact with Visa on their behalf for the transaction.
- Paymetric XiIntercept eComm Visa Checkout WebAPI Controller uses the incoming 'callid' sent by the merchant and makes another GetPaymentData call on behalf of the merchant requesting the FULL data packet to obtain all of the user information including the raw card number.
- Paymetric tokenizes the card number so that it never reaches the Merchant's environment nor website. The token along with all of the customer information is passed back to the merchant in the same format as the Content Media-Type in the request packet, either "application/xml" or "application/json".

### 13.1 Visa Checkout Requirements

- Paymetric Merchant GUID (global unique identifier used to authenticate to the XiIntercept eComm platform)
- Paymetric Shared Key (a character string used for encoding the XiIntercept eComm Request and Response packets as well as confirming the signature of both packets)
- Visa API Key (a character string provided by Visa to authenticate to their platform)
- Visa Shared Secret (character string associated with the API Key provided by Visa)
- Visa Checkout alternative payment method must be implemented with one of the following integration types: Form Post, Ajax or Iframe.

### 13.2 Visa Checkout Integration Workflow

An AccessToken is required to initiate an XiIntercept eComm session. The AccessToken is referenced in the Form Post, Ajax, and Iframe integration type workflows. A server-side call is made to get the Access Token.

### A. Implement Chosen Standard Integration Type

Before the Visa Checkout workflow is applicable, you will implement either the Form Post, Ajax, or Iframe integration type for standard payments. After obtaining the AccessToken, the workflow will be as follows if the User chooses Visa Checkout as the payment method.

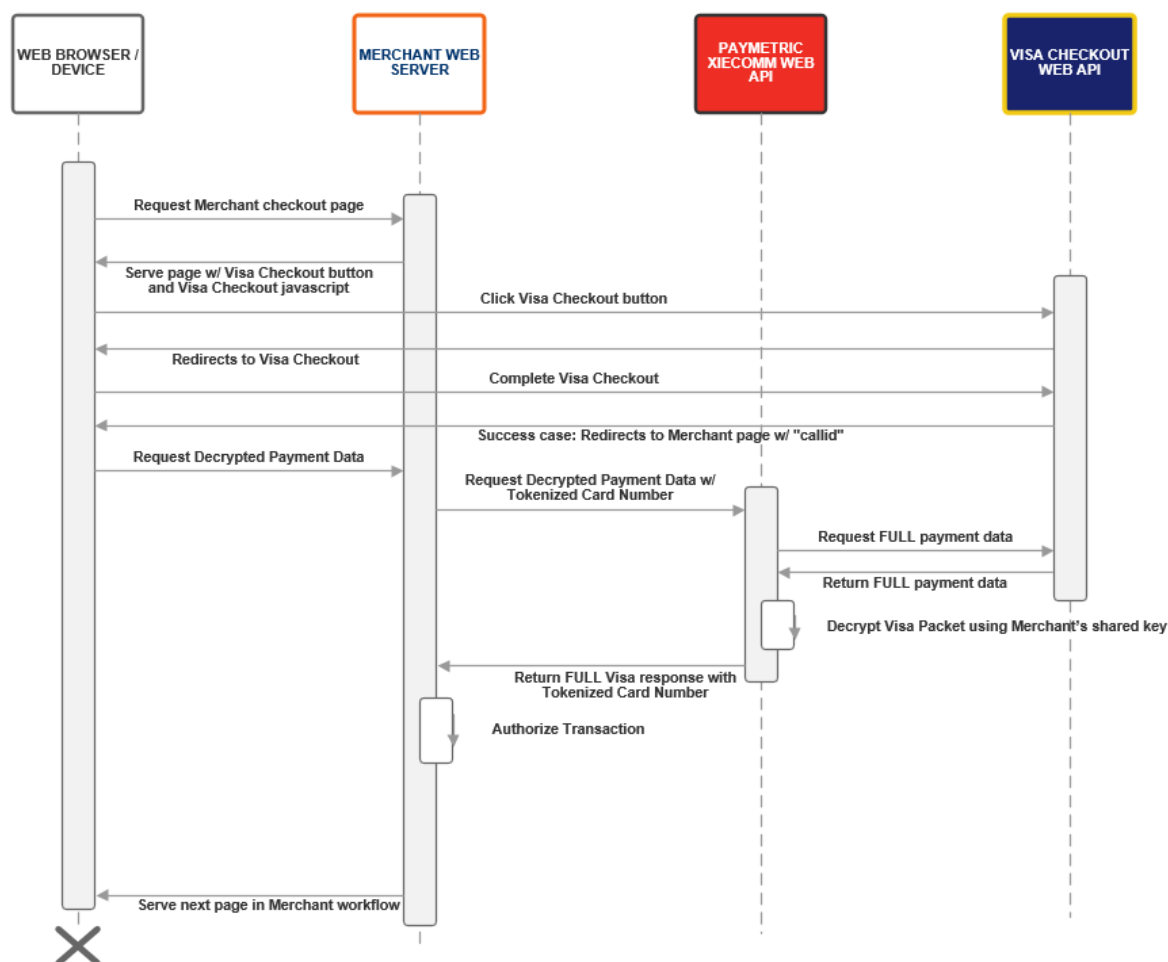


Figure 8: VisaCheckout Sequence Diagram

The following two sections detail the sequence diagram workflow:

### B. Perform Get Callid Directly from Visa

Refer to Visa's documentation for more information on implementing the Visa Checkout button and obtaining the 'callid' ([https://arch.developer.visa.com/vme/merchant/documents/Visa\\_Checkout\\_JavaScript\\_Integration\\_Guide/Get\\_Payment\\_Data.html](https://arch.developer.visa.com/vme/merchant/documents/Visa_Checkout_JavaScript_Integration_Guide/Get_Payment_Data.html)). Following are the high-level steps:

1. User accesses Merchant's checkout page or the payment selection page

2. Merchant provides the user with the Visa Checkout button with the dataLevel query parameter set to 'Summary' ("datalevel=Summary")
3. User clicks on the Visa Checkout button and logs into his/her Visa checkout account
4. On Success, Visa sends back basic info to the Merchant in JSON which includes a 'callid'. This is the 'callid' that merchant will provide to Paymetric when getting the Payment Data.

### C. Perform Get Payment Data Call to Paymetric

XiIntercept eComm Visa Checkout endpoint addresses:

QA: <https://cert-xiecomm.paymetric.com/DleComm/visacheckoutgetpaymentdata>

Production: <https://xiecomm.paymetric.com/DleComm/visacheckoutgetpaymentdata>

1. Merchant sends a POST request with the 'callid' to Paymetric's GetPaymentData Controller using the XiIntercept eComm URL. E.g. <https://cert-xiecomm.paymetric.com/diecomm/visacheckoutgetpaymentdata>
  - a. Incoming POST request must meet the following conditions:
    - i. Have a Request Header with the following name and value set: "xi-version" : "1.0"
    - ii. Have a Request Content with MediaType of either: "application/json" or "application/xml"
    - iii. Have a complete Request Body either in JSON or XML with the following fields set:
      1. MerchantGUID – Provided by Paymetric. Can obtain from Merchant Portal or your Implementation Consultant.
      2. Signature - HMAC SHA256 Hash of MerchantGUID + CallId + APIKey + SharedSecret + TimeStamp in that order signed by <SharedKey> provided by Paymetric. Can obtain the Shared Key from Merchant Portal. (See [Visa Checkout HMAC-SHA256 C# Sample](#) <sup>86</sup> if needed)
      3. CallId - The callid obtained by the merchant in response back from Visa (in steps 1-4 above).
      4. TimeStamp - This must be same as was used in forming the Signature. Timestamp is the UTC Timestamp following ISO 8601 format: yyyy-MM-ddTHH:mm:ssZ, e.g. 2016-08-25T19:26:40Z
      5. AccessToken – Unique one-time token Merchant retrieves during the first step in the XiIntercept eComm 3.0 workflow.
  - b. COMPLETE REQUEST Sample:

```
Method: POST, RequestUri: 'http://localhost/DIEComm/visacheckoutgetpaymentdata',
Version: 1.1, Content:
System.Web.Http.WebHost.HttpControllerHandler+LazyStreamContent, Headers:
{ Connection: Keep-Alive Expect: 100-continue Host: localhost xi-version: 1.0
Content-Length: 200 Content-Type: application/json}
Request Body: {"XiVCGetPaymentRequest":{"MerchantGuid":"6fcc767a-890d-407a-87f5-
898368fe5171","Signature":"sqMlkNqcN8tw93mPry57FBe/4RI+vc7GeZdSx0HCOwU=", "CallId":
"4753436069251330801","TimeStamp":"2016-08-25T19:46:22Z","AccessToken":"de542dac-
1466-454f-b741-1c6bedfe5c84"}}
```

2. Paymetric performs the following:

- a. Validates the signature, ensures the request is coming from a known merchant, and makes the Get Payment Data call to Visa.
- b. Decrypts the response from Visa using the Shared Secret key (configured in Merchant Portal), locates the accountNumber and tokenizes it.
- c. Sends the tokenized, decrypted response to the Merchant in either XML or JSON, as per the Content Media-Type present in the request, and the Visa response will be wrapped and have the following fields:
  - i. MerchantGUID -
  - ii. Signature - HMAC SHA256 Hash of MerchantGUID + CallId + APIKey + SharedSecret in that order, signed by the Shared Key given by Paymetric. This response signature is different from the incoming signature and it is provided to the merchant in case they want to do a check if the response was from Paymetric. Note: in case of incorrect request, the error response sent to the merchant might not have any signature at all.
  - iii. StatusCode - 1 for Pass or any other appropriate failure code. See the [Visa Checkout Status Codes](#) section for a list of all codes.
  - iv. TimeStamp – The TimeStamp obtained by computing time till UTC now.
  - v. VisaResponse - A node that consists of the response obtained from Visa (successful or error).
  - vi. Sample Response:

```
{
  "XiVCGetPaymentResponse": {
    "MerchantGuid": "6fcc767a-890d-407a-87f5-898368fe5171",
    "Signature": "DbraXtLPcKCRfwklnAks01M8CUwvhroKg84ggZnDfqY=",
    "StatusCode": "1",
    "TimeStamp": "2016-08-25T19:46:28Z",
    "VisaResponse": {
      "paymentRequest": {
        "currencyCode": "USD",
        "total": "21"
      },
      "userData": {
        "userFirstName": "John",
        "userLastName": "N",
        "userFullName": "John N",
        "userName": "jdoe@paymetric.com",
        "userEmail": "jdoe@paymetric.com",
        "encUserId": "N7KP3+fBSpiv4fthE/zxij9odHSiUdj3TwDb2wyU998=",
        "creationTimeStamp": "1469043515494",
        "paymentInstrument": {
          "id": "pie23+5NG0/UgHyijj116BaSSI4rltlm0pBhufM1m48=",
          "lastFourDigits": "4242",
          "binSixDigits": "424242",
          "accountNumber": "-E803-4242-J54PQNWFQJ73B",
          "billingAddress": {
            "personName": "John N",
            "firstName": "John",
            "lastName": "N",
            "line1": "South Congress Avenue",
            "city": "Austin",
            "stateProvinceCode": "TX",
            "postalCode": "78704",
            "countryCode": "US",
            "phone": "5122377520",
            "default": false,
            "verificationStatus": "VERIFIED",
            "expired": false,
            "cardArts": {
              "cardArt": [
                {
                  "baseImageFileName": "https://sandbox.secure.checkout.visa.com/VmeCardArts/wv87HR3X3jq1NXNJU_6YtLQyyO7mpu2aU6Yo3VWGKKM.png",
                  "height": 105,
                  "width": 164
                }
              ],
              "isSuerBid": "null",
              "nickName": "PMTest Visa ",
              "nameOnCard": "John N",
              "cardFirstName": "John",
              "cardLastName": "N",
              "expirationDate": {
                "month": "09",
                "year": "2020"
              },
              "paymentType": {
                "cardBrand": "VISA",
                "cardType": "CREDIT"
              },
              "shippingAddress": {
                "id": "na/woORRCpJUGijhyAEsFMnoU7A8nYkZDNzPMJsXYMI=",
                "verificationStatus": "CONSUMER_OVERRIDE",
                "personName": "John N",
                "firstName": "John",
                "lastName": "N",
                "line1": "South Congress Avenue",
                "city": "Austin",
                "stateProvinceCode": "TX",
                "postalCode": "78704",
                "countryCode": "US",
                "phone": "5122377520",
                "default": false,
                "riskData": {
                  "advice": "UNAVAILABLE",
                  "score": 0,
                  "avsResponseCode": "0",
                  "cvvResponseCode": "0",
                  "ageOfAccount": 57,
                  "visaCheckoutGuest": false,
                  "partialShippingAddress": {
                    "countryCode": "US",
                    "postalCode": "78704"
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

3. Merchant performs authorization (and any other business logic) and serves next page in their workflow.
4. End workflow.

#### NOTES:

- Default Encoding on XML or JSON response will be UTF8.
- Visa Checkout API Key and Shared Secret as well as the Paymetric Shared Key are all stored in an encrypted format.

## 13.3 Visa Checkout Status Codes

Following are the potential Status Codes returned in the response packet for the Visa Checkout workflow.

| Code                                  | Description             |
|---------------------------------------|-------------------------|
| Global XiIntercept eComm Status Codes |                         |
| 1                                     | Passed                  |
| 100                                   | Internal DI eComm Error |

| Code                                | Description                                    |
|-------------------------------------|------------------------------------------------|
| 101                                 | Merchant GUID Invalid                          |
| 102                                 | Signature Validation Failed                    |
| 104                                 | DleComm Session Expired                        |
| 120                                 | XiSecure Error                                 |
| Visa Checkout Specific Status Codes |                                                |
| 701                                 | Missing or Invalid Request Format              |
| 702                                 | Incorrect Request Data Received.               |
| 703                                 | Missing or Invalid Signature in the request    |
| 704                                 | Missing or invalid TimeStamp in the request    |
| 705                                 | Missing or Invalid CallId in the request       |
| 706                                 | Missing AccessToken in the request             |
| 707                                 | Incoming request took too long to reach        |
| 708                                 | Error communicating with VisaCheckout service. |

## 13.4 Visa Checkout Sample Responses

### Get Payment Data Successful Responses

#### XML Successful Response

```
<?xml version="1.0"?>
<XiVCGetPaymentResponse>
  <MerchantGuid>6fcc767a-890d-407a-87f5-898368fe5171</MerchantGuid>
  <Signature>DbraXtLPcKCRfwklnAks01M8CUwvhroKg84ggZnDfqY=</Signature>
  <StatusCode>1</StatusCode>
  <TimeStamp>2016-08-25T20:46:56Z</TimeStamp>
  <VisaResponse>
    <ns2:getPaymentDataResponse
xmlns:ns4="http://www.visa.com/vme/walletservices/internal/xo"
xmlns:ns3="http://www.visa.com/vme/walletservices/internal/common"
xmlns:ns2="http://www.visa.com/vme/walletservices/external/payment"
xmlns="http://www.visa.com/vme/walletservices/external/common">
      <partialShippingAddress>
        <countryCode>US</countryCode>
        <postalCode>78704</postalCode>
      </partialShippingAddress>
    </ns2:getPaymentDataResponse>
  </VisaResponse>
</XiVCGetPaymentResponse>
```

## XML Successful Response

```

<ns2:paymentRequest>
  <currencyCode>USD</currencyCode>
  <total>21</total>
</ns2:paymentRequest>
<ns2:userData>
  <ns2:userFirstName>John</ns2:userFirstName>
  <ns2:userLastName>N</ns2:userLastName>
  <ns2:userFullName>John N</ns2:userFullName>
  <ns2:userName>rnigam@paymetric.com</ns2:userName>
  <ns2:userEmail>rnigam@paymetric.com</ns2:userEmail>
  <ns2:encUserId>N7KP3+fBSpiv4fthE/zxij9odHSiUdj3TwDb2wyU998=</ns2:encUserId>
>
</ns2:userData>
<ns2:creationTimeStamp>2016-07-20T19:38:35.494Z</ns2:creationTimeStamp>
<ns2:paymentInstrument>
  <nickName>PMTest Visa </nickName>
  <nameOnCard>John N</nameOnCard>
  <cardFirstName>John</cardFirstName>
  <cardLastName>N</cardLastName>
  <expirationDate>
    <month>09</month>
    <year>2020</year>
  </expirationDate>
  <paymentType>
    <cardBrand>VISA</cardBrand>
    <cardType>CREDIT</cardType>
  </paymentType>
  <id>pie23+5NG0/UgHyijj116BaSSI4rltlm0pBhufM1m48=</id>
  <lastFourDigits>4242</lastFourDigits>
  <binSixDigits>424242</binSixDigits>
  <accountNumber>-E803-4242-J54PQNWFGQJ73B</accountNumber>
  <billingAddress>
    <personName>John N</personName>
    <firstName>John</firstName>
    <lastName>N</lastName>
    <line1>South Congress Avenue</line1>
    <city>Austin</city>
    <stateProvinceCode>TX</stateProvinceCode>
    <postalCode>78704</postalCode>
    <countryCode>US</countryCode>
    <phone>5122377520</phone>
    <default>>false</default>
  </billingAddress>
  <verificationStatus>VERIFIED</verificationStatus>
  <expired>>false</expired>
  <cardArts>
    <cardArt>
      <baseImageFileName>https://sandbox.secure.checkout.visa.com/VmeCardArts/wv87HR3X3jq1NXNJJu_6YtLQyyO7mpu2aU6Yo3VWGKKM.png</baseImageFileName>
      <height>105</height>
      <width>164</width>
    </cardArt>
  </cardArts>

```

## XML Successful Response

```
</cardArts>
<issuerBid>null</issuerBid>
</ns2:paymentInstrument>
<ns2:shippingAddress>
  <personName>John N</personName>
  <firstName>John</firstName>
  <lastName>N</lastName>
  <line1>South Congress Avenue</line1>
  <city>Austin</city>
  <stateProvinceCode>TX</stateProvinceCode>
  <postalCode>78704</postalCode>
  <countryCode>US</countryCode>
  <phone>5122377520</phone>
  <default>false</default>
  <id>na/woORRCpJUGijhyAEsFMnoU7A8nYkZDNzPMJsXYMI=</id>
  <verificationStatus>CONSUMER_OVERRIDE</verificationStatus>
</ns2:shippingAddress>
<ns2:riskData>
  <advice>UNAVAILABLE</advice>
  <score>0</score>
  <avsResponseCode>0</avsResponseCode>
  <cvvResponseCode>0</cvvResponseCode>
  <ageOfAccount>57</ageOfAccount>
</ns2:riskData>
<ns2:visaCheckoutGuest>false</ns2:visaCheckoutGuest>
</ns2:getPaymentDataResponse>
</VisaResponse>
</XiVGetPaymentResponse>
```

## JSON Successful Response

```
{
  "XiVCGetPaymentResponse": {
    "MerchantGuid": "6fcc767a-890d-407a-87f5-898368fe5171",
    "Signature": "DbraXtLPcKCRfwklnAks01M8CUwvhroKg84ggZnDfqY=",
    "StatusCode": "1",
    "TimeStamp": "2016-08-25T20:53:02Z",
    "VisaResponse": {
      "paymentRequest": {
        "currencyCode": "USD",
        "total": "21"
      },
      "userData": {
        "userFirstName": "John",
        "userLastName": "N",
        "userFullName": "John N",
        "userName": "rnigam@paymetric.com",
        "userEmail": "rnigam@paymetric.com",
        "encUserId": "N7KP3+fBSpiv4fthE/zxij9odHSiUdj3TwDb2wyU998=",
        "creationTimeStamp": "1469043515494",
        "paymentInstrument": {
          "id": "pie23+5NG0/UgHyijj1l6BaSSI4rltlm0pBhufMlm48=",
          "lastFourDigits": "4242",
          "binSixDigits": "424242",
          "accountNumber": "-E803-4242-J54PQNWFGQJ73B",
          "billingAddress": {
            "personName": "John N",
            "firstName": "John",
            "lastName": "N",
            "line1": "South Congress Avenue",
            "city": "Austin",
            "stateProvinceCode": "TX",
            "postalCode": "78704",
            "countryCode": "US",
            "phone": "5122377520",
            "default": false
          },
          "verificationStatus": "VERIFIED",
          "expired": false,
          "cardArts": {
            "cardArt": [
              {
                "baseImageFileName": "https://sandbox.secure.checkout.visa.com/VmeCardArts/wv87HR3X3jqLNXNJ_u6YtLQyyO7mpu2aU6Yo3VWGKKM.png",
                "height": 105,
                "width": 164
              }
            ],
            "issuerBid": "null",
            "nickName": "PMTTest Visa ",
            "nameOnCard": "John N",
            "cardFirstName": "John",
            "cardLastName": "N",
            "expirationDate": {
              "month": "09",
              "year": "2020"
            },
            "paymentType": {
              "cardBrand": "VISA",
              "cardType": "CREDIT"
            },
            "shippingAddress": {
              "id": "na/woORRCpJUGijhyAESFMnoU7A8nYkZDNzPMJsXYMI=",
              "verificationStatus": "CONSUME R_OVERRIDE",
              "personName": "John N",
              "firstName": "John",
              "lastName": "N",
              "line1": "South Congress Avenue",
              "city": "Austin",
              "stateProvinceCode": "TX",
              "postalCode": "78704",
              "countryCode": "US",
              "phone": "5122377520",
              "default": false
            },
            "riskData": {
              "advice": "UNAVAILABLE",
              "score": 0,
              "avsResponseCode": "0",
              "cvvResponseCode": "0",
              "ageOfAccount": 57
            },
            "visaCheckoutGuest": false,
            "partialShippingAddress": {
              "countryCode": "US",
              "postalCode": "78704"
            }
          }
        }
      }
    }
  }
}
```

## GetPaymentData Failure Response from Visa

The following responses will occur if the call to Paymetric was successful but there is an error returned from Visa.

## XML Failure Response from Visa

```
<?xml version="1.0"?>
<XiVCGetPaymentResponse>
  <MerchantGuid>6fcc767a-890d-407a-87f5-898368fe5171</MerchantGuid>
  <Signature>kXW7Jan19MH3r3eEuqlkYZ6zKoKNXPxHlug5DYvtzTc=</Signature>
  <StatusCode>1</StatusCode>
  <TimeStamp>2016-08-25T20:54:36Z</TimeStamp>
  <VisaResponse>
    <ns2:getPaymentDataResponse
      xmlns:ns4="http://www.visa.com/vme/walletservices/internal/xo"
      xmlns:ns3="http://www.visa.com/vme/walletservices/internal/common"
      xmlns:ns2="http://www.visa.com/vme/walletservices/external/payment"
      xmlns="http://www.visa.com/vme/walletservices/external/common">
```

**XML Failure Response from Visa**

```
<responseStatus>
  <status>404</status>
  <code>1010</code>
  <severity>ERROR</severity>
  <message>Expected resource is not available. Intent not found.</message>
</responseStatus>
</ns2:getPaymentDataResponse>
</VisaResponse>
</XiVCGetPaymentResponse>
```

**JSON Failure Response from Visa**

```
{ "XiVCGetPaymentResponse": { "MerchantGuid": "6fcc767a-890d-407a-87f5-898368fe5171", "Signature": "kXW7Jan19MH3r3eEuqlkYZ6zKoKNXPxHlug5DYvtzTc=", "StatusCode": "1", "TimeStamp": "2016-08-25T20:53:46Z", "VisaResponse": { "responseStatus": { "status": 404, "code": 1010, "severity": "ERROR", "message": "Expected resource is not available. Intent not found." } } } }
```

**GetPaymentData Failure from Paymetric**

The following responses will occur if there is a failure by Paymetric before it is sent Visa.

**XML Failure Response from Paymetric**

```
<?xml version="1.0"?>
<XiVCGetPaymentResponse>
  <MerchantGuid>6fcc767a-890d-407a-87f5-898368fe5171</MerchantGuid>
  <Signature />
  <StatusCode>104</StatusCode>
  <TimeStamp>2016-08-25T20:55:11Z</TimeStamp>
</XiVCGetPaymentResponse>
```

**JSON Failure Response from Paymetric**

```
{ "XiVCGetPaymentResponse": { "MerchantGuid": "6fcc767a-890d-407a-87f5-898368fe5171", "Signature": "", "StatusCode": "104", "TimeStamp": "2016-08-25T20:55:37Z" } }
```

## 13.5 Visa Checkout HMAC-SHA256 C# Sample

Following is the basic HMAC + SHA256 sample used to generate a hash of a string input using a key:

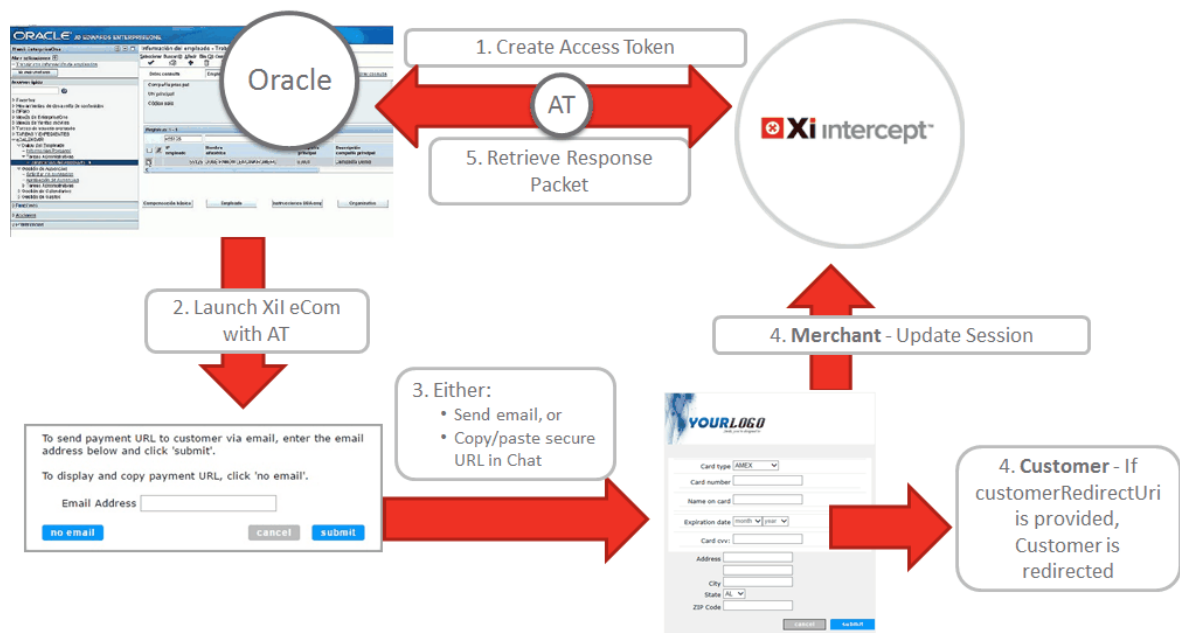
```
public static string GetHashHmacSha256(string text, string key)
{
    HMAC hashAlg = HMAC.Create("HMACSHA256");
    hashAlg.Key = System.Text.Encoding.Default.GetBytes(key);
    byte[] bytes = Encoding.ASCII.GetBytes(text);
    byte[] hash = hashAlg.ComputeHash(bytes);
    return Convert.ToBase64String(hash);
}
```

NOTE: You may want to add some additional logic to handle error scenarios.

Allows the Merchant Representative to send an email that contains a secure link to the XiIntercept interface hosted in Paymetric's SaaS environment. The email recipient clicks the link to launch XiIntercept and then enters the credit card information. The credit card details and a token are returned directly to the ERP system.

Optionally, the Merchant Representative can choose to obtain the secure Paymetric hosted URL and copy/paste it into a chat window or other secure form of communication with the Customer.

For Oracle, other web stores or non-SAP ERP systems, Securelink is integrated by adding the `<emailPaymentOption>` element in the XML packet for the get access token call that is made to XiIntercept eComm.



1. A direct server-side POST is required to retrieve the Access Token. How this POST is performed depends on the language you are using. The content type must be "application/x-www-form-urlencoded".
2. XiIntercept eCommerce Hosted UI is launched in the default browser with the AccessToken and the Merchant Representative is presented with two options to obtain the payment information:
  - A. A field to enter an email address for the Customer
  - B. A button to generate the secure URL that can be copied
3. Depending upon which option is chosen in Step 2,
  - A. The email is sent with the link that launches XiIntercept hosted page. Then Customer enters payment details.
  - B. Or, the Merchant representative sends URL via chat or other method. Then, Customer enters payment details.

4. The token is obtained.
5. Response Packet is obtained via a server-side GET call back into Oracle.

Note that other non-SAP ERP or applications/systems have the same workflow as defined for Oracle.

## 14.1 emailPaymentOption Element Attributes

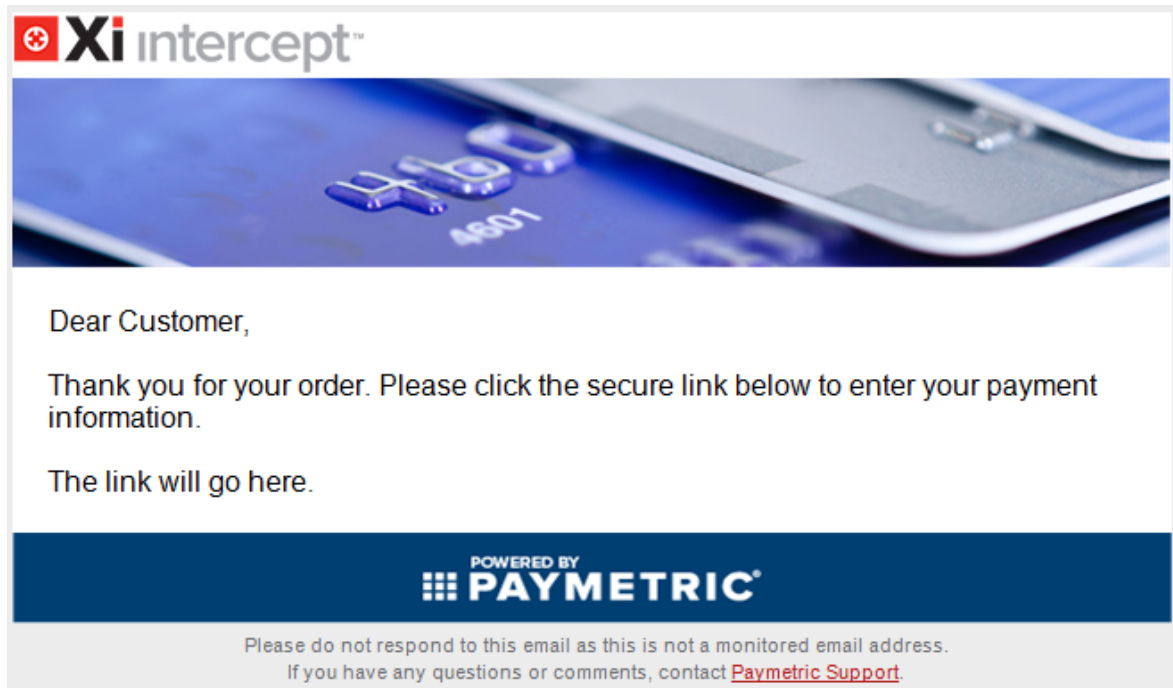
The following table defines the available `<emailPaymentOption>` attributes:

Table 14: Securelink Parameter Descriptions for Get Access Token Call

| Element Attributes  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |                                                                                          |               |                                                                                                                                                                                        |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| emailTemplate       | <p>The text for the email body. Use [BR] to insert a line break as needed.</p> <p>The following two variables are available to add the URL link in the body. Choose which to use depending upon how much control you want over the actual link.</p> <table><tr><td>[PAYMENTURL]</td><td>Is replaced with the appropriate QA or PROD URL based upon which system you are calling.</td></tr><tr><td>[PAYMENTLINK]</td><td>Is replaced with the following where PAYMENTURL is QA or PROD based on which system you are calling.<br/><code>&lt;a href='[PAYMENTURL] '&gt;Enter Payment Information&lt;/a&gt;</code></td></tr></table> | [PAYMENTURL] | Is replaced with the appropriate QA or PROD URL based upon which system you are calling. | [PAYMENTLINK] | Is replaced with the following where PAYMENTURL is QA or PROD based on which system you are calling.<br><code>&lt;a href='[PAYMENTURL] '&gt;Enter Payment Information&lt;/a&gt;</code> |
| [PAYMENTURL]        | Is replaced with the appropriate QA or PROD URL based upon which system you are calling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |                                                                                          |               |                                                                                                                                                                                        |
| [PAYMENTLINK]       | Is replaced with the following where PAYMENTURL is QA or PROD based on which system you are calling.<br><code>&lt;a href='[PAYMENTURL] '&gt;Enter Payment Information&lt;/a&gt;</code>                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                                                                                          |               |                                                                                                                                                                                        |
| merchantLogoURL     | Enter the URL for your custom logo. If this attribute is not provided, XiIntercept logo displays by default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |                                                                                          |               |                                                                                                                                                                                        |
| fromEmailAddress    | Defines the from address for the email. Defaults to <a href="mailto:noreply@paymetric.com">noreply@paymetric.com</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                                                                                          |               |                                                                                                                                                                                        |
| customerRedirectUri | The URL to which the Customer is redirected once he/she clicks submit after the entering the payment information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |                                                                                          |               |                                                                                                                                                                                        |

## 14.2 Default Email Text

If you do not define any of the `<emailPaymentOption>` attributes, following is an example of the default email that will be send if you choose the email option.



## 14.3 XML Packet Example with emailPaymentOption Element

The endpoint addresses to call SecureLink are as follows:

- QA: <https://cert-xiecomm.paymetric.com/DleComm/paymentpage/startemailpayment/<merchantGuid>/<accessToken>>
- PROD: <https://xiecomm.paymetric.com/DleComm/paymentpage/startemailpayment/<merchantGuid>/<accessToken>>

Following is an example of the XML packet with the `emailPaymentOption` element highlighted.

```
<?xml version="1.0" encoding="utf-8"?>
<merchantHtmlPacketModel>
  <paymentPagePacket>
    <redirectUri>http://dh-ecomm-w2k12.pmdev.local/XiInterceptDemo/Status.aspx</redirectUri>
    <cssUri>http://dh-ecomm-w2k12.pmdev.local/DleComm/Content/PaymentStyleSheet.css</cssUri>
    <emailPaymentOption>
```

```

emailTemplate="Thank you for your interest.[BR]Please click the following
link to continue.[BR][PAYMENTLINK][BR][BR]If you are unable to click the
link, copy/paste this URL into your browser: [PAYMENTURL][BR]Regards,[BR]
My Company, Inc.[BR] 1-800-555-5555"
merchantLogoUrl="http://img13.mysite.net/images/mylogo.png"
fromEmailAddress="noreply@company.com"
customerRedirectUri="https://mycompany.com" />
</paymentPagePacket>
<merchantHtml>
  <htmlSection class="payment-content"
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="billing-content">
      <tag name="div" class="billing-header">Billing
Information</tag>
      <tag name="div" class="billing-info">
        <paymentOptionRadioSection>
          <tag name="div">
            <tag name="div">choose payment type</tag>
            </tag>
            <label text="echeck" for="echeck" />
            <radPaymentOption for="echeck" />
            <label text="credit card" for="card" />
            <radPaymentOption for="card" />
          </paymentOptionRadioSection>
          <cardDropdownSection>
            <tag name="div">Enter your credit card
information</tag>
            <tag name="div">
              <label for="cardType" text="select a card" />
              <ddlCardType id="cd">
                <items>
                  <item for="american express" />
                  <item for="mastercard" />
                  <item for="visa" />
                  <customItem for="MSC" text="My Super
Card" />
                </items>
              </ddlCardType>
            </tag>
            <tag name="div">
              <label for="cardNumber" text="card number" />
              <tboxCardNumber tokenize="true" />
            </tag>
            <tag name="div">
              <label for="cardholderName" text="name on
card" />
              <tboxCardHolderName />
            </tag>
            <tag name="div">
              <label for="expMonth" text="exp date" />
              <ddlExpMonth default-text="month"
class="merchant_combos" />

```

```

                                <ddlExpYear default-text="year"
class="merchant_combos" years-to-display="10" />
                                </tag>
                                <tag name="div">
                                    <label for="cvv" text="card cvv:" />
                                    <tboxCvv />
                                    <htmlCvvHelp name="div" raise-click="1"/>
                                </tag>
                            </cardDropDownSection>
                            <echeckRadioSection>
                                <tag name="div">
                                    <label for="checking" />
                                    <radAccountType for="checking" />
                                    <label for="savings" />
                                    <radAccountType for="savings" />
                                </tag>
                                <tag name="div">
                                    <label for="accountNumber" text="account
number" />
                                    <tboxAccountNumber tokenize="true" />
                                </tag>
                                <tag name="div">
                                    <label for="routingNumber" text="routing
number" />
                                    <tboxRoutingNumber />
                                </tag>
                                <tag name="div">
                                    <label for="nameOnAccount" text="name on
account" />
                                    <tboxNameOnAccount />
                                </tag>
                                <tag name="div">
                                    <label for="bankName" text="bank name" />
                                    <tboxBankName />
                                </tag>
                            </echeckRadioSection>
                        </tag>
                        <tag name="div" class="payment-button">
                            <cancelButton text="cancel" class="cancel-button" />
                            <submitButton text="submit" class="submit-button" />
                        </tag>
                    </tag>
                </htmlSection>
            </merchantHtml>
        </merchantHtmlPacketModel>

```

For Iframe and Hosted integration types, there are three main options associated with localization as follows:

- Do not specify any localization which then defaults to U.S. English
- Specify to use the Browser locale; if it is a language not supported by XiIntercept eComm, then it defaults to U.S. English
- Hard code the locale using the XiIntercept eComm supported languages/culture codes

### Using Browser Locale

In the Request XML Packet, include the following after the `<merchantHtmlPackeModel>` element:

```
<localization>
  <browserLocalization />
</localization>
```

Note that hard-coded label text and messages in custom HTML will override both `merchantLocalization` and `browserLocalization`.

See the Supported Languages below.

### Hard Code Locale

In the Request XML Packet, include the following after the `<merchantHtmlPackeModel>` element:

```
<localization>
  <merchantLocalization>
    <cultureName>ja</cultureName>
  </merchantLocalization>
</localization>
```

Note that `merchantLocalization` overrides `browserLocalization` if both are set for some reason. Hard-coded label text and messages in custom HTML will override both `merchantLocalization` and `browserLocalization`.

See the Supported Languages below which includes the `cultureName` values for each.

### Supported Languages

The following table defines the languages currently supported by XiIntercept eComm along with the culture codes to use if you are hard coding the localization element. Reminder, when using browser localization, if the browser locale is not in the following list, then the XiIntercept eComm labels and messaging will default to U.S. English unless hard-coded in custom HTML text or message attributes.

| Language              | "cultureName"<br>Value |
|-----------------------|------------------------|
| Czech                 | cs                     |
| Danish                | da                     |
| German                | de                     |
| Spanish (European)    | es                     |
| Spanish (Latin)       | es-es                  |
| Finnish               | fi                     |
| French (Native)       | fr                     |
| French (Canadian)     | fr-ca                  |
| Hungarian             | hu                     |
| Italian               | it                     |
| Japanese              | ja                     |
| Korean                | ko                     |
| Dutch                 | nl                     |
| Flemish               | nl-be                  |
| Norwegian             | no                     |
| Polish                | pl                     |
| Portuguese (Portugal) | pt                     |
| Portuguese (Brazil)   | pt-br                  |
| Russian               | ru                     |
| Slovak                | sk                     |
| Slovenian             | sl                     |
| Swedish               | sv                     |
| Turkish               | tr                     |
| Chinese (Simplified)  | zh                     |
| Chinese (Traditional) | zh-tw                  |

Refer to the *XiIntercept eCommerce Language Translation* spreadsheet for the labels and validation message translations of all supported languages. The spreadsheet can be downloaded from Merchant Portal Resources page.

## Localization Example

Following is an example of using the browser locale in the IFrame Template XML packet:

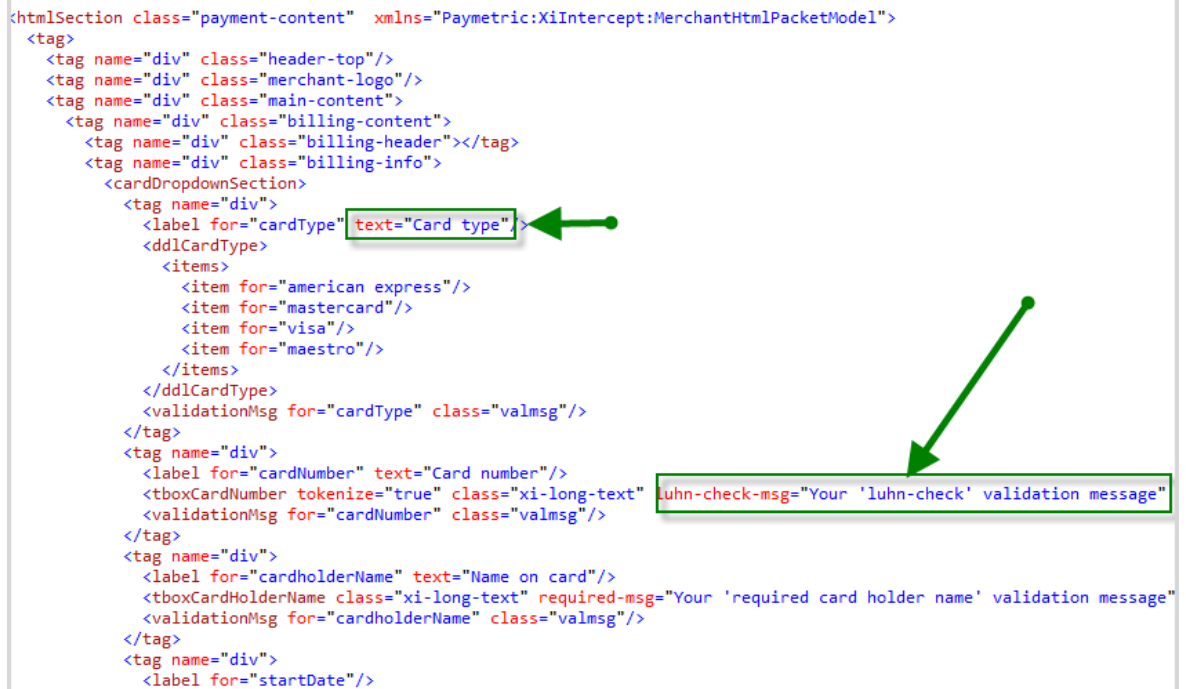
```
<?xml version="1.0" encoding="utf-8"?>
<merchantHtmlPacketModel
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <localization>
    <browserLocalization />
  </localization>
  <iFramePacket>
    <hostUri>{http://Yourdomain.here}</hostUri>
    <cssUri>http://dasvm121/DIEComm</cssUri>
    Sample/Content/IFrameTemplateStyleSheet.css</cssUri>
  </iFramePacket>
  <templateHtml name="CreditCard">
    <paymentTypes>
      <paymentType type="american express" />
      <paymentType type="mastercard" />
      <paymentType type="maestro" />
      <paymentType type="visa" />
      <paymentType type="custom" value="HD" text="Home Depot" />
    </paymentTypes>
  </templateHtml>
</merchantHtmlPacketModel>
```

## Localization and Custom HTML

Note that if you have hard-coded label text or messages using custom HTML, that value will override the localization setting. For example, in the following excerpt below from [Sample 13: Custom Messages](#), the values entered for the cardType label "text" and the tboxCardNumber "Luhn-check-msg" attributes will be used instead of any translated values based on browser or hard-coded locale.

If you are hard-coding text in the HTML, you must ensure that the text matches the locale you plan to use or this could result in mismatched languages rendered on the page.

```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"/>
        <tag name="div" class="billing-info">
          <cardDropdownSection>
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType>
                <items>
                  <item for="american express"/>
                  <item for="mastercard"/>
                  <item for="visa"/>
                  <item for="maestro"/>
                </items>
              </ddlCardType>
              <validationMsg for="cardType" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="cardNumber" text="Card number"/>
              <tboxCardNumber tokenize="true" class="xi-long-text"
                luhn-check-msg="Your 'luhn-check' validation message"
              >
                <validationMsg for="cardNumber" class="valmsg"/>
              </tag>
            </tag>
            <tag name="div">
              <label for="cardholderName" text="Name on card"/>
              <tboxCardHolderName class="xi-long-text" required-msg="Your 'required card holder name' validation message"
                >
                <validationMsg for="cardholderName" class="valmsg"/>
              </tag>
            </tag>
            <tag name="div">
              <label for="startDate"/>
            </tag>
          </cardDropdownSection>
        </tag>
      </div>
    </div>
  </tag>
</htmlSection>
```



The following table defines the supported card types and the values returned from XiIntercept eComm:

| Card Type        | Returned Value |
|------------------|----------------|
| Visa             | VI             |
| MasterCard       | MC             |
| American Express | AX             |
| Diner's          | DC             |
| Discover         | DI             |
| JCB              | JC             |
| Maestro          | SW             |

All other card types would be considered custom cards. The values are set in the XML.

| Error Code | Description                                                                                                                  |
|------------|------------------------------------------------------------------------------------------------------------------------------|
| -100       | Indicates an unexpected error occurred processing the server request                                                         |
| -101       | Indicates the web request could not be created                                                                               |
| -102       | Indicates the attempt to retrieve the response failed                                                                        |
| -103       | Indicates no response packet was returned from the server                                                                    |
| -104       | Indicates the response packet was invalid and could not be deserialized                                                      |
| 100        | Internal DI eComm Error                                                                                                      |
| 101        | Merchant GUID is Invalid                                                                                                     |
| 102        | Signature Validation Failed                                                                                                  |
| 103        | Request Packet is Invalid                                                                                                    |
| 104        | DI eComm Session has Expired                                                                                                 |
| 105        | DI eComm Session Error                                                                                                       |
| 106        | Error Decrypting Data                                                                                                        |
| 107        | Get Access Token Error                                                                                                       |
| 108        | Requested type hasn't been purchased                                                                                         |
| 120        | XiSecure Error                                                                                                               |
| 121        | XiSecure No Token                                                                                                            |
| 122        | XiSecure Certificate Expired UID: {0}, Expiration Date {1}                                                                   |
| 140        | Error Accessing Database                                                                                                     |
| 150        | The payment form has been submitted multiple times with the same access token. (I.e. User clicking submit payment to twice.) |
| 300        | Cardinal Commerce Response: {message provided by Cardinal}                                                                   |
| 301        | No Cardinal Commerce Configuration                                                                                           |

| Error Code | Description                                                |
|------------|------------------------------------------------------------|
| 302        | The Cardinal Commerce configuration is missing information |
| 401        | An attempt was made to load the payment page in an lframe  |

Section Contents:

- [Form Post and Ajax API Reference](#) <sup>99</sup>
- [Iframe API Reference](#) <sup>106</sup>

## 18.1 XIPlugin - Form Post and Ajax API Reference

The XIPlugin JavaScript plugin is used to reach out to the XiI eComm service by Ajax or Form POST. XIPlugin can either be referenced directly from the XiIntercept service or downloaded and then referenced from your server.

Reference or Download the following:

<https://xiecomm.paymetric.com/DIEComm/Scripts/XIPlugin/XIPlugin-1.0.0.js>

### 18.1.1 Formatted JS Data Structure

To perform the Ajax or Form POST call to XiIntercept eComm, you must first create your data packet. The XIPlugin can assist in preparing this data packet.

Section Contents:

- [\\$XIPlugin.createJSRequestPacket\(merchantGuid, accessToken\)](#) <sup>99</sup>
- [\\$XIPlugin.createField\(fieldName, tokenize, value\)](#) <sup>100</sup>
- [XiRequestPacket - Ajax and Form Post](#) <sup>100</sup>

#### 18.1.1.1 \$XIPlugin.createJSRequestPacket(merchantGuid, accessToken)

\$XIPlugin.createJSRequestPacket creates an empty data packet (XiRequestPacket).

##### Parameters:

`merchantGuid`

The Merchant GUID generated by Paymetric when the XiIntercept for eCommerce Onboarding request is processed. Login to Merchant Portal and navigate to Settings | XiIntercept for eCommerce to obtain the GUID.

`accessToken`

The Access Token obtained during the Get Access Token call.

##### Returned from the Function:

An instance of the XiRequestPacket data packet is returned.

## 18.1.1.2 \$XIPlugin.createField(fieldName, tokenize, value)

**Parameters:**

fieldName

Name of the field associated to the value. This will identify the field in the return packet.

tokenize

The Access Token obtained during the Get Access Token call. Boolean value that specifies that this value needs to be tokenized

value

The value associated with the field name

**Returned from the Function:**

An instance of the XiField data field is returned to add to a data packet.

## 18.1.1.3 XiRequestPacket - Ajax and Form Post

The XiRequestPacket is the data packet sent to Ajax or Post methods of XiPlugin.

`XiRequestPacket.addField`

This adds an XiField object to the data packet.

## 18.1.2 Communication Method Details - Ajax and Form Post

## Section Contents:

- [\\$XIPlugin.ajax](#)<sup>100</sup>
- [\\$XIPlugin.submit](#)<sup>103</sup>

## 18.1.2.1 \$XIPlugin.ajax

```
$XIPlugin.ajax({
    url: '<% XiIntercept Url%>' + "/Ajax",
    type: "POST", //<GET/POST>
    data: '<% Formatted JS Data - see Packaging above %>',
    success: function(a)
    {
    },
    error: function(a)
```

```

        {
        },
        beforeSend: function (a)
        {
        },
        complete: function (a)
        {
        },
        validation: function (a)
        {
            return true; //should return true or false.
        }
        intcp3DSecure: function (a)
        {
        },
    });

```

**Parameters:**

```
url: '<% XiIntercept Url%>' + "/Ajax",
```

Define XiIntercept eComm url and append "/Ajax". Use the appropriate QA or Production URL.

QA: <https://cert-xiecomm.paymetric.com/DleComm>

Production: <https://xiecomm.paymetric.com/DleComm>

```
type: "POST",
```

Call Type should be POST.

```
data: '<% Formatted JS Data - see Packaging above %>',
```

Set the Data element as defined in the [Packaging](#) section above.

```
success: function (a)
```

The function that is triggered if the call is successful.

```
error: function(a)
```

The function that is triggered if the call is NOT successful.

```
beforeSend: function(a)
```

The function that is triggered before the call is sent to XiIntercept eComm.

```
complete: function(a)
```

The function that is triggered after the Ajax call is complete whether is successful or not .

```
validation: function(a)
{
    return true; //should return true or false.
}
```

The function used to perform custom validations. It returns either true false.

If True, will go through with ajax call.

If False, will not go through with ajax call.

```
intcp3DSecure: function(a)
```

Function triggered when a card is enrolled in 3DSecure for customizing Cardinal display in an Iframe or Pop up instead of redirecting to Cardinal site. If this function is not intercepted then default behavior is invoked in which the site is redirected to Cardinal site.

Function Parameters:

- args has necessary argument to post the form to //3DSecure cardinal.
- UrlsAndPayloadsFor3Ds.CentinelAcsUrl → form action
- UrlsAndPayloadsFor3Ds.CentinelPayload → hidden //input name: PaReq
- UrlsAndPayloadsFor3Ds.CentinelTermUrl → hidden //input name: TermUrl
- e.UrlAndPayloadsFor3Ds.PaymetricPayload → hidden //input name: MD

## 18.1.2.2 \$XIPlugin.submit

```
$XIPlugin.submit({  
  
    url: '<% XiIntercept Url%>' + "/Form",  
    type: "POST", //<GET/POST>  
    data: '<% Formatted JS Data - See Packaging above %>',  
    validation: function(a)  
        {  
            return true; //should return true or  
            false.  
        }  
  
});
```

**Parameters:**

```
url: '<% XiIntercept Url%>' + "/Form",
```

Define XiIntercept eComm url and append "/Form". Use the appropriate QA or Production URL.

QA: <https://cert-xiecomm.paymetric.com/DleComm>

Production: <https://xiecomm.paymetric.com/DleComm>

```
type: "POST",
```

Call Type should be POST.

```
data: '<% Formatted JS Data - see Packaging above %>',
```

Set the Data element as defined in the [Packaging](#) <sup>99</sup> section above.

```
validation: function(a)  
    {  
        return true; //should return true or false.  
    }
```

The function used to perform custom validations. It returns either true false.

If True, will go through with post.

If False, will not go through with post.

### 18.1.3 Examples

```
var cc = document.getElementById('PaymentCreditCard').value;
var name = document.getElementById('Name').value;
```

//Getting Value from TextBox with JavaScript code

```
var myData = $XIPlugin.createJSRequestPacket(merchantGuid, accessToken);
```

//Create a JavaScript object to be used in Ajax or Form Post with parameter

- MerchantGuid
- AccessToken

```
var myField = $XIPlugin.createField('creditcard', true, cc);
```

//Create a JavaScript object to be added to myData (JSRequestPacket)

- First Param : Field Name (This is up to the Merchant. Any possible name.)
- Second Param : true (To Tokenize Value)
- Third Param : Value from TextBox

```
myData.addField(myField);
```

//Adding myField to myData (JSRequestPacket) Object

### Ajax Call

```
$XIPlugin.ajax({
    url: '<% XiIntercept Url %>' + "/Ajax",
    type: "POST", //<GET/POST>
    data: '<% Formatted JS Data - see above %>',
    success: function(e)
    {
        //function triggered on success.
    },
    error: function(e)
    {
        //function triggered on failure.
    },
    beforeSend: function(e)
    {
        //function triggered before ajax call.
    },
    complete: function(e)
    {
    }
});
```

```

        //function triggered after ajax call.
        // After success or failure
    },
    validation: function(e)
    {
        //Run custom validation
        return true; //should return true or false.

        // return true will go through with ajax
        // return false will not go through with
    },
    intcp3DSecure: function(e)
    {
        //Function triggered when a card is
        enrolled in 3DSecure for customizing
        Cardinal display in an Iframe or Pop up
        instead of redirecting to Cardinal site.
        If this function in not intercepted then
        default behavior is invoked in which site
        is redirected to Cardinal site.

        //e - args has necessary argument to post
        the form to 3DSecure cardinal.

        //e.UrlAndPayloadsFor3Ds.CentinelAcUrlàfo
        rm action

        //e.UrlAndPayloadsFor3Ds.CentinelPayloadàh
        idden input name: PaReq

        //e.UrlAndPayloadsFor3Ds.CentinelTermUrlàh
        idden input name: TermUrl

        //e.UrlAndPayloadsFor3Ds.PaymetricPayloadàh
        idden input name: MD
    }

});

```

### Form Post Call

```

$XIPlugin.submit({
    url: '<% XiIntercept Url%>' + "/Form",
    data: '<% Formatted JS Data - See above %>',
    validation: function(a)

```

```

    {
        //Run custom validation
        return true; //should return true or false.

        // return true will go through with post.
        // return false will not go through with
        post.
    }

});

```

## 18.2 XIFrame - IFrame API Reference

This javascript plugin is used for reaching out XiIntercept service by embedded IFrame in your website. XIFrame can either be referenced directly from XiIntercept service or downloaded from the XiIntercept service to your server and then referenced from your server.

Reference or download the following:

<https://xiecomm.paymetric.com/DIEComm/Scripts/XIFrame/XIFrame-1.1.0.js>

Section Contents:

- [\\$XIFrame.onload](#) 106
- [\\$XIFrame.submit](#) 110

### 18.2.1 \$XIFrame.onload

Resizing of the IFrame can be handled by a stylesheet. Should you need to resize based upon content within the IFrame, include the following javascript function on IFrame onload function.

\$XIFrame.onload must be called after the IFrame renders the content from XiIntercept eComm. It initializes the communication from the parent page to the IFrame.

```

function IFrame_OnLoad(){
    $XIFrame.onload ({
        iFrameId: <% Iframe Name or ID %>,
        targetUrl: <%XiIntercept Url%>/View/IFrame/<%MerchantGuid%>/<%AccessToken%>
        autosizeheight: true,
        autosizewidth: true,
        storeCardAmount: true,
        onCardTypeChange: function (e){},
        onValidate: function (e){},
        onLoadSuccess: function (){},
        onError: function(msg) {},
        onInvalidHandler: function (e){},
    });
}

```

```
        amount: 30.99 //Can pass amount outside of iframe.  
    });  
}
```

**\$XIFrame.onload Parameters:**

iFrameID

Name or ID of the Iframe.

targetUrl

Name or ID of the Iframe.

autosizeheight

Specifies whether to resize the height of the Iframe if the content in the Iframe increases. If this is set to false, the Iframe must be large enough to handle all content even with validation messages.

autosizewidth

Specifies whether to resize the width of the Iframe if the content in the Iframe increases. If this is set to false, the Iframe must be large enough to handle all content even with validation messages

storeCardAmount

Boolean specifying to return Amount from Iframe to parent window when \$XIFrame.getCardAmount() is invoked.

`onCardTypeChange`

Callback function triggered when Card Type is changed amount.

`onValidate`

Function that should be explicitly called to perform any validation defined within the IFramePacket with (e) as the argument if iframe passes or fails validation.

`onLoadSuccess`

Callback function triggered when the Iframe is correctly rendered.

`onError`

Callback function triggered if there are any problems rendering the content of the Iframe.

`onInvalidHandler`

This function results in reports on controls failing any XieComm validations with errors.

`amount`

Used to change default amount passed to 3D Secure for validation. It allows you to pass the amount outside of the Iframe.

### Best Practice Related to Page / Iframe Onload

To ensure the Iframe loads the correct URL with a valid AccessToken, see the following functions. Note that this prevents problems that can occur when a User clicks the back button after the Submit.

This function should reside on the page frontend outside of any other functions:

```
window.onload = function () {  
    var frameElement = document.getElementsByName('<%  
Iframe Name or ID %>')[0];  
    frameElement.contentWindow.location.href =  
frameElement.src;  
};  
window.onunload = function () { }
```

This function should reside on the page backend.

```
protected override void OnInit(EventArgs e)
{
    Response.Cache.SetCacheability(System.Web.HttpCacheability.NoCache);
    Response.Cache.SetNoStore();
    base.OnInit(e);
}
```

### 18.2.2 \$XIFrame.submit

```
$XIFrame.submit({
    iFrameId: <% Iframe Name or ID %>,
    targetUrl: <%XiIntercept Url%>/View/Iframe/<%
MerchantGuid%>/<%AccessToken%>
    onSuccess: function (e) {},
    onError: function (e) {},
    intcp3DSecure: function(e) {},

});
```

#### Parameters:

iFrameID

Name or ID of the Iframe.

targetUrl

Name or ID of the Iframe.

success: function (a)

The function that is triggered if the call is successful.

error: function (a)

The function that is triggered if the call is NOT successful.

### 18.2.3 Multiple IFrame Example

If you are using multiple IFrames, then you will need to create the same number of \$XIFrame instances. The following example demonstrates using 2 IFrames to split a payment:

```
var iFrameName1 = 'IFrame1';
var iFrameName2 = 'IFrame2';
var iframe1 = document.getElementsByName(iFrameName1);
var iframe2 = document.getElementsByName(iFrameName2);
```

//hold on to the XIFrame instance 1

```
var iframe1Plugin = $XIFrame({
  iFrameId: iFrameName1,
  targetUrl: $(iframe1).attr("src"),
  onCardTypeChange: IFrame1CardChange,
  autosizeheight: true,
  autosizewidth: true,
  onError: function(msg) { alert(msg); },
  onValidate: Validation,
  storeCardAmount: true,
  intcp3DSecure: Custom1Handling
});
```

//hold on to the XIFrame instance 2

```
var iframe2Plugin = $XIFrame({
  iFrameId: iFrameName2,
  targetUrl: $(iframe2).attr("src"),
  autosizeheight: true,
  autosizewidth: true,
  onError: function (msg) { alert(msg); },
  onValidate: Validation,
  storeCardAmount: true,
  intcp3DSecure: Custom2Handling
});
```

Holding on to the instances assist in making the future call as follows:

1. `iframe1Plugin.onload();` or `iframe1Plugin.submit({onSuccess: function (e) { //trigger on iframe1 submit success }}});`
2. `iframe2Plugin.onload();` or `iframe2Plugin.submit({onSuccess: function (e) { //trigger on iframe1 submit success }}});`

### 18.2.4 \$XIFrame.validate

The main reason to use this function is if you are using multiple IFrames.

```
function submitform() {
```

```
var iframe =
document.getElementsByName('dieCommFrame');
if (iframe) {
    $XIFrame.validate({
        iFrameId: 'dieCommFrame',
        targetUrl: iframe[0].getAttribute("src"),
        autosizeheight: true,
        autosizewidth: true,
        onValidate: function(e) {}
    });
}
```

**Parameters:**

iFrameID

Name or ID of the Iframe.

targetUrl

Name or ID of the Iframe.

autosizeheight

Specifies whether to resize the height of the Iframe if the content in the Iframe increases. If this is set to false, the Iframe must be large enough to handle all content even with validation messages.

autosizewidth

Specifies whether to resize the width of the Iframe if the content in the Iframe increases. If this is set to false, the Iframe must be large enough to handle all content even with validation messages

onValidate

Function that should be explicitly called to perform any validation defined within the IFramePacket with (e) as the argument if iframe passes or fails validation.

onError

Callback function triggered if there are any problems rendering the content of the Iframe.

### Example Using onValidate

```
function submitform() {
    var iframe =
document.getElementsByName('dieCommFrame');
    if (iframe) {
        $XIFrame.validate({
            iFrameId: 'dieCommFrame',
            targetUrl: iframe[0].getAttribute("src"),
            autosizeheight: true,
            autosizewidth: true,
            onValidate: function(e) {
                if (e) {
                    $XIFrame.submit({
                        iFrameId: 'dieCommFrame',
                        targetUrl:
iframe[0].getAttribute("src"),
                        onSuccess: function(msg) {
                            var message =
JSON.parse(msg);
                            if (message &&
message.data.HasPassed) {
                                var accessToken =
document.getElementById('<%= AccessToken.ClientID %>');
                                var signedToken =
document.getElementById('<%= SignedToken.ClientID %>');
                                window.location =
"Status.aspx?id=" + accessToken.value + "&s=" +
signedToken.value;
                            } else {
                                alert(message.data.Message);
                            }
                        },
                        onError: function(msg) {
                            alert("Error function : " +
msg);
                        }
                    });
                }
            }
        });
    }
}
```

```
        }  
        else {  
            alert('Iframe Validation failed');  
        }  
    },  
    onError: function(msg) {  
        alert("Error function : " + msg);  
    }  
    });  
}
```

Merchants can define custom html layouts for the XiIntercept for eCommerce (XiIntercept eComm) Iframe and Hosted solutions. This is done by creating XML that adheres to the Paymetric schema *MerchantHtmlPacketModel.xsd*.

## 19.1 Element: <htmlFieldValues>

The <htmlFieldValues> element is used to define default values for textboxes.

Child Elements:

- field: applicable to textboxes only such as cardholder name or echeck. This is limited to the textboxes:
  - cardHolderName (credit card)
  - cardAmount (credit card)
  - bankName (echeck)
  - nameOnAccount (echeck)
- additionalField: can be used for any Merchant-defined custom textbox

## 19.2 Element: <merchantHtml>

The XML consists of a main outer container defined by the <merchantHtml> element. Depending on the merchant's needs, this element can contain one or more of the following types of child sections:

- Payment Option Section – This allows the web user to choose between eCheck and Credit Card payment types.
- Credit Card Section – This provides inputs for credit card information.
- eCheck Section – This provides inputs for eCheck information
- Additional HTML Section – This provides inputs for additional data the merchant wants collected. For example, address data can be collected in this type of section.

Each section above defines elements for labels and inputs to capture data.

Default validations (such as required field, Luhn check for credit card numbers) are defined for the input elements. Additionally, merchants can override these defaults and/or provide custom validations for any input element.

Merchants can make use of common HTML elements such as DIVs to further control the layout of any section, label or input in the xml.

The following sections provide descriptions of the elements defined in the schema.

### 19.2.1 Common Elements and Attributes

These are descriptions of the common elements and attributes used throughout the schema

### 19.2.1.1 Common Attributes:

These are common attributes used throughout the custom html:

- **name:** Unless otherwise specified, this is the name of the html element to generate. Supported values are “div”, “span” and “section”. The default value is “div”.
- **class:** This is the CSS class for the element.
  - NOTE: The class attribute should be used to style elements. Other element attributes (such as id, name, etc) are internally generated Paymetric values that are subject to change in the future. The merchant’s CSS should not reference those attributes.
- **for:** This attribute is used to indicate a relationship between elements. For example, a “label” has a “for” attribute indicating what input it is linked to. An input has a “for” attribute indicating what data the input is for.
- **is-default:** For radio and dropdown inputs, if “1” then this item is the default value.
- **text:** For labels, indicates the text to display. If no text is displayed, the Paymetric default is used for that label. Note this is not currently supported for default text in an input text box.
- **default-text:** For dropdown inputs, indicates the default text to display. For example, in a credit card type dropdown, if “display-text” is “select a card”, then “select a card” is the text in the dropdown when the page loads.
- **tokenize:** Indicates if the data collected in the input should be tokenized. For credit card number and eCheck account number, this must be “true”.

### 19.2.1.2 Common Attributes for Validation

In addition to the attributes above, these are attributes for input elements that control validation:

- **error-class:** The CSS class to use on the input element if validation fails.
  - NOTE: The error-class attribute should be used to style elements. Other element attributes (such as id, name, etc) are internally generated Paymetric values that are subject to change in the future. The merchant’s CSS should not reference those attributes.
- **error-msg-style:** Supported values are:
  - **text:** The error message displays as text
  - **imageTooltip:** The error message displays an image with a tooltip
  - **inputTooltip:** The error message displays as a tooltip when hovering over the input
- **digits-only:** If “true”, only digits can be entered
- **digits-only-msg:** The error message displayed for digits-only validation
- **luhn-check:** If “true”, a luhn check is performed. Applicable to card numbers only.

- `luhn-check-msg`: The error message displayed for luhn check validation.
- `max`: If “true”, the maximum numeric value allowed
- `max-msg`: The error message displayed for max validation
- `min`: The minimum numeric value allowed
- `min-msg`: The error message displayed for min validation
- `maxlength`: The maximum text length allowed
- `maxlength-msg`: The error message displayed for maxlength validation
- `minlength`: The minimum text length allowed
- `minlength-msg`: The error message displayed for minlength validation
- `number`: If “true”, only a number can be entered
- `number-msg`: The error message for number validation
- `pattern`: A regular expression used to validate the input
- `pattern-msg`: The error message for pattern validation
- `range`: A numeric range for the input
- `range-msg`: The error message for range validation
- `rangelength`: A range indicating the length of the input
- `rangelength-msg`: The error message for range-length validation
- `required`: If “true”, indicates input is required (cannot be blank)
- `required-msg`: The error message for required validation

#### 19.2.1.3 Element: `<tag>`

The `<tag>` element is used to indicate an HTML tag should be written. The most common use is to use a `<tag>` anywhere an HTML “div” element is desired.

Attributes:

- `name` (“div”, “span”, or “section”)
- `class`

Child elements allowed:

- `<tag>`'s can contain any child element supported by its **parent** section.
- `<tag>`'s can also contain inner text.

#### 19.2.1.4 Element: `<label>`

The `<label>` element is used to define a label associated with an input.

Attributes:

- name: This is the name of the html element to generate. Supported values are “div”, “span” and “label”. The default is “label”.
- class
- text: Indicates what text is applied to the generated html element. If no text is provided, the Paymetric default is used.
- for: Indicates which input is linked to the label.

Child elements allowed:

- None.

#### 19.2.1.5 Element: <voidTag>

The <voidTag> element is used to indicate an HTML void tag should be written. A void tag is a tag that cannot contain content.

Attributes:

- name: The only allowed value is “br” (generates an html <br/> element).

Child elements allowed:

- None.

#### 19.2.1.6 Elements: <submitButton> and <cancelButton>

These elements are used by the Hosted solution to either submit or cancel the input and redirect to the merchant.

These button elements *cannot* be used in an iFrame solution. The merchant is required to provide their own submit/cancel framework using the JavaScript plugin.

Attributes:

- text: the button text displayed
- class

Child elements allowed:

- None

#### 19.2.1.7 Element: <validationMsg>

The <validationMsg> element provides control over the layout and display of validation error messages. If no <validationMsg> is used, then the default (jQuery) behavior is used to display the message immediately following the input control.

Use this element to create a separate html element for displaying validation messages or error images.

Attributes:

- name (“div”, “span”, or “section”)
- class \*\*
- for (the input the validation is for. The sections below define the valid values for this)

- error-class: the CSS class used on the validation message when a validation error exists \*\*

Child elements allowed:

- None

\*\*NOTE: The class and error-class attributes should be used to style elements. Other element attributes (such as id, name, etc) are internally generated Paymetric values that are subject to change in the future. The merchant's CSS should not reference those attributes.

## 19.2.2 Main Container Section

### 19.2.2.1 Element: <htmlSection>

This is the outermost container (div) for the custom defined HTML. All other sections and elements in the XML are defined within this section.

Attributes:

- name
- class
- xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel"

Child elements allowed:

- <xiProperties>
- <paymentOptionRadioSection>
- <paymentOptionExpandedRadioSection>
- <paymentOptionDropdownSection>
- <paymentOptionExpandedDropdownSection>
- <cardRadioSection>
- <cardDropdownSection>
- <cardNoTypeSection>
- <echeckRadioSection>
- <echeckDropdownSection>
- <cancelButton> (Hosted page only)
- <submitButton> (Hosted page only)
- <tag>
- <voidTag>
- <additionalHtmlSection>
- <textbox>
- <label>

## 19.2.3 XiProperties Section

This section defines global properties that can be applied to all elements.

### 19.2.3.1 Element: <xiProperties>

Container element for global xiProperties

Attributes:

- error-class: global CSS error class to use on validation messages
  - NOTE: The class attribute should be used to style elements. Other element attributes (such as id, name, etc) are internally generated Paymetric values that are subject to change in the future. The merchant's CSS should not reference those attributes.
- inline-validation: provides the ability to perform inline validation as soon as lframe loads

Child elements allowed:

- <errorTooltip>

#### 19.2.3.1.1 Child Element: <errorTooltip>

This defines effects and attributes when displaying an error tooltip

Attributes:

- class
- hide-duration: A string or number (in ms) determining how long the animation will run. Refer to <http://api.jqueryui.com/hide/> for detailed information.
- hide-effect: Refer to <http://api.jqueryui.com/category/effects/> for supported jquery vales
- show-duration: A string or number (in ms) determining how long the animation will run. Refer to <http://api.jqueryui.com/hide/> for detailed information.
- show-effect: Refer to <http://api.jqueryui.com/category/effects/> for supported jquery values
- position-at: Defines which position on the target element to align the positioned element against. Refer to <http://api.jqueryui.com/position> for supported jquery values
- position-collision
- position-my: Defines which position on the element being positioned to align with the target element. Refer to <http://api.jqueryui.com/position> for supported jquery values

Child elements allowed:

- None

### 19.2.4 Payment Option Section

The payment option section is used to allow the web user to select the type of payment method - eCheck or Credit Card. If the merchant only supports eCheck or Credit Card (not both) then this section should be omitted and only the eCheck or Credit Card section needs to be defined in the XML.

There are four types of payment option sections as follows:

- [Element: <paymentOptionRadioSection>](#)<sup>121</sup>
- [Element: <paymentOptionDropdownSection>](#)<sup>121</sup>
- [Element: <paymentOptionExpandedRadioSection>](#)<sup>123</sup>
- [Element: <paymentOptionExpandedDropdownSection>](#)<sup>123</sup>

#### 19.2.4.1 Element: <paymentOptionRadioSection>

This displays the payment options of “echeck” and “credit card” using radio buttons.

Attributes:

- name
- class

Child elements allowed:

- <label>
- <radPaymentOption>
- <tag>
- <voidTag>

##### 19.2.4.1.1 Child Element: <radPaymentOption>

This defines a radio button within the <paymentOptionRadioSection>

Attributes:

- for (**required**) : value can be “echeck” or “card”
- class
- is-default

Child elements allowed:

- None

##### 19.2.4.1.2 Child Element: <label>

This defines a label within the <paymentOptionRadioSection>

Attributes:

- for (**required**) : value can be “echeck” or “card”
- text
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.4.2 Element: <paymentOptionDropdownSection>

This displays the payment options of “echeck” and “credit card” using a select dropdown.

Attributes:

- name
- class

Child elements allowed:

- <label>
- <ddlPaymentOption>
- <tag>
- <voidTag>

#### 19.2.4.2.1 Child Element: <ddlPaymentOption>

This defines a select dropdown within the <paymentOptionDropdownSection>

Attributes:

- class
- default-text
- required
- required-msg
- error-class
  - NOTE: The class attribute should be used to style elements. Other element attributes (such as id, name, etc) are internally generated Paymetric values that are subject to change in the future. The merchant's CSS should not reference those attributes.
- error-msg-style

Child elements allowed:

- <items>: Contains a list of <item> elements
  - <item>
    - Attributes:
      - for (**required**): value can be "echeck" or "card"
      - is-default
      - text
    - Child elements: None

#### 19.2.4.2.2 Child Element: <label>

This defines a label within the <paymentOptionDropdownSection>

Attributes:

- for (**required**) : value can be "paymentOptionType"
- text
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.4.3 Element: <paymentOptionExpandedRadioSection>

This displays the payment options of “echeck” and *specific* card types using radio buttons. For example, these payment options could be defined: “echeck”, “mastercard” and “visa”.

Attributes:

- name
- class

Child elements allowed:

- <label>
- <radPaymentOption>
- <tag>
- <voidTag>

##### 19.2.4.3.1 Child Element: <radPaymentOption>

This defines a radio button within the <paymentOptionExpandedRadioSection>

Attributes:

- for (**required**) : value can be “echeck”, “american express”, “discover”, “jcb”, “maestro”, “mastercard” or “visa” or “diners”
- is-default
- required
- required-msg

Child elements allowed:

- None

##### 19.2.4.3.2 Child Element: <label>

This defines a label within the <paymentOptionExpandedRadioSection>

Attributes:

- for (**required**) : value can be “echeck”, “american express”, “discover”, “jcb”, “maestro”, “mastercard” or “visa” or
- text
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.4.4 Element: <paymentOptionExpandedDropdownSection>

This displays the payment options of “echeck” and *specific* card types using a select dropdown. For example, these payment options could be defined: “echeck”, “mastercard” and “visa”.

Attributes:

- name

- class

Child elements allowed:

- <label>
- <ddlPaymentOption>
- <tag>
- <voidTag>

#### 19.2.4.4.1 Child Element: <ddlPaymentOption>

This defined a select dropdown within the <paymentOptionExpandedDropdownSection>

Attributes:

- class
- default-text
- required
- required-msg
- error-class
  - NOTE: The class attribute should be used to style elements. Other element attributes (such as id, name, etc) are internally generated Paymetric values that are subject to change in the future. The merchant's CSS should not reference those attributes.
- error-msg-style

Child elements allowed:

- <items>: Contains a list of **<item>** elements
  - <item>
    - Attributes:
      - for: value can be "echeck", "american express", "discover", "jcb", "maestro", "mastercard" or "visa" or "diners"
      - is-default
      - text
    - Child elements: None

#### 19.2.4.4.2 Child Element: <label>

This defined a label within the <paymentOptionExpandedDropdownSection>

Attributes:

- for (**required**): value can be "echeck", "american express", "discover", "jcb", "maestro", "mastercard" or "visa" or "diners"
- text
- class
- name (div, span or label)

Child elements allowed:

- None

### 19.2.5 Credit Card Sections

The credit card section is used to allow the web user to input credit card information.

There are three types of credit card sections. Each of these sections contains common credit card elements defined in the next section.

#### 19.2.5.1 Default Card Validations

By default, the following client-side validations will be performed against credit card data. Any or all of these can be overridden or disabled in the XML.

Card number checks:

- Luhn check
- Amex: Card number matches regex “`^(34|37)([0-9]{13})$`”
- Discover: Card number matches regex “`^(64|65|66)([0-9]{12}|[0-9]{14})$)|  
^(622|624|625|626|628)([0-9]{11}|[0-9]{13})$)|^(6011)([0-9]{10}|[0-9]{12})$)$`”
- JCB: Card number matches regex “`^(353|355|356|357|358)([0-9]{13})$)|^(3528|3529)([0-9]{12})$)`”
- MasterCard: Card number matches regex “`^(36|51|52|53|54|55)([0-9]{12}|[0-9]{14})$`”
- Visa: Card number matches regex “`^(4)([0-9]{12}|[0-9]{15})$`”

Expiration and Start Dates:

Validation to verify the card expiration and start dates are valid is enabled by default.

#### 19.2.5.2 Element: <cardRadioSection>

This displays the credit card section with credit card types rendered using radio buttons.

Attributes:

- name
- class

Child elements allowed:

- <radCardType>
- <radCustomCardType>
- <label>
- <customLabel>
- <tag>
- <voidTag>
- <validationMsg> - “for” attribute can be any supported input (“cardType”, “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear”, “cvv”), or the merchant can provide a custom value.
- **See Common Credit Card Elements for descriptions of the following:**
  - <tbxCardNumber>

- <tbxCardHolderName>
- <tbxStartMonth>
- <tbxStartYear>
- <ddlStartMonth>
- <ddlStartYear>
- <tbxExpMonth>
- <tbxExpYear>
- <ddlExpMonth>
- <ddlExpYear>
- <tbxCvv>

#### 19.2.5.2.1 Child Element: <radCardType>

This defines a radio button for a Paymetric supported card type within the <cardRadioSection>

The <radCardType> must be declared as an empty element. <radCardType for="visa"/> is valid. <radCardType for="visa"></radCardType> is not at this time.

Attributes:

- for (**required**) : value can be "american express", "discover", "jcb", "maestro", "mastercard" or "visa" or "diners"
- class
- is-default
- error-class
- error-msg-style
- required
- required-msg

Child elements allowed:

- <validateFor> - **See Common Credit Card Elements**

#### 19.2.5.2.2 Child Element: <radCustomCardType>

This defines a radio button for a custom (merchant-defined) card type within the <cardRadioSection>

Attributes:

- for (**required**) : merchant defines the value
- class
- is-default
- error-class
- error-msg-style
- required
- required-msg
- show-exp-date: If "true" or not supplied, expiration date fields are displayed. If "false", expiration date fields are hidden.
- show-start-date: If "true", start date fields are displayed. If "false" or missing, start date fields are hidden.

- show-cvv: If “true” or not supplied, cvv fields are displayed. If “false”, cvv fields are hidden.
- show- cardholder-name: If “true” or not supplied, cardholder name fields are displayed. If “false”, cardholder name fields are hidden.

Child elements allowed:

- <validateFor> - **See Common Credit Card Elements**

#### 19.2.5.2.3 Child Element: <label>

This defines a label within the <cardRadioSection>

Attributes:

- for (**required**) : value can be “american express”, “discover”, “jcb”, “maestro”, “mastercard”, “visa”, “diners”, “cardType”, “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear” or “cvv”
- text
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.5.2.4 Child Element: <customLabel>

This defines a label to be used with a custom card type within the <cardRadioSection>

Attributes:

- for: Merchant defined value. This should match the “for” used in the <customItem> defined in <radCustomCardType>
- text (**required**)
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.5.3 Element: <cardDropDownSection>

This displays the credit card section with credit card types rendered using a select dropdown.

Attributes:

- name
- class

Child elements allowed:

- <label>
- <ddlCardType>
- <tag>
- <voidTag>

- <validationMsg> - “for” attribute can be any supported input (“cardType”, “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear”, “cvv”), or the merchant can provide a custom value.
- **See Common Credit Card Elements for descriptions of the following:**
  - <tboxCardNumber>
  - <tboxCardHolderName>
  - <tboxStartMonth>
  - <tboxStartYear>
  - <ddlStartMonth>
  - <ddlStartYear>
  - <tboxExpMonth>
  - <tboxExpYear>
  - <ddlExpMonth>
  - <ddlExpYear>
  - <tboxCvv>

#### 19.2.5.3.1 Child Element: <ddlCardType>

This defines a select dropdown for card type within the <cardDropdownSection>

Attributes:

- class
- default-text
- required
- required-msg
- error-class
- error-msg-style

Child elements allowed:

- <items>: Contains a list of **<item>** elements
  - <item>
    - Attributes:
      - for (**required**): value can be “american express”, “discover”, “jcb”, “maestro”, “mastercard” or “visa” or “diners”
      - is-default
      - text
    - Child elements:
      - <validateFor>: **See Common Credit Card Elements**
  - <customItem> - Merchant’s can use this to define custom card types
    - Attributes:
      - for (**required**): Merchant supplied value (code) for the custom card
      - is-default
      - text
      - show-exp-date: If “true” or not supplied, expiration date fields are displayed. If “false”, expiration date fields are hidden.

- show-start-date: If “true”, start date fields are displayed. If “false” or missing, start date fields are hidden.
- show-cvv: If “true” or not supplied, cvv fields are displayed. If “false”, cvv fields are hidden.
- show-cardholder-name: If “true” or not supplied, cardholder name fields are displayed. If “false”, cardholder name fields are hidden.
- Child elements:
  - <validateFor>: **See Common Credit Card Elements**

#### 19.2.5.3.2 Child Element: <label>

This defines a label within the <cardRadioSection>

Attributes:

- for (**required**) : value can be “cardType”, “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear” or “cvv”
- text
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.5.3.3 Child Element: <customLabel>

This defines a label for custom card types within the <cardRadioSection>

Attributes:

- for: Merchant defined value. This should match the “for” defined in the <customItem> in <ddlCardType>
- text (**required**)
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.5.4 Element: <cardNoTypeSection>

This displays the credit card section without a control for the card type. Merchants will use this in conjunction with the <paymentOptionExpandedRadio> or <paymentOptionExpandedDropdown> section which defines the card type outside the Card Section.

Attributes:

- name
- class

Child elements allowed:

- `<label>`
- `<tag>`
- `<voidTag>`
- `<validationMsg>` - “for” attribute can be any supported input (“cardType”, “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear”, “cvv”), or the merchant can provide a custom value.
- **See Common Credit Card Elements for descriptions of the following:**
  - `<tbxCardNumber>`
  - `<tbxCardHolderName>`
  - `<tbxStartMonth>`
  - `<tbxStartYear>`
  - `<ddlStartMonth>`
  - `<ddlStartYear>`
  - `<tbxExpMonth>`
  - `<tbxExpYear>`
  - `<ddlExpMonth>`
  - `<ddlExpYear>`
  - `<tbxCvv>`

#### 19.2.5.4.1 Child Element: `<label>`

This defines a label within the `<cardNoTypeSection>`

Attributes:

- **for (*required*)** : value can be “cardType”, “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear” or “cvv”
- **text**
- **class**
- **name** (div, span or label)

Child elements allowed:

- None

#### 19.2.5.5 Element: `<cardIndicatorSection>`

This displays the credit card section with credit card types rendered using a select dropdown.

Attributes:

- **name**
- **class**

Child elements allowed:

- `<label>`
- `<cardTypeIndicator>`

- <tag>
- <voidTag>
- <validationMsg> - “for” attribute can be any supported input (“cardType”, “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear”, “cvv”), or the merchant can provide a custom value.
- **See *Common Credit Card Elements for descriptions of the following:***
  - <tbxCardNumber>
  - <tbxCardHolderName>
  - <tbxStartMonth>
  - <tbxStartYear>
  - <ddlStartMonth>
  - <ddlStartYear>
  - <tbxExpMonth>
  - <tbxExpYear>
  - <ddlExpMonth>
  - <ddlExpYear>
  - <tbxCvv>

#### 19.2.5.5.1 Child Element: <cardTypeIndicator>

This defines a select dropdown for card type within the <cardTypeIndicator>

Attribute:

- class

Child elements allowed:

- <items>: Contains a list of **<item>** elements
  - <item>
    - Attributes:
      - for (**required**): value can be “american express”, “discover”, “jcb”, “maestro”, “mastercard” or “visa” or “diners”
      - class (**required**): represents the class in the CSS file containing the unselected/selected image; for the selected image, the CSS should contain a class with the same name + “ .selected”; for unselected image, the CSS should contain a class with the same name + “ .unselected”

EXAMPLE in CSS file:

```
.mcIndicator {
    background-image:
    url("../Images/mc.png");
    width: 33px;
    height: 26px;
    vertical-align: middle;
    margin-left: 40px;
    opacity: .3;
}
.mcIndicator.selected {
    opacity: 1;
```

}

- bin-range-expression: represents the card number BIN range regular expression that selects the appropriate card type image
- <customItem> - Merchant's can use this to define custom card types
  - Attributes:
    - for (**required**): Merchant supplied value (code) for the custom card
    - class (**required**): represents the class in the CSS file containing the unselected image; for the selected image, the CSS should contain a second class with the same name + ".selected"
    - bin-range-expression (**required**): represents the card number BIN range regular expression that selects the appropriate card type image
    - show-exp-date: If "true" or not supplied, expiration date fields are displayed. If "false", expiration date fields are hidden.
    - show-start-date: If "true", start date fields are displayed. If "false" or missing, start date fields are hidden.
    - show-cvv: If "true" or not supplied, cvv fields are displayed. If "false", cvv fields are hidden.
    - show-cardholder-name: If "true" or not supplied, cardholder name fields are displayed. If "false", cardholder name fields are hidden.

#### 19.2.5.5.2 Card Type Indicator Validations

If bin-range-expression is not included, the following values are used by default to determine the card type:

- Amex - starts with 34 or 37
- Visa - starts with 4
- MC - starts with 222, 2720,36, or 51 thru 55
- Discover - starts with 64-66, 622, 624-626,628, 6011
- JCB - starts with 353-358, 3528, 3529
- Diners - starts with 30+[0-5]+[0-9], 3095+[0-9], 36+[0-9], or 3+[8-9]+[0-9]
- Maestro - we will not provide a default match. Merchant must send in they're regex match pattern in the 'bin-range-expression' field

#### 19.2.6 Common Credit Card Elements

Below is a list of common inputs used by all 3 types of Credit Card section elements.

- <tbxCardNumber>
- <tbxCardHolderName>
- <tbxStartMonth>
- <tbxStartYear>
- <ddlStartMonth>
- <ddlStartYear>

- <tbxExpMonth>
- <tbxExpYear>
- <ddlExpMonth>
- <ddlExpYear>
- <tbxCvv>

#### 19.2.6.1 Textbox Elements: <tbx{XXX}>

<tbxCardNumber>, <tbxCardHolderName>, <tbxStartMonth>, <tbxStartYear>, <tbxExpMonth>, <tbxExpYear>, <tbxCvv>

Attributes: All of these elements support these common attributes:

- class
- error-class
- error-msg-style
- text
- tokenize (*required for <tbxCardNumber>*)
- is-numeric: prevents any character other than numeric; mobile device displays numeric keyboard
- html-required: required if Merchant wants to use floating labels
- placeholder: provides the default text for textbox
- any of the *Validation Attributes*, with the exception that only <tbxCardNumber> supports "luhn-check" and "luhn-check-msg")

Child elements allowed:

- None

#### <tbxCardNumber> only

In addition to the above attributes, tbxCardNumber has the following attribute as well:

- mask-number
  - Displays the credit card number that is entered by the user with spaces based on one of the following:
    - [<cardTypeIndicator>](#)<sup>131</sup> - based off BIN range
    - [<cardDropdownSection>](#)<sup>127</sup> - based off selected card type in dropdown
    - [<cardRadioSection>](#)<sup>126</sup> - based off selected card type radio button
  - Spacing pattern for Amex is 4 6 5, for example:
    - 34XX XXXXXX XXXXX
  - Spacing pattern for all other card types are in groups of 4, for example:
    - XXXX XXXX XXXX
    - XXXX XXXX XXXX X
    - XXXX XXXX XXXX XXX

- XXXX XXXX XXXX XXXX
- Spacing pattern is controlled by XiIntercept eCommerce; users are not allowed to enter spaces.
- This attribute is ignored if XiIntercept eCommerce is configured to accept token entry. You can see this setting in Merchant Portal. Navigate to Settings | Services | XiIntercept eCommerce.

#### 19.2.6.2 Elements: < ddIStartMonth>, <ddIStartYear>

These elements render select dropdowns for card start month and year. These dropdowns only display if Maestro is selected.

Attributes:

- class
- default-text
- display-format: valid values for <ddIStartMonth> are “M”, “MM”, “MMM” or “MMMM” (e.g. ‘9’, ‘09’, ‘Sep’ or ‘September’); valid values for <ddIStartYear> are “yy” or “yyyy” (e.g. ‘14’ or ‘2014’) The defaults are “MMM” and “yyyy”
- error-class
- error-msg-style
- required
- required-msg
- start-date: if “true”, then a single validation is performed against both the month and year. Use this to simplify validation by having a single message for the web user rather than separate messages for month and date. If this is used, set the **required** attributes on <ddIStartMonth> and <ddIStartYear> to “**false**”
- start-date-msg: the message to display if the selected month and year are not a valid start date
- years-to-display (**<ddIStartYear> only**): The number of years to display in the dropdown. The default is 10.

Child elements allowed: For <ddIStartMonth> only

- <months>: this is an **optional** element that contains child elements <jan>, <feb>, <mar>, <apr>, <may>, <jun>, <jul>, <aug>, <sep>, <oct>, <nov> and <dec>. Each of these child elements can have a single attribute **text** to define the text to use for that month if the default text generated is not sufficient.

#### 19.2.6.3 Elements: < ddIExpMonth>, <ddIExpYear>

These elements render select dropdowns for card expiration month and year.

Attributes:

- class
- default-text

- display-format: valid values for <ddlExpMonth> are “M”, “MM”, “MMM” or “MMMM” (e.g. ‘9’, ‘09’, ‘Sep’ or ‘September’); valid values for <ddlExpYear> are “yy” or “yyyy” (e.g. ‘14’ or ‘2014’) The defaults are “MMM” and “yyyy”
- error-class
- error-msg-style
- required
- required-msg
- exp-date: if “true”, then a single validation is performed against both the month and year. Use this to simplify validation by having a single message for the web user rather than separate messages for month and date. If this is used, set the **required** attributes on <ddlExpMonth> and <ddlExpYear> to “**false**”
- exp-date-msg: the message to display if the selected month and year are not a valid expiration date
- years-to-display (<ddlExpYear> **only**): The number of years to display in the dropdown. The default is 10.

Child elements allowed: For <ddlExpMonth> only

- <months>: this is an **optional** element that contains child elements <jan>, <feb>, <mar>, <apr>, <may>, <jun>, <jul>, <aug>, <sep>, <oct>, <nov> and <dec>. Each of these child elements can have a single attribute **text** to define the text to use for that month if the default text generated is not sufficient.

#### 19.2.6.4 Element: <validationFor>

The <validationFor> element is only used within a Credit Card section. This element is used to define a cross-control validation dependency between the selected card type and another credit card input field. For example, a “regex” validation dependency can be created between a specific card type and the card number textbox such that if that card type is selected, then card number must follow the regex pattern defined.

Use this element to create a separate html element for displaying validation messages or error images.

Attributes:

- None

Child elements allowed:

- <validateCardNumber>: This sets up a link between the card type and card number. Any of the **Validation Attributes** can be used on this element to add or remove validation.
- <validate>: This sets up a link between the card type and another input besides card number. The valid attributes for this element are:
  - for (**required**): The input this validation applies to. Valid values are “cardNumber”, “cardholderName”, “startDate”, “startMonth”, “startYear”, “expDate”, “expMonth”, “expYear” or “cvv”
  - Any of the **Validation Attributes** defined above, except for luhn-check and luhn-check-msg.

## 19.2.7 eCheck Sections

The eCheck section is used to allow the web user to input electronic check information.

There are two types of eCheck sections. Each of these sections contains common eCheck elements defined in the next section.

### 19.2.7.1 Element: <echeckRadioSection>

This displays the eCheck section with eCheck options rendered using radio buttons.

Attributes:

- name
- class

Child elements allowed:

- <radAccountType>
- <label>
- <tag>
- <voidTag>
- <validationMsg> - “for” attribute can be any supported input (“accountType”, “accountNumber”, “routingNumber”, “nameOnAccount”, or “bankName”)
- **See Common eCheck Elements for descriptions of the following:**
  - <tboxAccountNumber>
  - <tboxBankName>
  - <tboxNameOnAccount>
  - <tboxRoutingNumber>

#### 19.2.7.1.1 Child Element: < radAccountType >

This defines a radio button for Paymetric supported eCheck options within the <echeckRadioSection>

Attributes:

- for (**required**) : value can be “checking” or “savings”
- class
- is-default
- error-class
- error-msg-style
- required
- required-msg

Child elements allowed:

- None

#### 19.2.7.1.2 Child Element: <label>

This defines a label within the < echeckRadioSection >

Attributes:

- for (**required**) : value can be “checking”, “savings”, “accountNumber”, “routingNumber”, “nameOnAccount”, or “bankName”
- text
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.7.2 Element: <echeckDropdownSection>

This displays the eCheck section with eCheck options rendered using a select dropdown.

Attributes:

- name
- class

Child elements allowed:

- <ddlAccountType>
- <label>
- <tag>
- <voidTag>
- <validationMsg> - “for” attribute can be any supported input (“accountType”, “accountNumber”, “routingNumber”, “nameOnAccount”, or “bankName”)
- **See Common eCheck Elements for descriptions of the following:**
  - <tboxAccountNumber>
  - <tboxBankName>
  - <tboxNameOnAccount>
  - <tboxRoutingNumber>

##### 19.2.7.2.1 Child Element: <ddlAccountType >

This defines a select dropdown for Paymetric supported eCheck options within the <echeckDropdownSection >

Attributes:

- class
- default-text
- error-class
- error-msg-style
- required
- required-msg

Child elements allowed:

- <items>: Contains a list of **<item>** elements
  - <item>
    - Attributes:
      - for (**required**): value can be “checking” or “savings”
      - is-default

- text
- Child elements:
  - None

#### 19.2.7.2.2 Child Element: <label>

This defines a label within the <echeckDropDownSection >

Attributes:

- for (**required**) : value can be “accountType”, “accountNumber”, “routingNumber”, “nameOnAccount”, or “bankName”
- text
- class
- name (div, span or label)

Child elements allowed:

- None

### 19.2.8 Common eCheck Elements

Below is a list of common inputs used by both eCheck section elements.

- <tbxAccountNumber>
- <tbxBankName>
- <tbxNameOnAccount>
- <tbxRoutingNumber>

#### 19.2.8.1 Textbox Elements: < tbx{XXX} >

<tbxAccountNumber>, <tbxBankName >, <tbxNameOnAccount >, <tbxRoutingNumber >

Attributes: All of these elements support these common attributes:

- class
- error-class
- error-msg-style
- text
- tokenize (**required for <tbxAccountNumber>**)
- is-numeric: prevents any character other than numeric; mobile device displays numeric keyboard
- html-required: required if Merchant wants to use floating labels
- placeholder: provides the default text for textbox
- any of the **Validation Attributes**, except for “luhn-check” and “luhn-check-msg”

Child elements allowed:

- None

### 19.2.9 Additional HTML Section

This section is used to contain any additional data input that the merchant wants to collect. For example, to collect address information add this section to the XML.

### 19.2.9.1 Element: <additionalHtmlSectionElements>

Attributes:

- class
- name

Child elements allowed:

- <tag>
- <label>
- <textbox>
- <ddlUsStateDropdown>
- <validationMsg>

#### 19.2.9.1.1 Child Element: <label>

This defines a label within the <additionalHtmlSectionElements>

Attributes:

- for: Merchant defined value
- text
- class
- name (div, span or label)

Child elements allowed:

- None

#### 19.2.9.1.2 Child Element: <textBox>

This defines an input textbox within the <additionalHtmlSectionElements >

Attributes:

- class
- text
- name (**required**): Merchant defined value. This HTML input generated will have a "name" attribute with this value.
- is-numeric: prevents any character other than numeric; mobile device displays numeric keyboard
- html-required: **required** if Merchant wants to use floating labels
- placeholder: provides the default text for textbox
- any of the **Validation Attributes**, except for "luhn-check" and "luhn-check-msg"

Child elements allowed:

- None

#### 19.2.9.1.3 Child Element: <ddlUsStateDropdown>

This defines a select dropdown of US states within the <additionalHtmlSectionElements >

Attributes:

- class
- default-text

- error-class
- error-msg-style
- required
- required-msg

Child elements allowed:

- None

### 19.2.10 Sample Merchant XMLs

Below are samples that show various ways to configure the Merchant Custom XML and the HTML output generated for the web user.

*These samples use Paymetric logos and copyright information. Merchant logos and information can be supplied thru the XML and CSS.*

#### 19.2.10.1 Sample 1: Credit Card (Dropdown)

```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"/>
        <tag name="div" class="billing-info">
          <cardDropDownSection>
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType>
                <items>
                  <item for="american express"/>
                </items>
              </ddlCardType>
            </tag>
          </cardDropDownSection>
        </tag>
      </div>
    </tag>
  </tag>
</htmlSection>
```

```

        <item for="mastercard"/>
        <item for="visa"/>
        <item for="maestro"/>
    </items>
</ddlCardType>
</tag>
<tag name="div">
    <label for="cardNumber" text="Card number"/>
    <tboxCardNumber tokenize="true" class="xi-long-text"/>
    <validationMsg for="cardNumber" class="valmsg"/>
</tag>
<tag name="div">
    <label for="cardholderName" text="Name on card"/>
    <tboxCardHolderName class="xi-long-text"/>
    <validationMsg for="cardholderName" class="valmsg"/>
</tag>
<tag name="div">
    <label for="startDate"/>
    <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
    <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
    <validationMsg for="startYear" class="valmsg"/>
</tag>
<tag name="div">
    <label for="expDate" text="Expiration date"/>
    <ddlExpMonth default-text="month" class="date_combos" required="false"/>
    <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
    <validationMsg for="expYear" class="valmsg"/>
</tag>
<tag name="div">
    <label for="cvv" text="Card cvv:"/>
    <tboxCvv class="xi-short-text"/>
    <validationMsg for="cvv" class="valmsg"/>
</tag>
</cardDropdownSection>
</tag>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button"/>
    <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">
            Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
            respective owners. Legal Notice.
        </tag>
    </tag>
</tag>
</tag>
</tag>

```

```
</htmlSection>
```

### 19.2.10.2 Sample 2: Credit Card (Dropdown) with Additional HTML section for capturing Address

The screenshot displays the 'Xi intercept' logo at the top. Below it is a form titled 'Pay with credit card'. The form contains the following fields:

- Card type: A dropdown menu with 'AMEX' selected.
- Card number: A text input field.
- Name on card: A text input field.
- Expiration date: Two dropdown menus for 'month' and 'year'.
- Card cvv: A text input field.
- Address information section:
  - Address: A text input field.
  - City: A text input field.
  - State: A dropdown menu with 'AL' selected.
  - ZIP Code: A text input field.

At the bottom of the form are 'cancel' and 'submit' buttons. Below the form is a copyright notice: 'Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice.' and the 'Powered By PAYMETRIC' logo.

```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"></tag>
        <tag name="div" class="billing-info">
          <cardDropdownSection>
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType>
                <items>
                  <item for="american express"/>
                  <item for="mastercard"/>
                  <item for="visa"/>
                  <item for="maestro"/>
                </items>
              </ddlCardType>
            </tag>
          </cardDropdownSection>
          <tag name="div">
            <div class="address">
              <label for="address" text="Address"/>
              <input type="text" value=""/>
              <div class="city">
                <label for="city" text="City"/>
                <input type="text" value=""/>
              </div>
              <div class="state">
                <label for="state" text="State"/>
                <select value="AL">
                  <option value="AL">AL</option>
                </select>
              </div>
              <div class="zip">
                <label for="zip" text="ZIP Code"/>
                <input type="text" value=""/>
              </div>
            </div>
          </tag>
        </div>
      </tag>
    </div>
  </tag>
</htmlSection>
```

```

        <label for="cardNumber" text="Card number"/>
        <tbxCardNumber tokenize="true" class="xi-long-text"/>
        <validationMsg for="cardNumber" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="cardholderName" text="Name on card"/>
        <tbxCardHolderName class="xi-long-text"/>
        <validationMsg for="cardholderName" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="startDate"/>
        <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
        <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
        <validationMsg for="startYear" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="expDate" text="Expiration date"/>
        <ddlExpMonth default-text="month" class="date_combos" required="false"/>
        <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
        <validationMsg for="expYear" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="cvv" text="Card cvv:"/>
        <tbxCvv class="xi-short-text"/>
        <validationMsg for="cvv" class="valmsg"/>
    </tag>
</cardDropdownSection>
</tag>
<tag name="div" class="address-header"></tag>
<tag name="div" class="address-info">
    <additionalHtmlSection class="address-content">
        <tag name="div">
            <label for="address1"/>
            <textBox name="address1" class="xi-long-text" required-msg="Please enter an
address"/>
        </tag>
        <tag name="div">
            <label for="address2"/>
            <textBox name="address2" class="xi-long-text" required="false"/>
            <validationMsg for="address1" class="valmsg"/>
        </tag>
        <tag name="div">
            <label for="city"/>
            <textBox name="city" class="xi-long-text" required-msg="Please enter the city"/>
            <validationMsg for="city" class="valmsg"/>
        </tag>
        <tag name="div">
            <label for="state"/>
            <ddlUsStateDropdown/>
            <validationMsg for="state" class="valmsg"/>
        </tag>
    </tag>
    <tag name="div">

```

```

        <label for="postalCode"/>
        <textBox name="postalCode" class="xi-medium-text" pattern="^\d{5}(-\d{4})?$"
required-msg="Please enter the postal code" pattern-msg="Please enter a valid postal code"/>
        <validationMsg for="postalCode" class="valmsg"/>
    </tag>
</additionalHtmlSection>
</tag>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button"/>
    <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">
            Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
            respective owners. Legal Notice.
        </tag>
    </tag>
</tag>
</tag>
</tag>
</tag>
</htmlSection>

```

### 19.2.10.3 Sample 3: Credit Card (Radio)

**Pay with credit card**

Master Card ☒ Visa ☐

Card number

Name on card

Expiration date

Card cvv:

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```

<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
    <tag>
        <tag name="div" class="header-top"/>
        <tag name="div" class="merchant-logo"/>
    </tag>

```

```

<tag name="div" class="main-content">
  <tag name="div" class="billing-content">
    <tag name="div" class="billing-header"></tag>
    <tag name="div" class="billing-info">
      <cardRadioSection>
        <tag name="div">
          <label for="mastercard"/>
          <radCardType for="mastercard"/>
          <label for="visa" text="Visa"/>
          <radCardType for="visa"/>
        </tag>
        <tag name="div">
          <label for="cardNumber" text="Card number"/>
          <tboxCardNumber tokenize="true" class="xi-long-text"/>
          <validationMsg for="cardNumber" class="valmsg"/>
        </tag>
        <tag name="div">
          <label for="cardholderName" text="Name on card"/>
          <tboxCardHolderName class="xi-long-text"/>
          <validationMsg for="cardholderName" class="valmsg"/>
        </tag>
        <tag name="div">
          <label for="startDate"/>
          <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
          <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
          <validationMsg for="startYear" class="valmsg"/>
        </tag>
        <tag name="div">
          <label for="expDate" text="Expiration date"/>
          <ddlExpMonth default-text="month" class="date_combos" required="false"/>
          <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
          <validationMsg for="expYear" class="valmsg"/>
        </tag>
        <tag name="div">
          <label for="cvv" text="Card cvv:"/>
          <tboxCvv class="xi-short-text"/>
          <validationMsg for="cvv" class="valmsg"/>
        </tag>
      </cardRadioSection>
    </tag>
    <tag name="div" class="payment-button">
      <cancelButton text="cancel" class="cancel-button"/>
      <submitButton text="submit" class="submit-button"/>
    </tag>
  </tag>
  <tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
      <tag name="div" class="middle-bottom-frame-content">

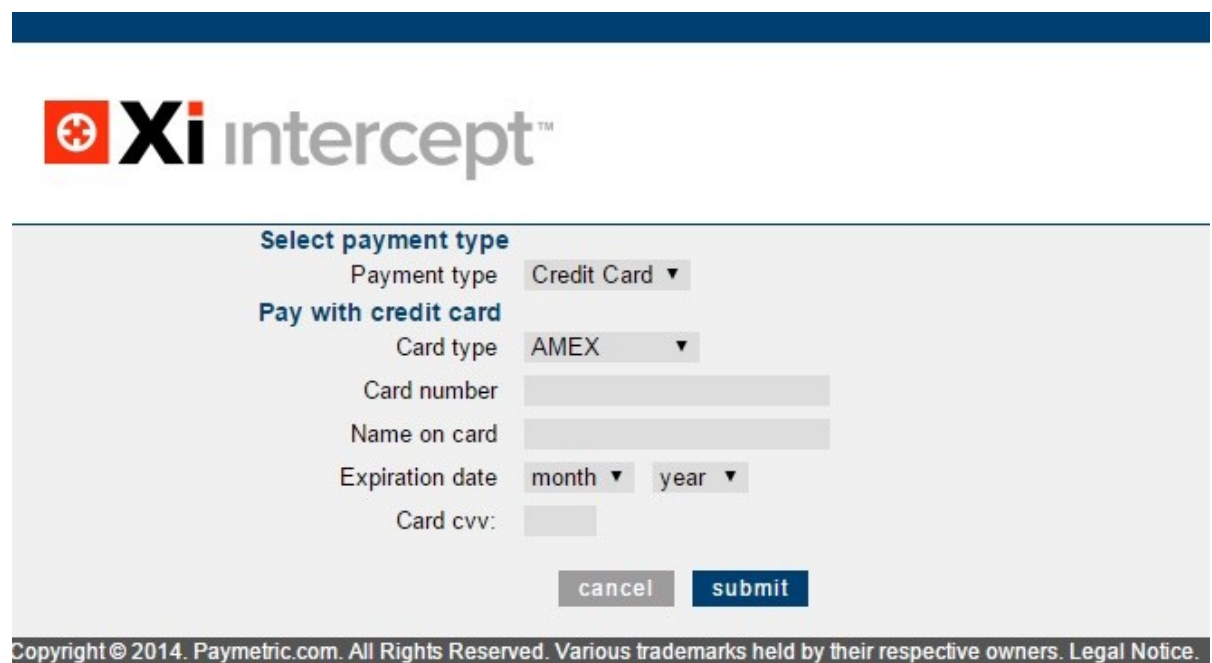
```

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```
</tag>
</tag>
</tag>
</tag>
</htmlSection>
```

#### 19.2.10.4 Sample 4: Credit Card (Dropdown) and eCheck (Dropdown) with Payment Option (Dropdown)

##### Credit Card View



The screenshot displays the 'Xi intercept' logo at the top. Below it, a form titled 'Select payment type' is shown. The 'Payment type' dropdown is set to 'Credit Card'. Under the heading 'Pay with credit card', the 'Card type' dropdown is set to 'AMEX'. The form includes input fields for 'Card number', 'Name on card', and 'Card cvv:'. The 'Expiration date' is selected using two dropdowns for 'month' and 'year'. At the bottom of the form are 'cancel' and 'submit' buttons. A footer at the bottom of the form area reads: 'Copyright © 2014. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice.'

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## eCheck View

**Select payment type**  
Payment type

**Pay with electronic check**  
Select Account  
Type

Account Number

Routing Number

Bank Name

Name on Account

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```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <paymentOptionDropdownSection>
        <tag name="div" class="billing-content">
          <tag name="div" class="payoption-header"></tag>
          <tag name="div" class="payoption-info">
            <tag>
              <label for="paymentOptionType" text="Payment type"/>
              <ddlPaymentOption>
                <items>
                  <item for="card"/>
                  <item for="echeck"/>
                </items>
              </ddlPaymentOption>
            </tag>
          </tag>
        </tag>
      </paymentOptionDropdownSection>
      <cardDropdownSection>
        <tag name="div" class="billing-content">
          <tag name="div" class="billing-header"></tag>
          <tag name="div" class="billing-info">

```

```

<tag name="div">
  <label for="cardType" text="Card type"/>
  <ddlCardType>
    <items>
      <item for="american express"/>
      <item for="mastercard"/>
      <item for="visa"/>
      <item for="maestro"/>
    </items>
  </ddlCardType>
</tag>
<tag name="div">
  <label for="cardNumber" text="Card number"/>
  <tboxCardNumber tokenize="true" class="xi-long-text"/>
  <validationMsg for="cardNumber" class="valmsg"/>
</tag>
<tag name="div">
  <label for="cardholderName" text="Name on card"/>
  <tboxCardHolderName class="xi-long-text"/>
  <validationMsg for="cardholderName" class="valmsg"/>
</tag>
<tag name="div">
  <label for="startDate"/>
  <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
  <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
  <validationMsg for="startYear" class="valmsg"/>
</tag>
<tag name="div">
  <label for="expDate" text="Expiration date"/>
  <ddlExpMonth default-text="month" class="date_combos" required="false"/>
  <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
  <validationMsg for="expYear" class="valmsg"/>
</tag>
<tag name="div">
  <label for="cvv" text="Card cvv:"/>
  <tboxCvv class="xi-short-text"/>
  <validationMsg for="cvv" class="valmsg"/>
</tag>
</tag>

</cardDropdownSection>
<echeckDropdownSection>
  <tag name="div" class="billing-content">
    <tag name="div" class="echeck-header"></tag>
    <tag name="div" class="billing-info">
      <tag name="div">
        <label for="accountType"/>
        <ddlAccountType>
          <items>
            <item for="checking"/>
            <item for="savings"/>
          </items>
        </ddlAccountType>
      </tag>
    </tag>
  </tag>
</echeckDropdownSection>

```

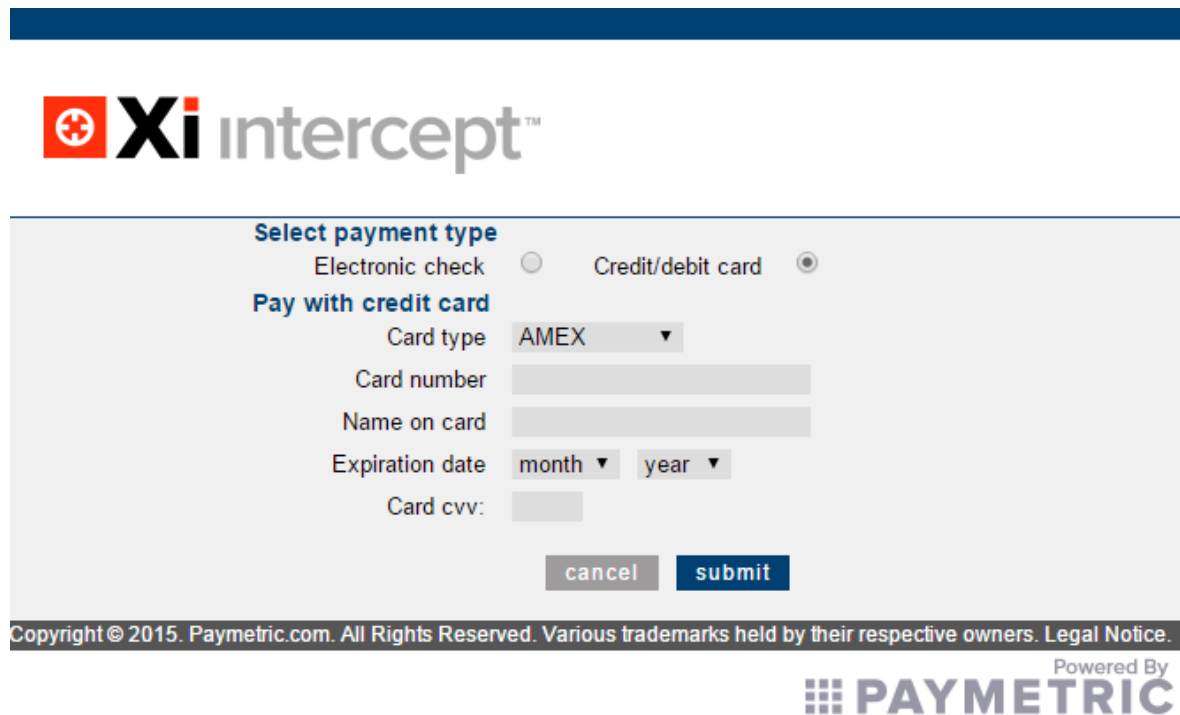
```

        </items>
      </ddlAccountType>
    </tag>
    <tag name="div">
      <label for="accountNumber"/>
      <tboxAccountNumber tokenize="true"/>
    </tag>
    <tag name="div">
      <label for="routingNumber"/>
      <tboxRoutingNumber/>
    </tag>
    <tag name="div">
      <label for="bankName"/>
      <tboxBankName/>
    </tag>
    <tag name="div">
      <label for="nameOnAccount"/>
      <tboxNameOnAccount/>
    </tag>
  </tag>
</echeckDropdownSection>
<tag name="div" class="payment-button">
  <cancelButton text="cancel" class="cancel-button"/>
  <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
  <tag name="div" class="footer-spacing"></tag>
  <tag name="div" class="top-bottom-frame"></tag>
  <tag name="div" class="middle-bottom-frame">
    <tag name="div" class="middle-bottom-frame-content">
      Copyright © 2014. Paymetric.com. All Rights Reserved. Various trademarks held by their
      respective owners. Legal Notice.
    </tag>
  </tag>
</tag>
</tag>
</htmlSection>

```

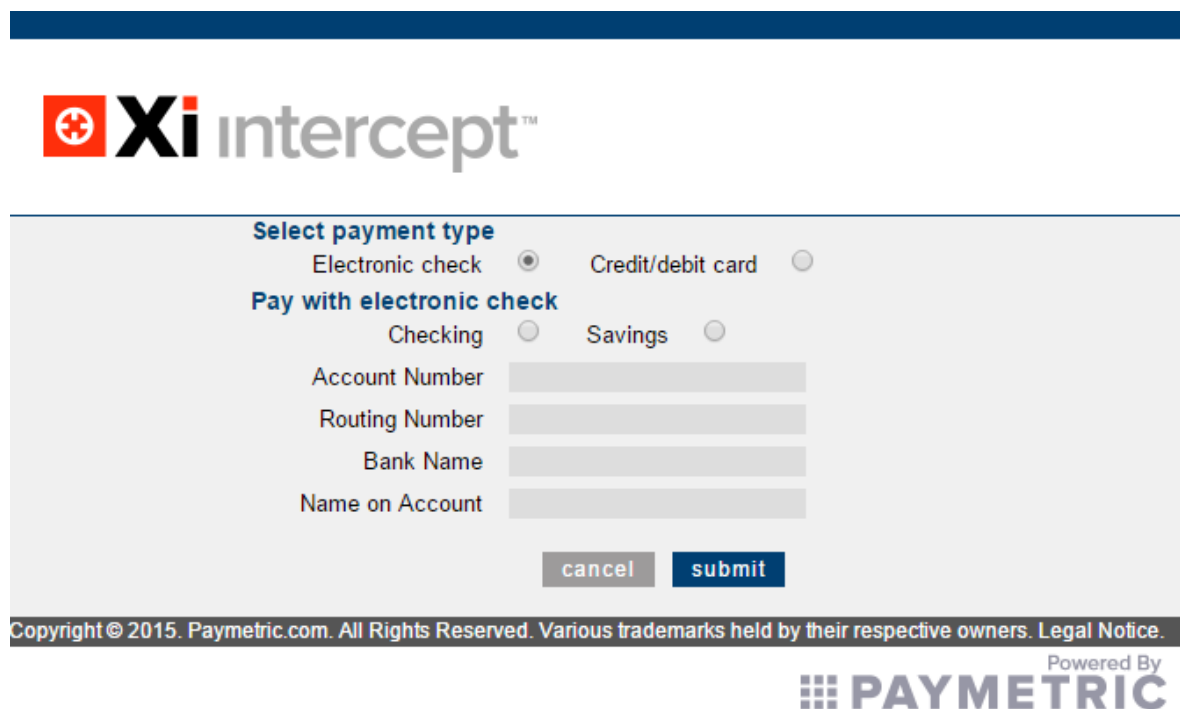
## 19.2.10.5 Sample 5: Credit Card (Dropdown) and eCheck (Radio) with Payment Option (Radio)

## Credit Card View



The screenshot shows the Xi Intercept payment interface for a credit card. At the top is the Xi Intercept logo. Below it, the 'Select payment type' section has two radio buttons: 'Electronic check' (unselected) and 'Credit/debit card' (selected). Under 'Credit/debit card', the 'Pay with credit card' section contains a 'Card type' dropdown menu set to 'AMEX', followed by input fields for 'Card number', 'Name on card', 'Expiration date' (with month and year dropdowns), and 'Card cvv'. At the bottom of this section are 'cancel' and 'submit' buttons. A footer contains the copyright notice 'Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice.' and the Paymetric logo with the text 'Powered By'.

## eCheck View



The screenshot shows the Xi Intercept payment interface for an eCheck. At the top is the Xi Intercept logo. Below it, the 'Select payment type' section has two radio buttons: 'Electronic check' (selected) and 'Credit/debit card' (unselected). Under 'Electronic check', the 'Pay with electronic check' section contains two radio buttons: 'Checking' (selected) and 'Savings' (unselected). Below these are input fields for 'Account Number', 'Routing Number', 'Bank Name', and 'Name on Account'. At the bottom of this section are 'cancel' and 'submit' buttons. A footer contains the copyright notice 'Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice.' and the Paymetric logo with the text 'Powered By'.

```

<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <paymentOptionRadioSection>
        <tag name="div" class="billing-content">
          <tag name="div" class="payoption-header"/>
          <tag name="div" class="payoption-info">
            <tag>
              <label for="echeck" text="Electronic check"/>
              <radPaymentOption for="echeck"/>
              <label for="card" text="Credit/debit card"/>
              <radPaymentOption for="card"/>
            </tag>
          </tag>
        </tag>
      </paymentOptionRadioSection>
      <cardDropdownSection>
        <tag name="div" class="billing-content">
          <tag name="div" class="billing-header"/>
          <tag name="div" class="billing-info">
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType>
                <items>
                  <item for="american express"/>
                  <item for="mastercard"/>
                  <item for="visa"/>
                  <item for="maestro"/>
                </items>
              </ddlCardType>
            </tag>
            <tag name="div">
              <label for="cardNumber" text="Card number"/>
              <tboxCardNumber tokenize="true" class="xi-long-text"/>
              <validationMsg for="cardNumber" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="cardholderName" text="Name on card"/>
              <tboxCardHolderName class="xi-long-text"/>
              <validationMsg for="cardholderName" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="startDate"/>
              <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
              <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
              <validationMsg for="startYear" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="expDate" text="Expiration date"/>
              <ddlExpMonth default-text="month" class="date_combos" required="false"/>

```

```

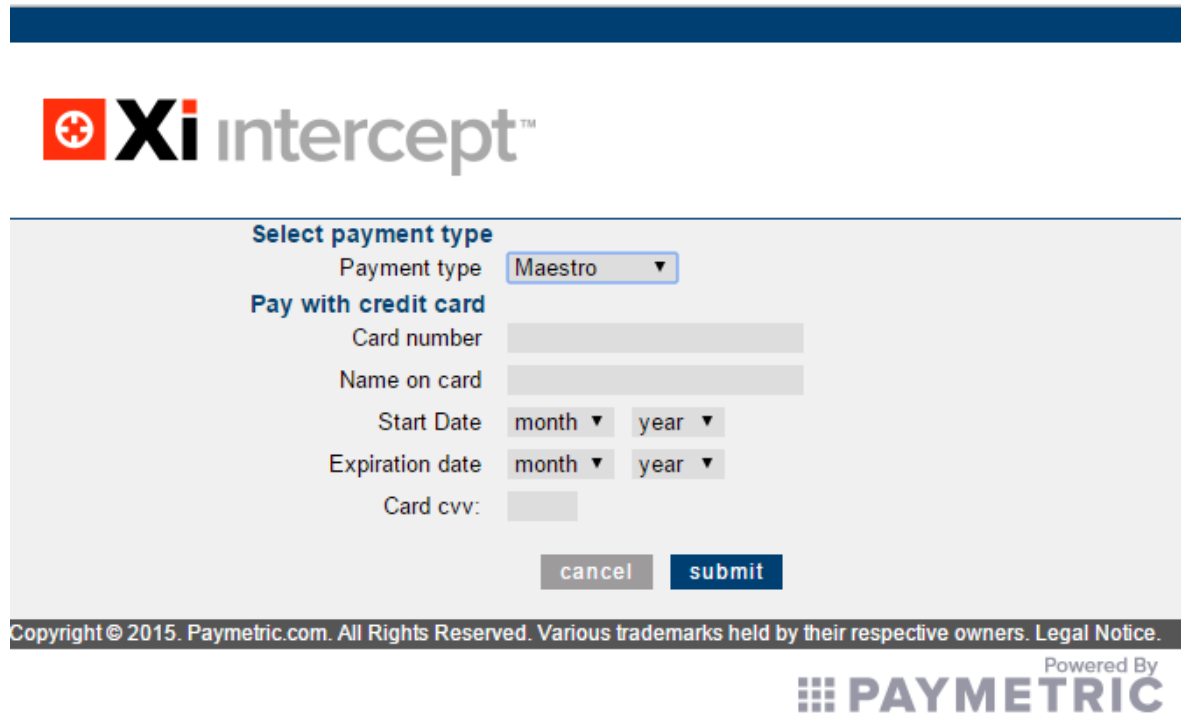
        <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
        <validationMsg for="expYear" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="cvv" text="Card cvv:"/>
        <tbxCvv class="xi-short-text"/>
        <validationMsg for="cvv" class="valmsg"/>
    </tag>
</tag>
</cardDropdownSection>
<echeckRadioSection>
    <tag name="div" class="billing-content">
        <tag name="div" class="echeck-header"></tag>
        <tag name="div" class="billing-info">
            <tag name="div">
                <label for="checking"/>
                <radAccountType for="checking"/>
                <label for="savings" class="ec-label"/>
                <radAccountType for="savings"/>
            </tag>
            <tag name="div">
                <label for="accountNumber"/>
                <tbxAccountNumber tokenize="true"/>
            </tag>
            <tag name="div">
                <label for="routingNumber"/>
                <tbxRoutingNumber/>
            </tag>
            <tag name="div">
                <label for="bankName"/>
                <tbxBankName/>
            </tag>
            <tag name="div">
                <label for="nameOnAccount"/>
                <tbxNameOnAccount/>
            </tag>
        </tag>
    </tag>
</echeckRadioSection>
    <tag name="div" class="payment-button">
        <cancelButton text="cancel" class="cancel-button"/>
        <submitButton text="submit" class="submit-button"/>
    </tag>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">
            Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
            respective owners. Legal Notice.
        </tag>
    </tag>

```

```
</tag>  
</tag>  
</tag>  
</htmlSection>
```

#### 19.2.10.6 Sample 6: Credit Card (NoType) and eCheck (Radio) with Payment Option (Expanded Dropdown)

##### Credit Card View



The screenshot displays the Xi Intercept payment interface. At the top, the Xi Intercept logo is shown. Below it, a form titled "Select payment type" features a dropdown menu currently set to "Maestro". Under the heading "Pay with credit card", there are input fields for "Card number", "Name on card", "Start Date" (with month and year dropdowns), "Expiration date" (with month and year dropdowns), and "Card cvv:". At the bottom of the form are "cancel" and "submit" buttons. A footer bar contains the copyright notice "Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice." and the Paymetric logo with the text "Powered By PAYMETRIC".

## eCheck View

**Select payment type**  
Payment type

**Pay with electronic check**  
☒ Checking    ☐ Savings

Account Number

Routing Number

Bank Name

Name on Account

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```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <paymentOptionExpandedDropdownSection>
        <tag name="div" class="billing-content">
          <tag name="div" class="payoption-header"></tag>
          <tag name="div" class="payoption-info">
            <tag>
              <label for="paymentOptionType" text="Payment type"/>
              <ddlPaymentOption>
                <items>
                  <item for="echeck"/>
                  <item for="american express"/>
                  <item for="discover"/>
                  <item for="jcb"/>
                  <item for="maestro"/>
                  <item for="mastercard"/>
                  <item for="visa"/>
                </items>
              </ddlPaymentOption>
            </tag>
          </tag>
        </tag>
      </paymentOptionExpandedDropdownSection>
      <cardNoTypeSection>
        <tag name="div" class="billing-content">
```

```

<tag name="div" class="billing-header"></tag>
<tag name="div" class="billing-info">
  <tag name="div">
    <label for="cardNumber" text="Card number"/>
    <tboxCardNumber tokenize="true" class="xi-long-text"/>
    <validationMsg for="cardNumber" class="valmsg"/>
  </tag>
  <tag name="div">
    <label for="cardholderName" text="Name on card"/>
    <tboxCardHolderName class="xi-long-text"/>
    <validationMsg for="cardholderName" class="valmsg"/>
  </tag>
  <tag name="div">
    <label for="startDate"/>
    <ddlStartMonth display-format="MMM" class="date_combos" required="false" default-
text="month"/>
    <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true" default-text="year"/>
    <validationMsg for="startYear" class="valmsg"/>
  </tag>
  <tag name="div">
    <label for="expDate" text="Expiration date"/>
    <ddlExpMonth default-text="month" class="date_combos" required="false"/>
    <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
    <validationMsg for="expYear" class="valmsg"/>
  </tag>
  <tag name="div">
    <label for="cvv" text="Card cvv:"/>
    <tboxCvv class="xi-short-text"/>
    <validationMsg for="cvv" class="valmsg"/>
  </tag>
</tag>
</cardNoTypeSection>
<echeckRadioSection>
  <tag name="div" class="billing-content">
    <tag name="div" class="echeck-header"></tag>
    <tag name="div" class="billing-info">
      <tag name="div">
        <label for="checking"/>
        <radAccountType for="checking"/>
        <label for="savings" class="ec-label"/>
        <radAccountType for="savings"/>
      </tag>
      <tag name="div">
        <label for="accountNumber"/>
        <tboxAccountNumber tokenize="true"/>
      </tag>
      <tag name="div">
        <label for="routingNumber"/>
        <tboxRoutingNumber/>
      </tag>
      <tag name="div">

```

```
        <label for="bankName"/>
        <tbxBankName/>
    </tag>
    <tag name="div">
        <label for="nameOnAccount"/>
        <tbxNameOnAccount/>
    </tag>
</tag>
</echeckRadioSection>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button"/>
    <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">
            Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
            respective owners. Legal Notice.
        </tag>
    </tag>
</tag>
</tag>
</tag>
</htmlSection>
```

## 19.2.10.7 Sample 7: Credit Card (NoType) and eCheck (Radio) with Payment Option (Expanded Radio)

## Credit Card View



**Select payment type**

Electronic check ☐

Amex ☒

Master Card ☐

VISA ☐

**Pay with credit card**

Card number

Name on card

Expiration date

Card cvv:

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## eCheck View

**Select payment type**

Electronic check ☒

Amex ☐

Master Card ☐

VISA ☐

**Pay with electronic check**

Checking ☐ Savings ☐

Account Number

Routing Number

Bank Name

Name on Account

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```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <paymentOptionExpandedRadioSection>
        <tag name="div" class="billing-content">
          <tag name="div" class="payoption-header"></tag>
          <tag name="div" class="payoption-info">
            <tag>
              <label for="echeck" text="Electronic check"/>
              <radPaymentOption for="echeck"/>
            </tag>
            <tag>
              <label for="american express" text="Amex"/>
              <radPaymentOption for="american express"/>
            </tag>
            <tag>
              <label for="mastercard"/>
              <radPaymentOption for="mastercard"/>
            </tag>
            <tag>
              <label for="visa"/>
              <radPaymentOption for="visa"/>
            </tag>
          </tag>
        </tag>
      </paymentOptionExpandedRadioSection>
    </tag>
  </tag>
</htmlSection>
```

```

        </tag>
    </tag>
</tag>
</paymentOptionExpandedRadioSection>
<cardNoTypeSection>
    <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"></tag>
        <tag name="div" class="billing-info">
            <tag name="div">
                <label for="cardNumber" text="Card number"/>
                <tboxCardNumber tokenize="true" class="xi-long-text"/>
                <validationMsg for="cardNumber" class="valmsg"/>
            </tag>
            <tag name="div">
                <label for="cardholderName" text="Name on card"/>
                <tboxCardHolderName class="xi-long-text"/>
                <validationMsg for="cardholderName" class="valmsg"/>
            </tag>
            <tag name="div">
                <label for="startDate"/>
                <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
                <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
                <validationMsg for="startYear" class="valmsg"/>
            </tag>
            <tag name="div">
                <label for="expDate" text="Expiration date"/>
                <ddlExpMonth default-text="month" class="date_combos" required="false"/>
                <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
                <validationMsg for="expYear" class="valmsg"/>
            </tag>
            <tag name="div">
                <label for="cvv" text="Card cvv:"/>
                <tboxCvv class="xi-short-text"/>
                <validationMsg for="cvv" class="valmsg"/>
            </tag>
        </tag>
    </tag>
</cardNoTypeSection>
<echeckRadioSection>
    <tag name="div" class="billing-content">
        <tag name="div" class="echeck-header"></tag>
        <tag name="div" class="billing-info">
            <tag name="div">
                <label for="checking"/>
                <radAccountType for="checking"/>
                <label for="savings" class="ec-label"/>
                <radAccountType for="savings"/>
            </tag>
            <tag name="div">
                <label for="accountNumber"/>
                <tboxAccountNumber tokenize="true"/>
            </tag>
        </tag>
    </tag>
</echeckRadioSection>

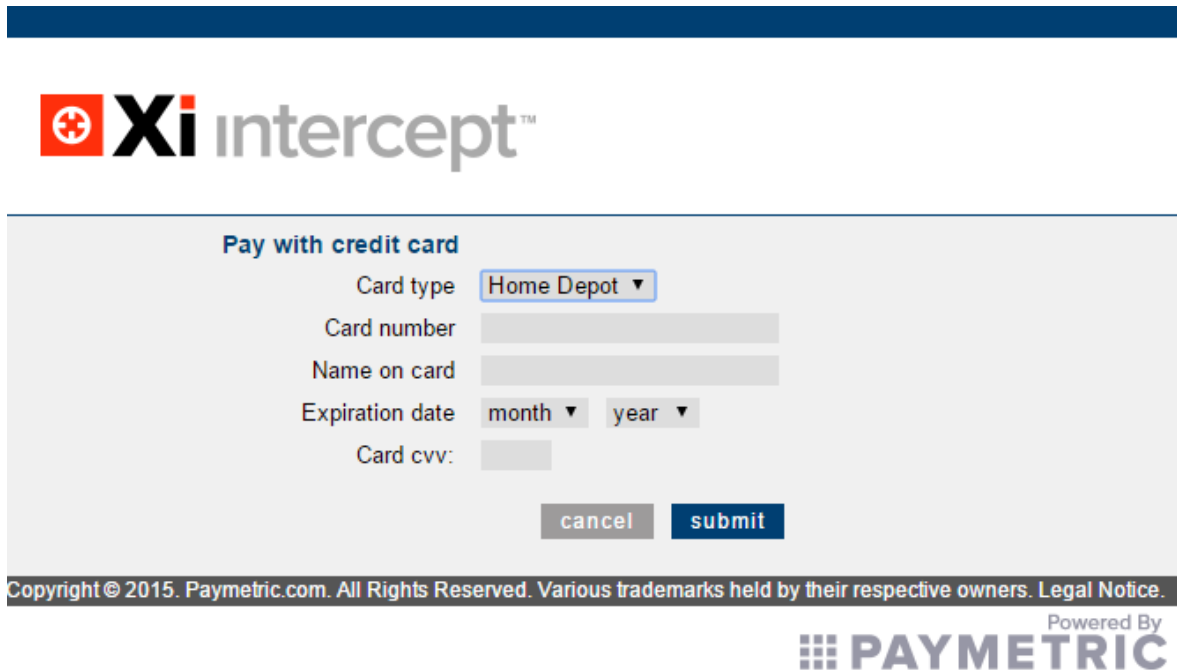
```

```

        <tag name="div">
            <label for="routingNumber"/>
            <tbxRoutingNumber/>
        </tag>
        <tag name="div">
            <label for="bankName"/>
            <tbxBankName/>
        </tag>
        <tag name="div">
            <label for="nameOnAccount"/>
            <tbxNameOnAccount/>
        </tag>
    </tag>
</echeckRadioSection>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button"/>
    <submitButton text="submit" class="submit-button"/>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">
            Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
            respective owners. Legal Notice.
        </tag>
    </tag>
</tag>
</htmlSection>

```

## 19.2.10.8 Sample 8: Credit Card (Dropdown) with Merchant Custom Card Type (Company A)



The screenshot shows a payment form titled "Pay with credit card". It includes a "Card type" dropdown menu currently set to "Home Depot". Below this are input fields for "Card number", "Name on card", "Expiration date" (with month and year dropdowns), and "Card cvv". At the bottom are "cancel" and "submit" buttons. A footer contains the copyright notice "Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice." and the Paymetric logo with the text "Powered By PAYMETRIC".

```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"></tag>
        <tag name="div" class="billing-info">
          <cardDropdownSection>
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType>
                <items>
                  <item for="american express"/>
                  <item for="mastercard"/>
                  <item for="visa"/>
                  <item for="maestro"/>
                  <customItem for="HD" text="Company A"/>
                </items>
              </ddlCardType>
            </tag>
            <tag name="div">
              <label for="cardNumber" text="Card number"/>
              <tbxCardNumber tokenize="true" class="xi-long-text"/>
              <validationMsg for="cardNumber" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="cardholderName" text="Name on card"/>
              <tbxCardHolderName class="xi-long-text"/>
              <validationMsg for="cardholderName" class="valmsg"/>
            </tag>
          </cardDropdownSection>
        </tag>
      </div>
    </tag>
  </tag>
</htmlSection>
```

```

    </tag>
    <tag name="div">
      <label for="startDate"/>
      <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
      <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
      <validationMsg for="startYear" class="valmsg"/>
    </tag>
    <tag name="div">
      <label for="expDate" text="Expiration date"/>
      <ddlExpMonth default-text="month" class="date_combos" required="false"/>
      <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
      <validationMsg for="expYear" class="valmsg"/>
    </tag>
    <tag name="div">
      <label for="cvv" text="Card cvv:"/>
      <tboxCvv class="xi-short-text"/>
      <validationMsg for="cvv" class="valmsg"/>
    </tag>
  </cardDropdownSection>
</tag>
<tag name="div" class="payment-button">
  <cancelButton text="cancel" class="cancel-button"/>
  <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
  <tag name="div" class="footer-spacing"></tag>
  <tag name="div" class="top-bottom-frame"></tag>
  <tag name="div" class="middle-bottom-frame">
    <tag name="div" class="middle-bottom-frame-content">
      Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
      respective owners. Legal Notice.
    </tag>
  </tag>
</tag>
</tag>
</htmlSection>

```

## 19.2.10.9 Sample 9: Credit Card (Radio) with Merchant Custom Card Type (Company A)

**Pay with credit card**

Master Card ☐

Visa ☐

Home Depot ☐

Card number

Name on card

Expiration date

Card cvv:

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```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"/>
        <tag name="div" class="billing-info">
          <cardRadioSection>
            <tag name="div">
              <label for="mastercard"/>
              <radCardType for="mastercard"/>
            </tag>
            <tag name="div">
              <label for="visa" text="Visa"/>
              <radCardType for="visa"/>
            </tag>
            <tag name="div">
              <customLabel for="HD" text="Company A"/>
              <radCustomCardType for="HD"/>
            </tag>
          </cardRadioSection>
          <voidTag name="br"/>
          <tag name="div">
            <label for="cardNumber" text="Card number"/>
            <tbxCardNumber tokenize="true" class="xi-long-text"/>
            <validationMsg for="cardNumber" class="valmsg"/>
          </tag>
        </div>
      </div>
    </div>
  </tag>
</htmlSection>
```

```

    </tag>
    <tag name="div">
      <label for="cardholderName" text="Name on card"/>
      <tboxCardHolderName class="xi-long-text"/>
      <validationMsg for="cardholderName" class="valmsg"/>
    </tag>
    <tag name="div">
      <label for="startDate"/>
      <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
      <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
      <validationMsg for="startYear" class="valmsg"/>
    </tag>
    <tag name="div">
      <label for="expDate" text="Expiration date"/>
      <ddlExpMonth default-text="month" class="date_combos" required="false"/>
      <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
      <validationMsg for="expYear" class="valmsg"/>
    </tag>
    <tag name="div">
      <label for="cvv" text="Card cvv:"/>
      <tboxCvv class="xi-short-text"/>
      <validationMsg for="cvv" class="valmsg"/>
    </tag>
  </cardRadioSection>
</tag>
<tag name="div" class="payment-button">
  <cancelButton text="cancel" class="cancel-button"/>
  <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
  <tag name="div" class="footer-spacing"></tag>
  <tag name="div" class="top-bottom-frame"></tag>
  <tag name="div" class="middle-bottom-frame">
    <tag name="div" class="middle-bottom-frame-content">
      Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
      respective owners. Legal Notice.
    </tag>
  </tag>
</tag>
</tag>
</htmlSection>

```

## 19.2.10.1 Sample 10: Credit Card with Card Indicator

**Pay with credit card**

Card number 4111111111111111

Name on card

Expiration date month year

Mastercard VISA American Express

cancel submit

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```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"/>
        <tag name="div" class="billing-info">
          <cardIndicatorSection>
            <tag name="div">
              <label for="cardNumber" text="Card number"/>
              <tbxCardNumber tokenize="true" class="xi-long-text"/>
              <validationMsg for="cardNumber" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="cardholderName" text="Name on card"/>
              <tbxCardHolderName class="xi-long-text"/>
              <validationMsg for="cardholderName" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="expDate" text="Expiration date"/>
              <ddlExpMonth default-text="month" class="date_combos" required="false"/>
              <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
              <validationMsg for="expYear" class="valmsg"/>
            </tag>
            <tag name="div">
              <cardTypeIndicator class="cardIndicator">
                <items>
                  <item for="mastercard" class="mcIndicator" bin-range-expression="^5" ></item>
                  <item for="visa" class="viIndicator"></item>
                  <item for="american express" class="axIndicator"></item>
                </items>
              </cardTypeIndicator>
            </tag>
          </cardIndicatorSection>
        </tag>
      </div>
    </div>
  </div>
</htmlSection>
```

```

        <tag name="div" class="payment-button">
            <cancelButton text="cancel" class="cancel-button"/>
            <submitButton text="submit" class="submit-button"/>
        </tag>
    </tag>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"/>
    <tag name="div" class="top-bottom-frame"/>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">Copyright © 2017. Paymetric.com. All
Rights Reserved. Various trademarks held by their respective owners. Legal Notice.</tag>
    </tag>
    </tag>
</tag>
</htmlSection>

```

### 19.2.10.11 Sample 11: Input Validations: Text

The screenshot displays a payment form titled "Pay with credit card". It includes the following fields and validation messages:

- Card type:** A dropdown menu with the text "select a card". Below it, a red message says "Please select the card type".
- Card number:** A text input field. Below it, a red message says "Please enter a card number".
- Name on card:** A text input field. Below it, a red message says "Please enter the card holder name".
- Expiration date:** Two dropdown menus for "month" and "year". Below them, a red message says "Please enter a valid expiration date".
- Card cvv:** A text input field. Below it, a red message says "Please enter the card security code".

At the bottom of the form are two buttons: "cancel" and "submit". Below the form is a copyright notice: "Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice." and the Paymetric logo with the text "Powered By PAYMETRIC".

```

<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
    <tag>
        <tag name="div" class="header-top"/>
        <tag name="div" class="merchant-logo"/>
        <tag name="div" class="main-content">
            <tag name="div" class="billing-content">
                <tag name="div" class="billing-header"></tag>
                <tag name="div" class="billing-info">
                    <cardDropdownSection>
                        <tag name="div">

```

```

<label for="cardType" text="Card type"/>
<ddlCardType default-text="select a card">
  <items>
    <item for="american express"/>
    <item for="mastercard"/>
    <item for="visa"/>
    <item for="maestro"/>
  </items>
</ddlCardType>
<validationMsg for="cardType" class="valmsg"/>
</tag>
<tag name="div">
  <label for="cardNumber" text="Card number"/>
  <tboxCardNumber tokenize="true" class="xi-long-text"/>
  <validationMsg for="cardNumber" class="valmsg"/>
</tag>
<tag name="div">
  <label for="cardholderName" text="Name on card"/>
  <tboxCardHolderName class="xi-long-text"/>
  <validationMsg for="cardholderName" class="valmsg"/>
</tag>
<tag name="div">
  <label for="startDate"/>
  <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
  <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
  <validationMsg for="startYear" class="valmsg"/>
</tag>
<tag name="div">
  <label for="expDate" text="Expiration date"/>
  <ddlExpMonth default-text="month" class="date_combos" required="false"/>
  <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true"/>
  <validationMsg for="expYear" class="valmsg"/>
</tag>
<tag name="div">
  <label for="cvv" text="Card cvv:"/>
  <tboxCvv class="xi-short-text"/>
  <validationMsg for="cvv" class="valmsg"/>
</tag>
</cardDropdownSection>
</tag>
<tag name="div" class="payment-button">
  <cancelButton text="cancel" class="cancel-button"/>
  <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
  <tag name="div" class="footer-spacing"></tag>
  <tag name="div" class="top-bottom-frame"></tag>
  <tag name="div" class="middle-bottom-frame">
    <tag name="div" class="middle-bottom-frame-content">

```

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```

    </tag>
  </tag>
</tag>
</tag>
</htmlSection>

```

### 19.2.10.12 Sample 12: Input Validations: Image with Tooltip

The screenshot displays the 'Xi intercept' payment interface. At the top is the 'Xi intercept' logo. Below it is a section titled 'Pay with credit card'. This section contains several input fields: 'Card type' (a dropdown menu with 'select a card' and an information icon), 'Card number' (a text input with an information icon), 'Name on card' (a text input with an information icon), 'Expiration date' (two dropdown menus for 'month' and 'year', both with information icons), and 'Card cvv' (a text input with an information icon). At the bottom of the form are 'cancel' and 'submit' buttons. Below the form is a copyright notice: 'Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice.' and the 'PAYMETRIC' logo with the text 'Powered By'.

```

<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <xiProperties error-class="input-error" inline-validation="true">
    <errorTooltip class="errorTooltip" show-effect="fadeIn" show-duration="5000" hide-
effect="fadeOut" hide-duration="5000" />
  </xiProperties>
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"></tag>
        <tag name="div" class="billing-info">
          <cardDropdownSection>
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType default-text="select a card" error-msg-style="imageTooltip">
                <items>
                  <item for="american express"/>
                  <item for="mastercard"/>
                  <item for="visa"/>
                  <item for="maestro"/>
                </items>
              </ddlCardType>
            </tag>
          </cardDropdownSection>
        </tag>
      </div>
    </tag>
  </tag>
</htmlSection>

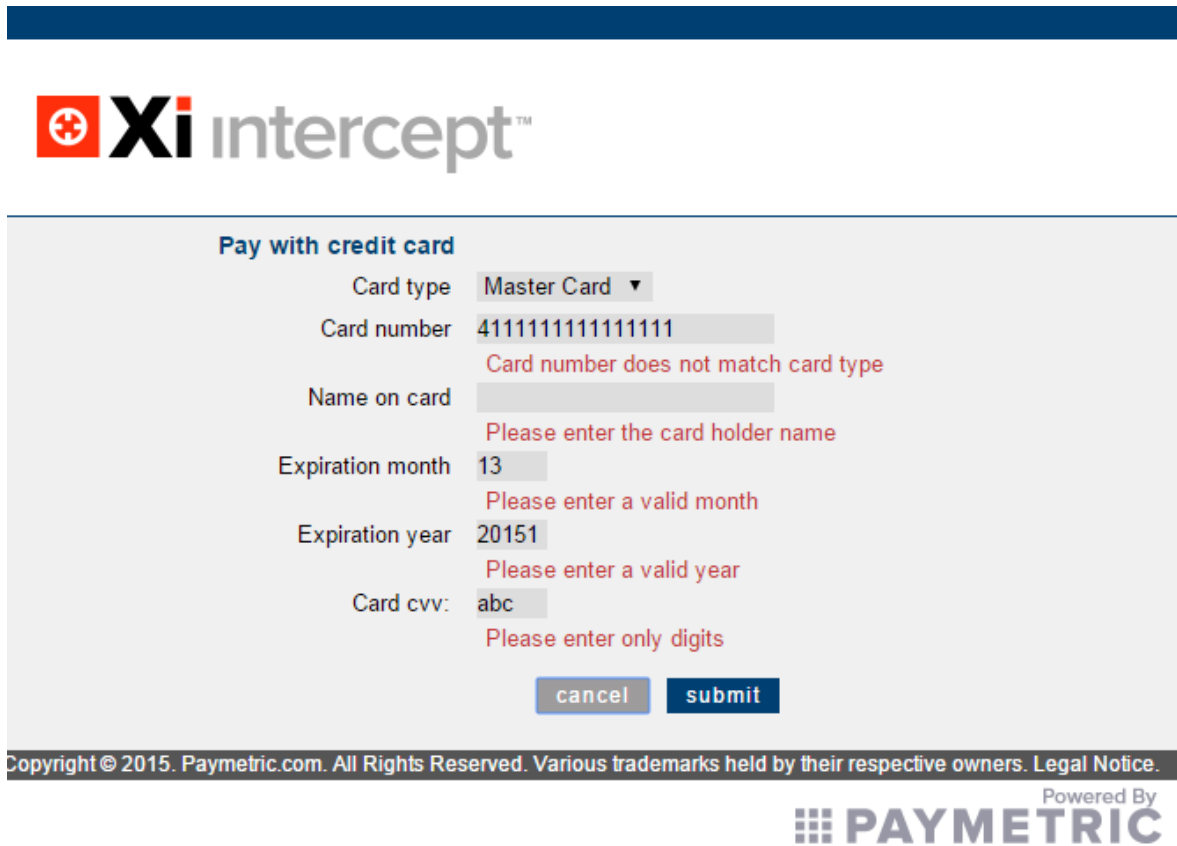
```

```

        </ddlCardType>
        <validationMsg for="cardType" error-class="valmsg-img"/>
    </tag>
    <tag name="div">
        <label for="cardNumber" text="Card number"/>
        <tboxCardNumber tokenize="true" class="xi-long-text" error-msg-
style="imageTooltip"/>
        <validationMsg for="cardNumber" error-class="valmsg-img"/>
    </tag>
    <tag name="div">
        <label for="cardholderName" text="Name on card"/>
        <tboxCardHolderName class="xi-long-text" error-msg-style="imageTooltip"/>
        <validationMsg for="cardholderName" error-class="valmsg-img"/>
    </tag>
    <tag name="div">
        <label for="startDate"/>
        <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
        <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true" error-msg-style="imageTooltip"/>
        <validationMsg for="startYear" error-class="valmsg-img"/>
    </tag>
    <tag name="div">
        <label for="expDate" text="Expiration date"/>
        <ddlExpMonth default-text="month" class="date_combos" required="false"/>
        <ddlExpYear default-text="year" class="date_combos" years-to-display="10"
required="false" exp-date="true" error-msg-style="imageTooltip"/>
        <validationMsg for="expYear" error-class="valmsg-img"/>
    </tag>
    <tag name="div">
        <label for="cvv" text="Card cvv:"/>
        <tboxCvv class="xi-short-text" error-msg-style="imageTooltip"/>
        <validationMsg for="cvv" error-class="valmsg-img"/>
    </tag>
</cardDropdownSection>
</tag>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button"/>
    <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">
            Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
            respective owners. Legal Notice.
        </tag>
    </tag>
</tag>
</tag>
</htmlSection>

```

## 19.2.10.1 Sample 13: Custom Validations



The screenshot shows a payment form titled "Pay with credit card". The form includes fields for Card type, Card number, Name on card, Expiration month, Expiration year, and Card cvv. The Card type is set to "Master Card". The Card number is "4111111111111111", which triggers the error "Card number does not match card type". The Name on card field is empty, triggering the error "Please enter the card holder name". The Expiration month is "13", triggering the error "Please enter a valid month". The Expiration year is "20151", triggering the error "Please enter a valid year". The Card cvv is "abc", triggering the error "Please enter only digits". At the bottom, there are "cancel" and "submit" buttons. The footer contains the copyright notice "Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice." and the Paymetric logo with the text "Powered By PAYMETRIC".

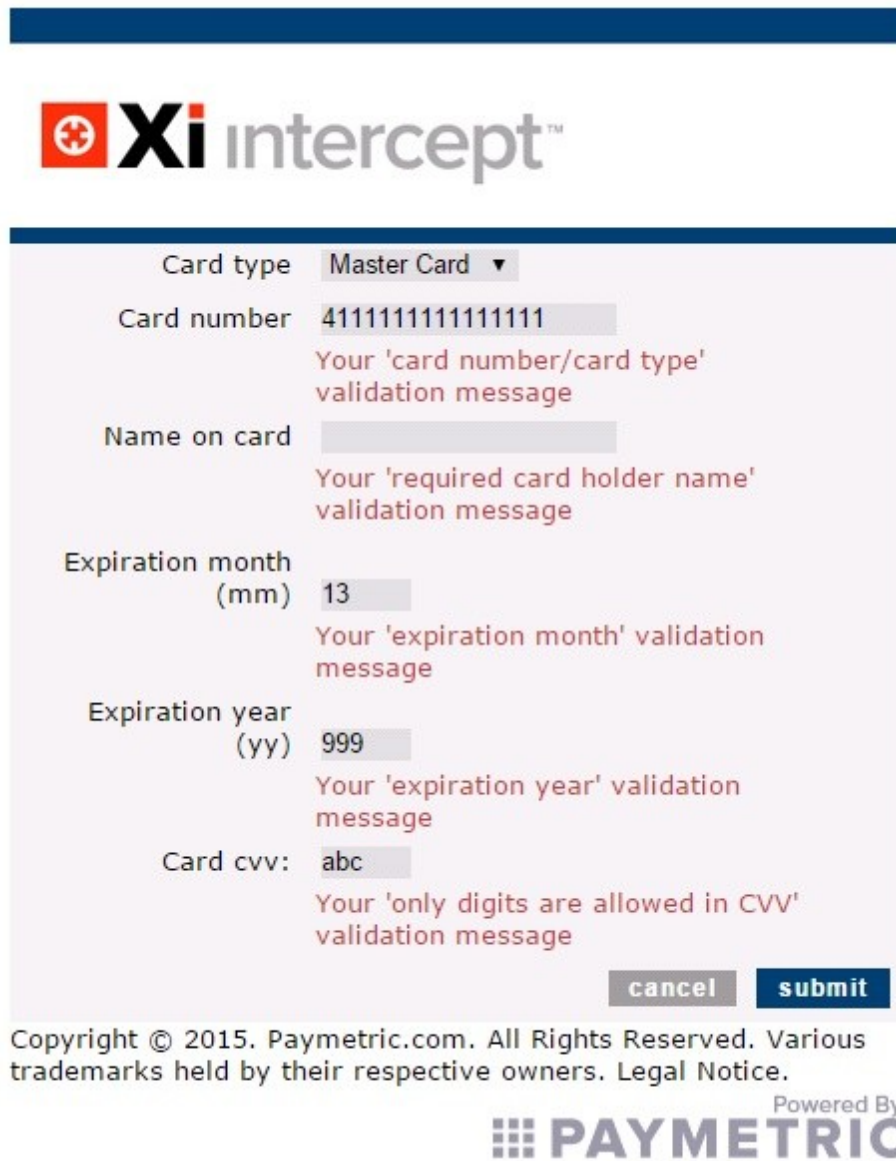
```
<htmlSection class="payment-content" xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"/>
        <tag name="div" class="billing-info">
          <cardDropdownSection>
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType default-text="select a card">
                <items>
                  <item for="american express"/>
                  <item for="mastercard"/>
                  <item for="visa"/>
                  <item for="maestro"/>
                </items>
              </ddlCardType>
              <validationMsg for="cardType" class="valmsg"/>
            </tag>
            <tag name="div">
              <label for="cardNumber" text="Card number"/>
              <tbxCardNumber tokenize="true" class="xi-long-text" luhn-check="false"/>
            </tag>
          </cardDropdownSection>
        </tag>
      </div>
    </tag>
  </htmlSection>
```

```

        <validationMsg for="cardNumber" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="cardholderName" text="Name on card"/>
        <tbxCardHolderName class="xi-long-text"/>
        <validationMsg for="cardholderName" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="startDate"/>
        <ddlStartMonth display-format="MMM" class="date_combos" required="false"/>
        <ddlStartYear class="date_combos" years-to-display="10" required="false" start-
date="true"/>
        <validationMsg for="startYear" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="expMonth" text="Expiration month"/>
        <tbxExpMonth class="xi-short-text"/>
        <validationMsg for="expMonth" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="expYear" text="Expiration year"/>
        <tbxExpYear pattern="[0-9]{4}" pattern-msg="Please enter a valid year" class="xi-
short-text"/>
        <validationMsg for="expYear" class="valmsg"/>
    </tag>
    <tag name="div">
        <label for="cvv" text="Card cvv:"/>
        <tbxCvv class="xi-short-text" minlength="3" maxlength="3" minlength-msg="CVV must
be 3 digits" maxlength-msg="CVV must be 3 digits" digits-only="true"/>
        <validationMsg for="cvv" class="valmsg"/>
    </tag>
    </cardDropdownSection>
</tag>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button"/>
    <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
<tag name="div" class="copyright-footer">
    <tag name="div" class="footer-spacing"></tag>
    <tag name="div" class="top-bottom-frame"></tag>
    <tag name="div" class="middle-bottom-frame">
        <tag name="div" class="middle-bottom-frame-content">
            Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their
            respective owners. Legal Notice.
        </tag>
    </tag>
</tag>
</tag>
</htmlSection>

```

## 19.2.10.14 Sample 14: Custom Messages



The screenshot displays the Xi Intercept payment interface. At the top is the Xi Intercept logo. Below it is a form with the following fields and validation messages:

- Card type:** A dropdown menu showing "Master Card".
- Card number:** A text field containing "4111111111111111". Below it is a red validation message: "Your 'card number/card type' validation message".
- Name on card:** A text field. Below it is a red validation message: "Your 'required card holder name' validation message".
- Expiration month (mm):** A text field containing "13". Below it is a red validation message: "Your 'expiration month' validation message".
- Expiration year (yy):** A text field containing "999". Below it is a red validation message: "Your 'expiration year' validation message".
- Card cvv:** A text field containing "abc". Below it is a red validation message: "Your 'only digits are allowed in CVV' validation message".

At the bottom right of the form are two buttons: "cancel" and "submit". Below the form is a copyright notice: "Copyright © 2015. Paymetric.com. All Rights Reserved. Various trademarks held by their respective owners. Legal Notice." and the Paymetric logo with the text "Powered By PAYMETRIC".

```
<htmlSection class="payment-content"
xmlns="Paymetric:XiIntercept:MerchantHtmlPacketModel">
  <tag>
    <tag name="div" class="header-top"/>
    <tag name="div" class="merchant-logo"/>
    <tag name="div" class="main-content">
      <tag name="div" class="billing-content">
        <tag name="div" class="billing-header"></tag>
        <tag name="div" class="billing-info">
          <cardDropdownSection>
            <tag name="div">
              <label for="cardType" text="Card type"/>
              <ddlCardType>
                <items>
                  <item for="american express"/>
                </items>
              </ddlCardType>
            </tag>
          </cardDropdownSection>
        </tag>
      </div>
    </tag>
  </htmlSection>
```

```

        <item for="mastercard"/>
        <item for="visa"/>
        <item for="maestro"/>
    </items>
</ddlCardType>
<validationMsg for="cardType" class="valmsg"/>
</tag>
<tag name="div">
    <label for="cardNumber" text="Card number"/>
    <tboxCardNumber tokenize="true" class="xi-long-text" luhn-check-
msg="Your 'luhn-check' validation message" pattern-msg="Your 'card number/card type'
validation message"/>
    <validationMsg for="cardNumber" class="valmsg"/>
</tag>
<tag name="div">
    <label for="cardholderName" text="Name on card"/>
    <tboxCardHolderName class="xi-long-text" required-msg="Your 'required
card holder name' validation message"/>
    <validationMsg for="cardholderName" class="valmsg"/>
</tag>
<tag name="div">
    <label for="startDate"/>
    <ddlStartMonth display-format="MMM" class="date_combos"
required="false"/>
    <ddlStartYear class="date_combos" years-to-display="10"
required="false" start-date="true"/>
    <validationMsg for="startYear" class="valmsg"/>
</tag>
<tag name="div">
    <label for="expMonth" text="Expiration month (mm)"/>
    <tboxExpMonth class="xi-short-text" range-msg="Your 'expiration month'
validation message"/>
    <validationMsg for="expMonth" class="valmsg"/>
</tag>
<tag name="div">
    <label for="expYear" text="Expiration year (yy)"/>
    <tboxExpYear range="0,99" range-msg="Your 'expiration year' validation
message" class="xi-short-text" exp-date="false"/>
    <validationMsg for="expYear" class="valmsg"/>
</tag>
<tag name="div">
    <label for="cvv" text="Card cvv:"/>
    <tboxCvv class="xi-short-text" minlength="3" maxlength="3" minlength-
msg="Your 'CVV must be 3 digits' validation message" maxlength-msg="Your 'CVV must be
3 digits' validation message"
        digits-only="true" digits-only-msg="Your 'only digits are
allowed in CWV' validation message"/>
    <validationMsg for="cvv" class="valmsg"/>
</tag>
</cardDropdownSection>
</tag>
<tag name="div" class="payment-button">
    <cancelButton text="cancel" class="cancel-button"/>
    <submitButton text="submit" class="submit-button"/>
</tag>
</tag>
</tag>

```

```
<tag name="div" class="copyright-footer">
  <tag name="div" class="footer-spacing"></tag>
  <tag name="div" class="top-bottom-frame"></tag>
  <tag name="div" class="middle-bottom-frame">
    <tag name="div" class="middle-bottom-frame-content">
      Copyright © 2016. Paymetric.com. All Rights Reserved. Various trademarks
      held by their respective owners. Legal Notice.
    </tag>
  </tag>
</tag>
</tag>
</tag>
</htmlSection>
```